

# Selection on Exit and Sticky Prices

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## Abstract

I study how selection on exit affects the transmission of monetary shocks. Using Dutch product-level scanner data, I document that low-sales products are more likely to exit the market and adjust their prices less often. I develop a menu-cost model with state-dependent exit, and I use it to simulate a counterfactual in which the total exit rate is unchanged, but turnover is random. Selection amplifies the real effects of aggregate nominal shocks: the composition of entry and exit raises the cumulative output response by about 15 percent, and the amplification reaches almost 35 percent in low-inflation regimes. Selective exit falls on firms whose price corrections were already near and spares stale prices elsewhere in the cross-section. At the same time, high exit risk shortens firms' horizons and weakens their incentive to adjust prices. The same mechanisms work under time-dependent pricing, but they mostly offset and leave a smaller amplification.

**JEL Codes:** E31, E50, L11.

**Keywords:** Selective exit, price stickiness, menu costs, monetary non-neutrality, firm dynamics, scanner data.

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# 1 Introduction

Selective exit changes how monetary shocks pass through the economy.<sup>1</sup> Products do not leave the market at random: using scanner data, I show that exit falls disproportionately on products with low sales. I show that this composition matters for monetary transmission. Holding the total amount of turnover fixed, an economy in which low-sales products are more likely to exit reacts more strongly to the same aggregate nominal shock than an economy in which exit is random. In a quantitative model calibrated to Dutch scanner data, the calibrated composition of entry and exit raises the one-year cumulative output response to a nominal shock by 15 percent relative to random turnover at the same exit rate. The amplification exceeds 35 percent in low-inflation environments.<sup>2</sup>

Exit replaces an old price with a new one. When exit is random, it often removes a product in the middle of a pricing spell, carrying a stale price that would otherwise remain in place. The entrant that replaces it posts a fresh price, so random turnover buys aggregate price flexibility. Selective exit works differently. Low-value firms are more likely to leave, and in the model these firms are often close to changing their prices after a bad idiosyncratic shock. Replacing them adds little flexibility, because their own price correction was already about to happen. At the same total exit rate, selection therefore redirects exit away from products whose stale prices would have survived. More mispricing remains in the cross-section, the aggregate price level catches up more slowly, and the same nominal shock has larger real effects. The amount of turnover stays fixed; its allocation changes how much mispricing survives.

Sticky-price models usually ask which firms change their prices after a nominal shock. This paper asks a prior question: which firms are still there to carry their old prices? Product markets constantly reshuffle. In United States barcode data, entry and exit of products run about four times the corresponding rates in labor markets (Broda and Weinstein, 2010), and about 8 percent of products turn over every quarter (Argente et al., 2018).<sup>3</sup> Some products disappear, new products enter, and each entrant arrives with a fresh price. Turnover therefore changes the set of prices that aggregate inflation must work through. If exit were random, this reshuffling would act like an extra source of price adjustment, because some stale prices would disappear before the firms changed them. But exit is not random in the data. Low-sales products are more likely to leave, and these products occupy a particular part of the pricing space. The pa-

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<sup>1</sup>Selection in this paper operates on the exit margin: which products leave the market. The menu-cost literature uses the same word for the adjustment margin, where the prices furthest from their desired level are the ones that change (Goloso and Lucas, 2007; Carvalho and Kryvtsov, 2021; Karadi et al., 2024). The two interact: selective exit removes the imminent adjusters that price selection works through.

<sup>2</sup>I use the terms “firm” and “product” interchangeably throughout the paper, as the model assumes one firm and one product. The empirical counterpart, however, is a product.

<sup>3</sup>The margin carries aggregate weight. In Swiss household scanner data, a cost-of-living index that accounts for preference shifts and product entry and exit differs from the conventional index by 2.6 percentage points of annual inflation on average, and by 3.7 percentage points after a large relative-price shock (Braun and Lein, 2021). Inflation also has a large firm-level component: firm-specific shocks account for 41 percent of its variance in advanced economies (Alvarez-Blaser et al., 2025).

per shows that this composition changes the amount of mispricing left in the economy after a shock. Monetary transmission depends not only on how often prices change, but also on which sticky prices survive.

I use Dutch retail scanner data from 2008–2020. The data cover 163.5 million non-promotion transactions and almost one million UPCs. I sort products by revenue and document a clear selection pattern. Low-sales products are much more likely to leave the market. Products in the bottom revenue decile exit within the next six months about 10 percent of the time, compared with less than 8 percent for products in the top half of the distribution. These same low-sales products change prices less often: bottom-decile products change prices in about 7 percent of months, compared with 10 percent in the top decile.<sup>4</sup> The gradient matches producer-price evidence: large firms change prices two to three times more often than small firms (Goldberg and Hellerstein, 2009), and firms with more products adjust more often (Bhattarai and Schoenle, 2014). The exit events also look like genuine product departures rather than local stockouts or measuring noise. Revenue-weighted, 82 percent of measured exits are national,<sup>5</sup> and exiting products already sell roughly 20 percent less than category peers one year before they disappear. These facts discipline the model. The same product state must account for sales, pricing, and survival.

I build a quantitative menu-cost model with firm heterogeneity and nominal aggregate shocks, following the menu-cost tradition of Golosov and Lucas (2007) and Nakamura and Steinsson (2010). The model has the minimum structure needed to match the facts and quantify the effect of selection. Productivity affects a firm’s revenue, its desired reset price, and its continuation value. Firms face a menu cost when they change prices, and a state-dependent exit hazard that depends on firm value. The hazard nests iid exit and a hard cutoff rule as in Hopenhayn (1992); at intermediate values exit is probabilistic, in line with Clementi and Palazzo (2016). This hazard matters twice: it changes firms’ pricing decisions, and it changes which firms remain in the cross-section. I normalize the hazard so the aggregate exit rate stays fixed. Selection therefore changes which firms leave, not how many firms leave. When I set entry and exit selectivity to zero, the model produces a random-turnover economy with the same total exit rate.

I calibrate the model to the scanner data. The calibration matches the main pricing moments and the relative size of exiting and entering products, which discipline selection on both margins. The model matches these targets and also reproduces, without targeting them directly, the sign and rank ordering of the empirical facts. I then compare the calibrated economy to a random-turnover counterfactual. In this comparison, nominal frictions and the total amount of

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<sup>4</sup>A three percentage-point gap in adjustment frequency is large in the metric that matters for monetary transmission, because the real effects of nominal shocks scale with the inverse of the frequency. If every price in the economy repriced at the top-decile frequency, the sufficient statistic approach of Alvarez et al. (2016) suggests that the real effects of a monetary shock would increase by roughly fifty percent, holding the price change distributions fixed.

<sup>5</sup>Dutch chains price nationally, in line with the uniform pricing that DellaVigna and Gentzkow (2019) document for United States chains.

turnover stay fixed, but exit and entry no longer select on firm state. The two economies have almost the same pricing moments, yet after a one-time nominal aggregate shock, the one-year cumulative impulse response (CIR) of output rises by about 15 percent in the calibrated economy. I also compare an economy with selection only on entry to one with selection only on exit. Moving selectivity from the entry margin to the exit margin raises the cumulative output response by about 18 percent. The amount of churn alone explains little; the composition of churn generates essentially the whole turnover effect on monetary transmission.

Selection works through two channels. First, firms price while taking their exit risk into account. A firm with a high exit hazard has a shorter expected horizon, so it gains less from paying the menu cost today. It lets its price move further away from its desired price before adjusting. I call this the option-value channel, in line with the menu-cost literature in which changing price can be thought of as exercising an option (Stokey, 2009).<sup>6</sup> Second, selection changes which prices disappear from the cross-section. Low-value firms often have bad idiosyncratic draws, high marginal costs, and prices that they would like to raise. Many of them are close to the boundary of their inaction region. When one of these firms exits, the price increase it was about to make disappears with it, and the entrant arrives with a fresh price. In this sense, the economy still gets a new price, but turnover replaces a price correction that was already about to happen. I call this the censoring channel, because selective exit censors imminent price adjustments.

I quantify the two channels with four counterfactuals. I let firms make pricing decisions under one perceived exit hazard, while the cross-section evolves under another realized exit hazard. This separates the effect of exit risk on firm behavior from the effect of exit on the distribution of prices. The option-value channel alone accounts for about one third of the gap between selective and random turnover. The censoring channel alone is small, because entrants replace most of the price correction that exiting firms would have made. The interaction is the largest component. High exit risk makes exposed firms wait longer and builds a larger stock of stale prices. Selective exit then protects that stock by moving exit away from firms in the interior of their pricing spells, where random turnover would have removed persistent mispricing.

I develop a simple analytical model to make this logic transparent. The price gap is the distance between a firm's posted price and the price it would choose if adjustment were free. I hold the inaction band fixed and study how exit changes the cross-section of gaps. At zero trend inflation, the cumulative output response equals the variance of price gaps divided by the variance of idiosyncratic shocks, as in Alvarez et al. (2016) and Alvarez et al. (2022). This formula gives a simple way to value exit at each point in the pricing spell. Exit has no aggregate effect at a fresh price, because the price is already correct. It also has little effect at the adjust-

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<sup>6</sup>Oikawa and Ueda (2018) derive a version of this effect with a hazard common to all firms: under positive trend inflation, a higher creative-destruction rate leads firms to postpone price resets, because exit may arrive soon after the menu cost is paid. A value-dependent hazard concentrates the effect on the firms most likely to leave.

ment boundary, because the firm was about to reset its price anyway. Exit matters most in the interior of the inaction region. There, it removes a stale price that would otherwise remain in place, so it works like a subsidy to price flexibility, similar to the role of entry in [Bilbiie \(2021\)](#). Random exit draws from the whole cross-section and often hits these interior firms. Selective exit draws more heavily from firms near the boundary. I derive a sufficient condition under which reallocating a fixed exit flow toward the boundary and away from the interior raises the cumulative output response. The exit hazard measured from the calibrated model satisfies this condition. The analytical model establishes the sign of the composition effect; the option-value channel and its interaction with censoring generate most of the quantitative magnitude.

The amplification is non-linear in trend inflation and depends on inflation regimes. At low trend inflation, especially around zero or mild deflation, few firms are close to changing prices on their own. Random exit then removes many stale prices that would otherwise remain in place, while selective exit mostly removes firms whose corrections were already near. This makes the gap between random and selective turnover large. At an annual trend inflation rate of 1.2 percent, the amplification reaches almost 35 percent. It declines as trend inflation rises and drops sharply once trend inflation exceeds 6 percent. Larger shocks have the same effect. At the calibrated annual trend inflation rate of 3.36 percent, raising the nominal shock to 5 percent cuts the amplification from 14.6 to 7.1 percent. High trend inflation ([Gagnon, 2009](#); [Alvarez et al., 2019](#)) and large shocks ([Karadi and Reiff, 2019](#); [Cavallo et al., 2024](#)) push more firms toward price adjustment, so random exit becomes more likely to hit firms that were already about to reprice. Random and selective exit then look more alike.

The main result does not rely on menu costs, though the mechanism changes under Calvo pricing. I replace the menu cost with a Calvo lottery and keep the household, productivity process, exit hazard, entry block, and adjustment frequency fixed. The selective-turnover benchmark raises the one-year cumulative output response from 0.552 to 0.565, an amplification of about 2.4 percent. The effect is smaller because two forces push in opposite directions. With exogenous adjustment timing, selective exit removes firms with large price gaps and lets entrants correct those prices early. This dampens the response. At the same time, high exit risk shortens firms' expected horizon. Firms facing high exit risk put more weight on current profits and set reset prices closer to their static optimum. This amplifies the response. The two forces almost offset. Turnover composition matters under both pricing technologies, but the firm-level mechanism depends on how firms change prices.

Turnover therefore acts as an indirect repricing technology. Random turnover spends replacement on stale prices and adds flexibility to an economy with sticky prices. Selective turnover spends replacement on corrections that were already due and leaves persistent mispricing elsewhere in the market. The frequency of price changes and the rate of turnover therefore do not summarize aggregate price flexibility. The allocation of replacement across firms matters too. Churn composition can change monetary transmission even when the exit rate and the price-adjustment frequency barely move. Menu-cost models usually abstract from turnover ([Golosov and Lucas, 2007](#); [Nakamura and Steinsson, 2010](#); [Midrigan, 2011](#)) or assign it a uniform hazard ([Oikawa and Ueda, 2015](#)), and measured adjustment frequencies exclude price

changes at product substitution (Nakamura and Steinsson, 2008; Klenow and Kryvtsov, 2008); Nakamura and Steinsson (2012) show the adjustment lost this way is large. Both approaches miss this margin precisely when it matters most: in ordinary low-inflation periods.

**Related literature.** This paper connects to four strands of literature. The first studies what determines the size of monetary non-neutrality in menu-cost economies. Golosov and Lucas (2007) calibrate a menu-cost model with idiosyncratic productivity shocks to the frequency and size of U.S. price changes. They find that money has small and short-lived real effects. The reason is selection on the adjustment margin: firms that pay the menu cost are the firms whose prices are furthest from their desired prices, so a few price changes absorb most of the shock. Carvalho and Kryvtsov (2021) measure this price selection in CPI microdata and find it strong within product categories; Karadi et al. (2024) find that misaligned prices adjust more often, yet selection after identified aggregate shocks is weak. Selection in those papers concerns which prices change and lowers non-neutrality when strong. Selection here concerns which products survive, and it raises non-neutrality. Later work restores larger real effects by changing the pricing problem or the structure of marginal cost, through variable factor utilization (Neiss and Pappa, 2005), sectoral heterogeneity and intermediate inputs (Nakamura and Steinsson, 2010), multiproduct scope economies (Midrigan, 2011), strategic complementarities (Mongey, 2021; Wang and Werning, 2022; Alvarez et al., 2023), variable demand elasticity (Aruoba et al., 2024), and an adjustment frequency that responds to inflation (Blanco et al., 2024, 2025). I take a different route. The pricing problem stays close to Golosov and Lucas (2007). Monetary non-neutrality changes because exit changes which firms remain in the cross-section. Exit lands on firms whose price changes were already near, spares firms whose wrong prices had more time left, and makes the real effects of money larger.

The second strand links firm dynamics and monetary transmission. One side of this literature studies entry and exit in economies with nominal rigidities. These papers make the number of producers a margin of aggregate transmission and study how monetary shocks move entry, exit, variety, demand, and markups (Bergin and Corsetti, 2008; Bilbiie et al., 2008; Lewis and Poilly, 2012; Bilbiie et al., 2014; Etro and Rossi, 2015; Uusküla, 2016; Bilbiie, 2021). Bilbiie (2021) shows that variety adjustment can substitute for posted-price adjustment: money can become neutral even when no incumbent price changes. Random replacement plays a similar role here because it removes stale prices and brings in new ones. The key object is how much of this flexibility survives once exit selects across firms. Oikawa and Ueda (2015, 2018) add entry and exit to menu-cost economies and vary the amount of turnover. My main experiment fixes the amount of turnover and varies its composition. Read (2020) studies the closest environment, with endogenous selective exit among heterogeneous firms in a New Keynesian model. In his model, exiting firms produce a competitive undifferentiated input, while sticky prices sit in a separate symmetric Rotemberg sector. In my model, continuation value, inherited price, and the exit hazard belong to the same product. This joint state opens the option-value and censoring channels and lets scanner data on exit and pricing discipline their size. Newer work asks how monetary policy changes turnover in New Keynesian economies with heterogeneous

firms: contractions cull low-productivity varieties, easings keep marginal firms alive, and policy uncertainty raises exit (Hamano and Zanetti, 2022; Colciago and Silvestrini, 2022; Hartwig and Lieberknecht, 2025; Fasani et al., 2023; Gamber, 2026). I ask the reverse question: how the selectivity of exit changes the effects of money. Answering this question requires a pricing block on the firms that enter and exit.

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The other side studies the composition of firm dynamics itself. Exit can cleanse firms with low productivity, destroy valuable matches, or move over the business cycle (Caballero and Hammour, 1994; Barlevy, 2002; Osotimehin and Pappadà, 2017; Lee and Mukoyama, 2015; Foster et al., 2016). Entry composition matters as well: differences at birth help determine which firms grow and survive (Sedláček and Sterk, 2017; Sterk et al., 2021). On the product side, sales fall through most of the life cycle as newer products crowd older ones out (Argente et al., 2024); the low-sales exiters in my data sit at the end of that decline. I use the same empirical object (which products leave and which products enter) and value it in units of aggregate price flexibility. Selection that raises average productivity can also leave more stale prices standing. On the empirical pricing side, the size gradients echo evidence that larger producers adjust prices more frequently (Goldberg and Hellerstein, 2009), and that firms selling more goods adjust more often (Bhattarai and Schoenle, 2014). Argente and Yeh (2022) show that young products

change prices more often and by more than mature products in U.S. scanner data. In their model, firms learn about demand by experimenting with prices, and this learning motive amplifies the real effects of nominal shocks. Their amplification comes from the pricing choices of young survivors at the start of a product's life. Mine comes from which products the economy deletes at the end of it, holding the death rate fixed.

The third strand compares state-dependent and time-dependent pricing. [Auclert et al. \(2024\)](#) prove a first-order equivalence between the menu-cost Phillips curve and a mixture of Calvo Phillips curves. [Alvarez et al. \(2016\)](#) show that, across a large class of pricing models, the cumulated output response to a small nominal shock equals the kurtosis of price changes divided by their frequency. [Alvarez et al. \(2017\)](#) extend the equivalence to whole impulse responses and show where it breaks: large shocks activate the nonlinear adjustment margin in state-dependent economies. [Alvarez et al. \(2022\)](#) extend the statistic to generalized hazards in the tradition of [Caballero and Engel \(1999\)](#), and [Baley and Blanco \(2021\)](#) extend it to lumpy adjustment more generally. [Alvarez et al. \(2024\)](#) and [Hong et al. \(2023\)](#) test these predictions in French CPI data and U.S. producer-price data. I add selective turnover to this comparison. Selection amplifies monetary shocks under both menu-cost and Calvo pricing, but the forces differ. Under menu costs, selection changes both the option value of price adjustment and the distribution of surviving price gaps. Under Calvo, the timing of adjustment is fixed, so selection works mainly through reset prices and through which gaps exit removes. The Calvo economy therefore gives a smaller amplification.

Finally, the fourth strand studies nonlinear monetary transmission. The real effects of nominal shocks depend on the inflation regime and on the size of the shock. Price-adjustment frequency rises with inflation ([Gagnon, 2009](#); [Alvarez et al., 2019](#)). Large cost shocks recruit extra adjusters and pass through quickly ([Karadi and Reiff, 2019](#); [Auer et al., 2021](#); [Cavallo et al., 2024](#)). Volatility activates the same margin in recessions ([Vavra, 2014](#)). Strong nominal pressure makes firms more likely to adjust, so the economy becomes more flexible when the trend or the shock is large. I show that the same logic changes the value of selective exit. Trend inflation and large shocks move more firms toward their adjustment boundaries. Random exit then hits more firms that were already close to repricing, so random and selective exit become more similar. The amplification from selection is therefore largest in low-inflation environments and falls when nominal pressure is high.

Section 2 documents the exit and pricing facts in the scanner data. Section 3 lays out the model. Section 4 reports the calibration, the aggregate result, and the decomposition into channels. Section 5 varies trend inflation and the size of the shock. Section 6 derives the mechanism in closed form in a reduced model. Section 7 extends the exercise under Calvo pricing. Section 8 concludes.

## 2 Empirical evidence

This section documents three cross-sectional patterns in product-level data, organized by product revenue. Smaller products exit the data more often, adjust their prices less often, and adjust

**Table 1: Descriptive statistics.**

| Complete panel                 | Value          |
|--------------------------------|----------------|
| Sample period                  | 2008m1–2020m12 |
| Transactions (non-promotion)   | 163,524,586    |
| Recorded purchase value (mn €) | 341            |
| Unique products (UPCs)         | 999,062        |
| Brands                         | 20,404         |
| Retail chains                  | 231            |
| Product categories (groups)    | 719 (86)       |

Notes: The complete panel counts all non-promotion transactions, 2008–2020.

by larger amounts when they do.

## 2.1 Data and coverage

I use Dutch retail scanner data for the years 2008–2020 provided by [AiMark](#). The dataset covers fast-moving consumer goods (FMCG): food, beverages, cleaning products, beauty, and over-the-counter drugs. The panel captures a substantial share of supermarket expenditure: about €1.8 billion of sector revenue in 2008, while total sector turnover is about €30 billion in the same year.<sup>7</sup> This setting makes it a good case to study price stickiness and product dynamics: the Netherlands is a fairly homogeneous economy with little cross-province variation, and the long structure of the panel allows me to obtain enough variation in the product lifecycles. Products are sold across chains: each observed store belongs to a national retail chain. For each purchase I observe the purchase date and province, the retail chain, the UPC barcode, units, whether the sale was on promotion, brand and group and category assignments, from which I construct unit values. Product groups are larger groups of similar products (e.g. *cat food* or *soft cheese*); categories represent a more granular subdivision within a group (e.g. *canned cat food*, *foreign cheese*). In total, I have 999,062 unique UPCs that fall into 86 product groups and 719 product categories. The goods belong to 20,404 unique manufacturing brands, and are sold across 231 different retail chains.<sup>8</sup>

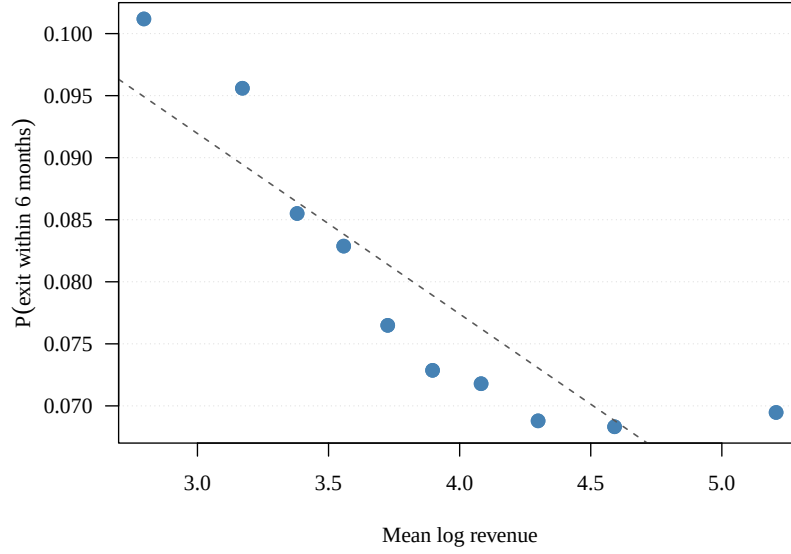
Table 1 describes the data. The raw panel records 163.5 million non-promotion transactions over 156 months. From these I build cells, a UPC in a chain in a province, collapse each cell’s transactions within a month to a modal posted price, and keep the cell-months and product lifecycles that satisfy coverage filters. Appendix A.1 details each step of the construction: the unit-value price, the coverage filters, the removal of unflagged sales, and the classification of price changes, entry, and exit.

I summarize the size of a product by its lagged log revenue: the natural log of the previous month’s revenue at the product-chain-province cell. Lagging by one month avoids the

<sup>7</sup><https://www.ad.nl/economie/cbl-omzetgroei-voor-supermarkten~a46bd8d9/>

<sup>8</sup>In Section A.2 in the Appendix I show that price changes of the same product are synchronized across chains and move in the same direction, so I read them as coming from the product rather than from the retailer.

**Figure 1: Six-month exit probability by size decile.**



Notes: For each decile of lagged log revenue, the figure plots the share of product-months for which the cell exits within the next six months. Each decile contains 25,074 product-months. The dashed line is the OLS fit through the ten bin means; no fixed effects.

mechanical contemporaneous link between today's price change and today's revenue. The headline scatters use this lagged log revenue as the size proxy. As a robustness check, Appendix A.3 reproduces the same exercises with a smoother twelve-month rolling-mean proxy that captures persistent product size at the cost of a slightly smaller sample.

## 2.2 Three patterns by product size

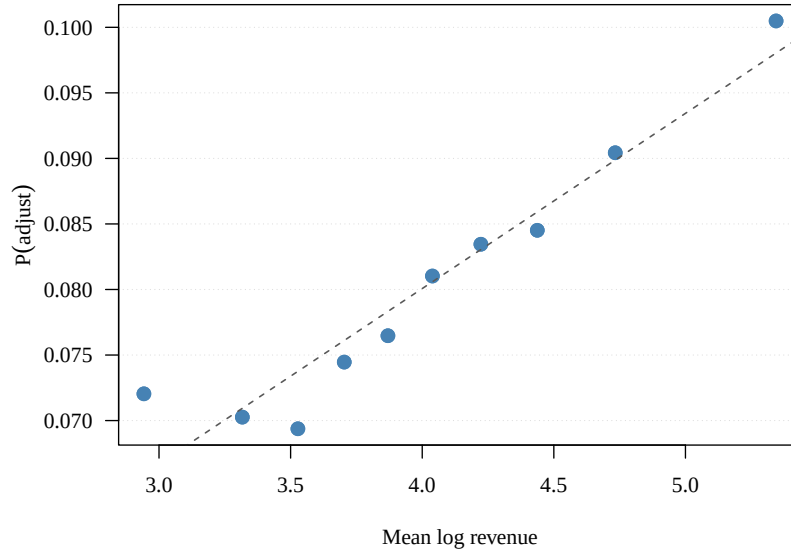
I sort product-months into deciles of lagged log revenue and report three bin scatters, each summarizing one cross-sectional moment of interest by size. The accompanying linear regressions, with and without category and month fixed effects, are reported alongside the figures.

**Fact 1.** *Products with lower revenue are more likely to exit.*

Figure 1 reports the share of product-months that exit within six months, by size decile. The pattern is monotone in size and clearly non-linear: the bottom decile exits at a six-month rate of about ten percent, the top half of the size distribution settles below eight percent, and most of the gradient is concentrated in the bottom three deciles. A pooled linear regression of the exit-within-six-months indicator on lagged log revenue, with no fixed effects and standard errors clustered by category, returns a slope of  $-0.014$  per log-unit of revenue ( $SE = 0.005$ ,  $p < 0.01$ ). The slope is small in absolute terms because the dependent variable is itself a small probability; multiplying by the bottom-to-top decile gap of about two and a half log-units of revenue gives the three-percentage-point spread visible in the figure. With category and month fixed effects the slope goes to  $-0.045$  ( $SE = 0.005$ ,  $p < 0.001$ ): even within a category and month, smaller products are more exit-prone, and the gradient is steeper than in the pooled case.

**Fact 2.** *Products with lower revenue are associated with less frequent price changes.*

**Figure 2: Adjustment frequency by size decile.**



Notes: For each decile of lagged log revenue, the figure plots the mean indicator that the cell adjusted its price in the current month. Each decile contains 16,908 product-months. The dashed line is the OLS fit through the ten bin means; no fixed effects.

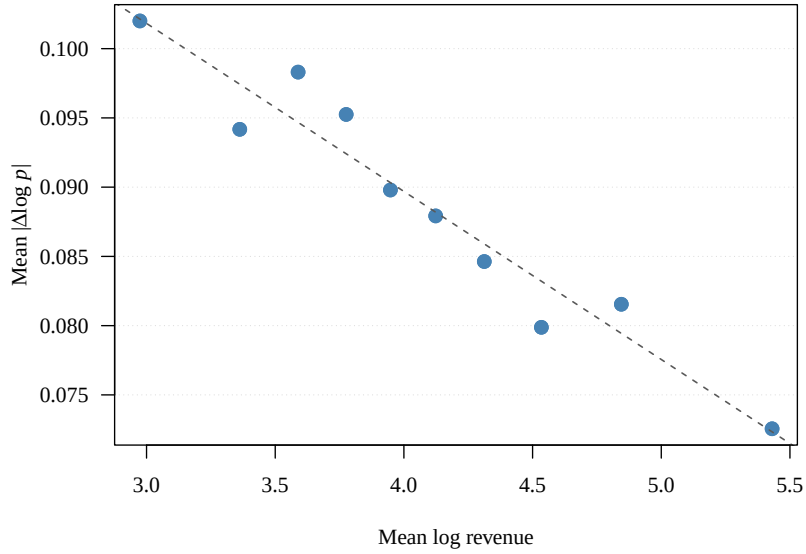
Figure 2 reports the mean monthly adjustment frequency by size decile. Bottom-decile cells adjust prices about seven percent of the time, top-decile cells about ten percent. The figure shows raw bin means with no fixed effects; the pooled linear regression of the adjustment indicator on lagged log revenue returns a slope of  $+0.013$  per log-unit of revenue ( $SE = 0.004$ ,  $p < 0.001$ ), which translates into the three-percentage-point bottom-to-top spread visible in the panel. Once category and month fixed effects absorb between-category and time variation, however, the slope flattens to  $-0.003$  and is not statistically distinguishable from zero. The frequency gradient is therefore primarily a between-category fact: high-revenue categories adjust more often, but within a given category there is little gradient by revenue. The exit gradient, by contrast, survives the inclusion of category fixed effects. Of the three patterns documented in this section, exit is the one that holds robustly within category; the frequency and size gradients reflect both within- and between-category variation, with the between component dominant.

**Fact 3.** *Products with lower revenue, when changing price, adjust by a larger amount.*

Figure 3 reports the mean absolute log price change conditional on a price change, by size decile. The pattern mirrors frequency: bottom-decile adjustments are about ten log points on average, top-decile adjustments about seven. The pooled slope is  $-0.012$  per log-unit of revenue ( $SE = 0.003$ ,  $p < 0.001$ ); with category and month fixed effects the slope is  $-0.002$  ( $SE = 0.002$ ), small but with the same sign and just outside conventional statistical significance.

Taken together, the three patterns identify the same set of smaller products as both more exit-prone (Figure 1) and less frequent, larger price-adjusters (Figures 2 and 3). The three patterns survive substituting a smoother twelve-month rolling proxy for size, and the exit gradient survives restricting the outcome to national exits and to exits after which the brand leaves the

**Figure 3: Mean absolute log price change by size decile.**



Notes: For each decile of lagged log revenue, the figure plots the mean of the absolute log price change among product-months that adjust. Each decile contains 1,357 adjustment events on average. The dashed line is the OLS fit through the ten bin means; no fixed effects.

category entirely; Appendix A.3 collects these checks.

The same joint signature shows up at the product-month level in regression form. Table 2 reports two linear probability models on the sample of non-adjusting product-months for which the next six months are observable: either the product is observed continuously through  $t + 6$ , or the product exits the data at some point in  $[t + 1, t + 6]$ . This filter ensures that the outcome of interest (whether the next event is a price change or an exit) can be classified for every observation. The dependent variables are indicators for the next event in  $[t + 1, t + 6]$  being a price change of either sign (column 1) and being exit (column 2); the right-hand-side variable is the lagged log revenue of the product-chain-province cell, with category and month fixed effects and standard errors clustered at the category level. A one-log-unit increase in lagged revenue (about  $2.7\times$  in level) raises the probability that the next event is a price change by 11.8 percentage points, against a base rate of about 25%. The same size dimension predicts a 3.6-percentage-point lower probability of exit-first, against a base rate of about 6%. Smaller, lower-revenue products both exit more and adjust less, in the same product-months.

Section A.3 defends these patterns against four possible confounders. First, they survive if I only look at national exit (products leaving every chain and every province) and at brand exits (the brand leaving the category entirely). Second, all three patterns are stable over time: they hold in the two halves of the sample, showing they are not the result of some one-time aggregate shock. Third, the exit pattern hinges on revenue, which mixes price and quantity: a small product could simply be a cheap one. This is not the case, as I show that these are low-volume/low-sales products, so exit is more likely among products with lower sales, and not cheaper ones. Fourth, the patterns are also robust to brand structure. The exit result holds also if I restrict multi-product brands to the highest-revenue product, showing that it is not just

**Table 2: Firm-level orthogonality test of exit and price-changing activity.**

|                               | (1) $1\{\text{first} = \text{change}\}$ | (2) $1\{\text{first} = \text{exit}\}$ |
|-------------------------------|-----------------------------------------|---------------------------------------|
| $\log \text{revenue}_{u,t-1}$ | +0.118***<br>(0.011)                    | −0.036***<br>(0.004)                  |
| Category FE                   | ✓ (185)                                 | ✓ (185)                               |
| Month FE                      | ✓ (153)                                 | ✓ (153)                               |
| Cluster                       | Category                                | Category                              |
| $N$                           | 233,146                                 | 233,146                               |

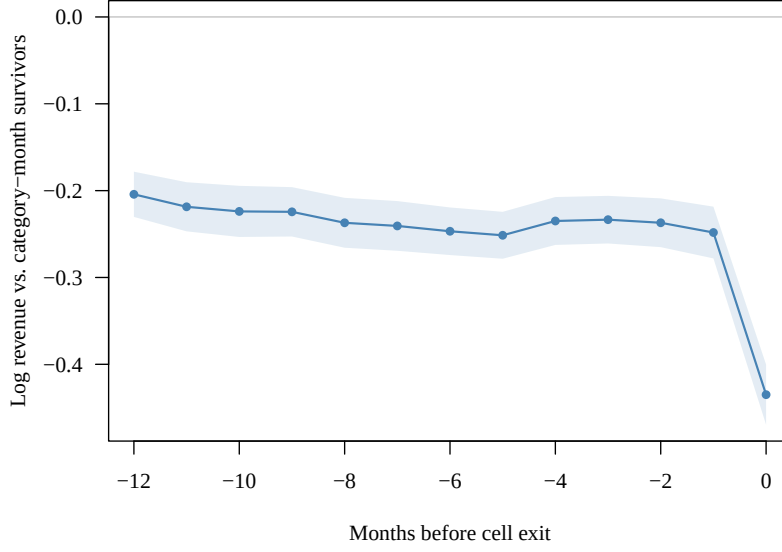
Notes: Linear probability models on the sample of non-adjusting product-months that either have six months of forward visibility or exit within six months. Dependent variable in column (1) is an indicator for the first event in  $[t + 1, t + 6]$  being a price change of either sign; in column (2), an indicator for the first event being exit. The right-hand-side variable is the natural log of the previous month’s revenue at the product-chain-province cell. Category and month fixed effects are additive; standard errors clustered at the category level in parentheses. \*\*\* denotes significance at the 0.1 percent level.

a confounder of smaller brands carrying smaller products.

Finally, the model ties the exit hazard to a product leaving the market (or the shelf) because of profitability reasons, so the exit margin in the data must record the death of a product, not a chain rotating its shelf or a brand relabeling a barcode. Three diagnostics, collected in Appendix A.4, support the former interpretation. First, exit is mainly national: weighted by revenue, around eighty percent of measured exit leaves the entire panel within six months, and among products sold in several chains, seventy percent. Second, exit ends a declining stream of revenues. Figure 4 follows exiting products over their final year: the average exiting product already sells about twenty percent below surviving products in its own category and month a full year before it disappears, and drifts further down as the end approaches. Third, relabeling accounts for little of measured exit. The concern is that a brand retires one barcode and launches a new one for various reasons (unrelated to the product’s sales), so the data record an exit where no product actually died. The exit itself seems to raise the chance of a nearby launch by only twelve percentage points, and the exiting product had been selling twenty percent below its peers for a year. Therefore, even if product exit in part reflects brands experimenting with new variations on existing products, the data suggest that any exit, regardless of its type, is driven by low sales.

Section 3 builds the minimal model that reconciles the three price patterns: a menu-cost economy with idiosyncratic productivity shocks and selective exit. The menu cost generates the frequency and size gradients: smaller products tend to be lower-productivity products, and they therefore adjust less often and by more when they do. Selective exit generates the exit gradient by tying the exit hazard to firm productivity: lower-value firms exit more, and lower-value firms are smaller. The menu cost is taken as a reduced-form wedge for any friction that makes price adjustment costly, rather than as a literal physical cost of repricing.

**Figure 4: Revenue before exit, all exiting products.**



*Notes:* Event-study coefficients from a regression of log monthly revenue on dummies for the twelve months up to a product’s exit, with category-month fixed effects, on the turnover sample of Table 1. The reference group is all product-months of non-exiting products, so each point is the exiting products’ log revenue relative to surviving products in the same category and month. The shaded band is a ninety-five percent confidence interval, standard errors clustered by category.

### 3 Model

This section builds a general equilibrium menu-cost model with selective firm turnover and aggregate risk. The pricing block is in the tradition of [Golosov and Lucas \(2007\)](#), [Burstein and Hellwig \(2007\)](#), [Nakamura and Steinsson \(2008\)](#), and [Aruoba et al. \(2024\)](#): a representative household, a competitive final-good producer, and a mass of intermediate producers that pay a menu cost to change their price. My contribution is in showing how selection through the entry and exit margins dampens, or amplifies, the real effects of an aggregate shock. Incumbents face a state-dependent exit hazard that decreases in the firm’s value, so low-value firms exit disproportionately. Selection operates on both the entry and exit margins, governed by two elasticities that I recover from the data and that determine how strong selection is. Setting both elasticities to zero collapses the model to uniform turnover, which I use to isolate the effect of selection. The model is then mapped to the Dutch scanner data, targeting key pricing moments; the two selectivity elasticities are calibrated to the relative size of exiting and entering products. The model recovers, untargeted, the three facts from Section 2.

#### 3.1 Household

A representative household supplies labor to the intermediate producers, holds a complete set of state-contingent securities, and owns the firms. Its problem is

$$\max_{C_t, h_t, B_{t+1}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [\log C_t - \chi h_t] \quad \text{s.t.} \quad P_t C_t + Q_t B_{t+1} \leq B_t + W_t h_t + \Pi_t, \quad (1)$$

where  $C_t$  is consumption of the final good,  $P_t$  its price,  $W_t$  the nominal wage,  $Q_t$  the price vector of the state-contingent securities  $B_{t+1}$  purchased at  $t$ ,  $B_t$  the realized payoff of the securities bought at  $t - 1$ , and  $\Pi_t$  aggregate profits rebated from the intermediate producers, net of menu costs. First-order conditions deliver labor supply

$$\frac{W_t}{P_t} = \chi C_t,$$

and the stochastic discount factor

$$\Xi_t = \beta^t \mathbb{E}_t \left( \frac{C_t}{C_{t+1}} \right).$$

### 3.2 Aggregate nominal demand

Monetary policy targets nominal expenditure  $S_t \equiv P_t C_t$ , which follows a random walk with drift,

$$\log S_t = \bar{\pi} + \log S_{t-1} + \sigma_S \eta_t, \quad \eta_t \sim N(0, 1). \quad (2)$$

With log-linear preferences, the labor supply equation implies  $W_t = \chi S_t$ , so firms' marginal cost moves one-for-one with nominal demand.

Monetary policy in this economy sets a path for nominal expenditure unlike standard New Keynesian models in which it targets the nominal interest rate.<sup>9</sup> The specification is the workhorse of the quantitative menu-cost literature (Golosov and Lucas, 2007; Nakamura and Steinsson, 2010; Midrigan, 2011; Aruoba et al., 2024). The regime also has normative standing of its own: Bullard and Di Cecio (2025) characterize nominal GDP targeting as optimal under household heterogeneity and nominal contracting frictions. Because (2) is common to every economy I solve, all of them face the same sequence of nominal demand shocks, and real outcomes differ only through the cross-section of firms. How selective exit interacts with a rule that responds to inflation and output is a question I leave for future research.

### 3.3 Final good producer

A perfectly competitive final-good producer combines a measure  $\mu_t$  of intermediate varieties via a CES technology, taking input prices as given and earning zero profits. Demand for variety  $i$  and the aggregate price are

$$y_{it} = Y_t \left( \frac{p_{it}}{P_t} \right)^{-\theta}, \quad P_t = \left( \int p_{it}^{1-\theta} d\mu_t \right)^{\frac{1}{1-\theta}}, \quad (3)$$

---

<sup>9</sup>The two setups differ in the instrument, not in the object: a price (the nominal interest rate) in the standard New Keynesian model, a quantity (nominal spending) here. Both pin down the path of nominal aggregate demand, and the real response is whatever the pricing friction determines. In a cashless economy the central bank can set the nominal interest rate so as to deliver a target path for nominal aggregate demand (Woodford, 2003).

where  $\theta > 1$  is the elasticity of substitution and  $\mu_t$  is the cross-sectional measure of operating firms.

### 3.4 Intermediate producers

There is an endogenous mass of intermediate producers indexed by  $i$ . Each operates a linear labor technology

$$y_{it} = z_{it} l_{it},$$

where  $z_{it}$  is firm-specific productivity that follows an AR(1) process in logs with conditional cdf  $F$ ,

$$\log z_{it} = \rho \log z_{i,t-1} + \sigma_\varepsilon \varepsilon_{it}, \quad \varepsilon_{it} \sim N(0, 1).$$

A firm's static real flow profit, given its inherited price  $p_{i,t-1}$  and its choice of price  $p_{it}$ , is

$$\pi(p_{it}, z_{it}, \mathcal{S}_t) = \frac{p_{it}}{P_t} y_{it} - \frac{W_t}{P_t} l_{it} - \frac{W_t}{P_t} f \mathbb{1}\{p_{it} \neq p_{i,t-1}\},$$

where  $f$  is a labor-denominated menu cost paid whenever the firm changes its price, and  $\mathcal{S}_t = (P_t, Y_t, W_t, \mu_t)$  collects the aggregate state, including the cross-sectional measure of firms.

**Incumbent's problem.** Let  $s = (p_{-1}, z, \mathcal{S})$  denote the firm's full state. The incumbent's value function, conditional on having survived this period's exit draw, is

$$V(s) = \max\{V_N(s), V_A(s)\}, \tag{4}$$

where  $V_N$  is the value of keeping the inherited price and  $V_A$  is the value of paying the menu cost and resetting:

$$\begin{aligned} V_N(p_{-1}, z, \mathcal{S}) &= \pi(p_{-1}, z, \mathcal{S}) + \mathbb{E}[\Xi (1 - \lambda(s')) V(p_{-1}, z', \mathcal{S}')], \\ V_A(z, \mathcal{S}) &= \max_p \left( \pi(p, z, \mathcal{S}) + \mathbb{E}[\Xi (1 - \lambda(s')) V(p, z', \mathcal{S}') \right]. \end{aligned}$$

Continuation values are discounted by the household's stochastic discount factor  $\Xi$  and by the next-period survival probability  $1 - \lambda(s')$ . Because the exit hazard depends on the firm's own continuation value through equation (5), the firm internalizes the effect of its pricing decision on its own future survival when solving the Bellman equation: a price that delivers higher continuation value also lowers the firm's exit probability. As is standard in this literature, expectations over  $\mathcal{S}'$  require firms to track the (infinite-dimensional) distribution of states. I follow [Krusell and Smith \(1998\)](#) and assume that firms forecast aggregates using a finite set of moments of the distribution, as detailed in Section 3.6.<sup>10</sup>

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<sup>10</sup>The selective exit hazard introduces an additional cross-sectional functional,  $\mathbb{E}_\mu[\exp(-\xi V)]$ , which in principle depends on the full distribution  $\mu$ . Under the same Krusell-Smith approximation that summarizes  $\mu$  by  $\mathcal{S}$ , this

**Exit hazard.** The exit hazard takes the form of an exponential tilt of the cross-sectional value distribution, normalized to deliver an aggregate exit rate of  $\delta$ :<sup>11</sup>

$$\lambda(s) = \delta \cdot \frac{\exp(-\xi V(s))}{\mathbb{E}_\mu[\exp(-\xi V)]}. \quad (5)$$

Two parameters control turnover. The first is  $\delta$ , the average exit rate: by construction,  $\mathbb{E}_\mu[\lambda] = \delta$ . The second is  $\xi \geq 0$ , which controls selectivity. With  $\xi = 0$  the hazard collapses to a constant  $\delta$  and exit is iid random, independent of the firm's state. With  $\xi > 0$  the hazard is decreasing in  $V(s)$ , and low- $V$  firms exit disproportionately; the tilt becomes steeper as  $\xi$  grows. Selective exit is therefore a single-parameter generalization of pure-replacement turnover. Two economies with the same  $\delta$  but different  $\xi$  have the same total exit rate and differ only in which firms leave.<sup>12</sup> Figure 5 plots the calibrated hazard as a function of firm productivity.

To a first order, exit depends on productivity: low-productivity firms exit at much higher rates than high-productivity ones, and the dependence on the inherited price is second order. In Figure 5, the solid line is the hazard for adjusters, who reset to the same flexible price regardless of  $p_{-1}$ , and the shaded band shows the variation in the hazard across the firm's inaction region. The band widens with the selectivity parameter  $\xi$ : by avoiding the menu cost, a low-productivity firm can lower its own exit probability to some extent, and this margin grows with the strength of selection.

**Entrants.** Entry is one-for-one with exit. Within each period, entrants follow a two-step procedure:

**Step 1.** An entrant draws an initial productivity  $z_1$  from the entry distribution

$$G_e(z) \propto G(z) \exp(-\tau \log z), \quad (6)$$

where  $G$  is the ergodic distribution of incumbent productivity and  $\tau \geq 0$  is the entry-selectivity parameter. At  $\tau = 0$ ,  $G_e = G$  and entrants are statistically identical to the incumbent pool. At  $\tau > 0$ , the entry distribution is tilted toward low-productivity types, a reduced-form representation of entrants, which tend to be smaller than the pool of incumbents. Conditional on  $z_1$ , the entrant sets a flexible price  $p_1 = \arg \max_p V_A(z_1, \mathcal{S})$  at no menu cost.

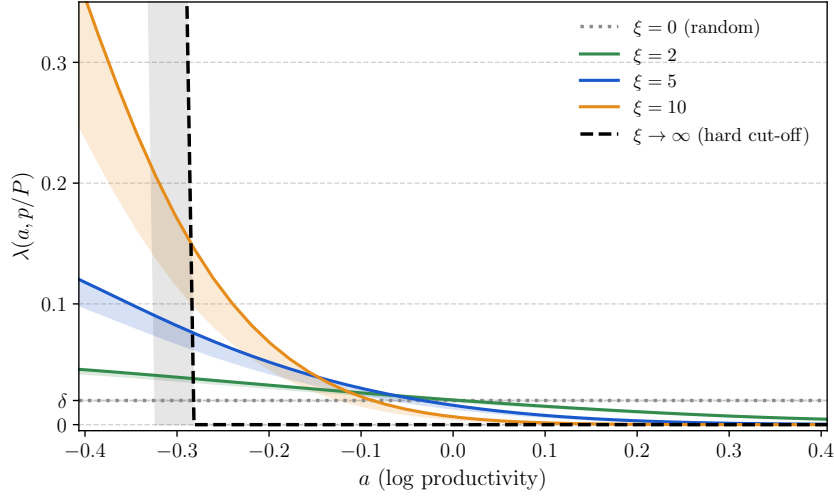
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functional is determined contemporaneously by integration once the value function is parametrized by  $\mathcal{S}$ , so it does not require a separate forecasting rule on top of the inflation rule of Section 3.6.

<sup>11</sup>In the numerical solution the hazard is also capped below one and the residual mass is redistributed across the remaining states so that the aggregate constraint binds with equality. At the calibrated values the cap rarely activates.

<sup>12</sup>The denominator  $\mathbb{E}_\mu[\exp(-\xi V)]$  acts as a cross-sectional normalizer that pins the average hazard at  $\delta$  regardless of the level of  $V$ . It is a contemporaneous functional of the cross-section, not a separate state of the firm's problem: once  $V$  is solved as a function of  $\mathcal{S}$ , the denominator is determined by integration. Section 3.6 returns to this point in the context of the solution method.

**Figure 5: Exit hazard by firm productivity.**



*Notes:* The figure plots the exit hazard  $\lambda(a, p/P)$ , for different values of the elasticities  $\xi$ . For each  $\xi$ , the solid line reports  $\lambda$  over the price-adjustment region conditional on  $a$ : since the hazard is flat in  $p/P$  within that region, the line captures the exit rate of adjusting firms. The shaded band shows the  $[\min, \max]$  range of  $\lambda$  within the no-adjustment region, where the hazard varies with  $p/P$ : low-productivity firms can effectively lower their exit probability by sticking to their old price, and save on the menu cost. Figure A.11 in the Appendix shows the exit hazard as a two-dimensional function of productivity and relative price.

**Step 2.** A within-period idiosyncratic shock then realizes,

$$\log z_2 = \rho \log z_1 + \sigma_\varepsilon \varepsilon, \quad \varepsilon \sim N(0, 1),$$

moving the entrant to its end-of-period productivity  $z_2$ . From this point forward the entrant behaves exactly as an incumbent: at the start of period  $t + 1$  it carries  $p_1$  as its inherited price and  $z_2$  as its inherited productivity, and faces the same exit and pricing decisions.

The two-step structure has two purposes. First, it ensures that entrants do not enter the cross-section already at their flex price: by the time the next exit draw arrives, an entrant's inherited price is generically misaligned with its current productivity, and from the start of the following period an entrant is observationally equivalent to an existing firm. Second, the within-period shock breaks the mechanical correlation between entry and immediate price adjustment that one would otherwise obtain. The mass of entrants in each period is closed by replacement,

$$M = \int \lambda(s) d\mu(s),$$

which equals total exit. No explicit free-entry condition is imposed.<sup>13</sup>

<sup>13</sup>A setting with free entry would tie the entrant mass to expected discounted profits and would let one study how entry conditions interact with pricing decisions and the aggregate state. I instead fix the entrant mass through one-for-one replacement, since the focus of this paper is on the composition of the surviving cross-section under

**Distribution of firms.** Combining the exit hazard, the incumbent transition kernel, and the entrant inflow yields the law of motion for the cross-sectional measure  $\mu$  over the state space:

$$\mu'(s') = \int (1 - \lambda(s)) Q(s, s') d\mu(s) + M \nu_e(s'), \quad (7)$$

where  $Q$  is the incumbent transition kernel induced by the price-adjustment policy, the AR(1) productivity process, and the law of motion of the aggregate state, and  $\nu_e$  is the entrant distribution over the state space induced by  $G_e$ , the flex-pricing step (1), and the within-period shock (2).

**Timing.** The within-period sequence of events is as follows:

1. The aggregate state  $\mathcal{S}_t$  is realized. Incumbents enter the period with their inherited price  $p_{i,t-1}$  and draw productivity  $z_{it} \sim F(z_{it} \mid z_{i,t-1})$ . Entrants execute their two-step block: in Step 1 they draw an initial productivity  $z_1 \sim G_e$  from (6) and set the flexible price  $p_1$  at no menu cost; in Step 2 a within-period idiosyncratic shock realizes that moves their productivity from  $z_1$  to  $z_2$ , with  $\log z_2 = \rho \log z_1 + \sigma_\varepsilon \varepsilon$ .
2. The exit hazard  $\lambda(s)$  fires on incumbents. Exiting incumbents leave the market and are replaced one-for-one by entrants, who join the cross-section at this step at state  $(p_1, z_2)$ .
3. Surviving incumbents and within-period entrants decide whether to keep their inherited price or to pay the menu cost and reset.
4. Production takes place and firms collect flow profits.

By the start of the following period, an entrant carries an inherited price and an inherited productivity and is statistically indistinguishable from any other firm. Figure 6 summarizes the timing.

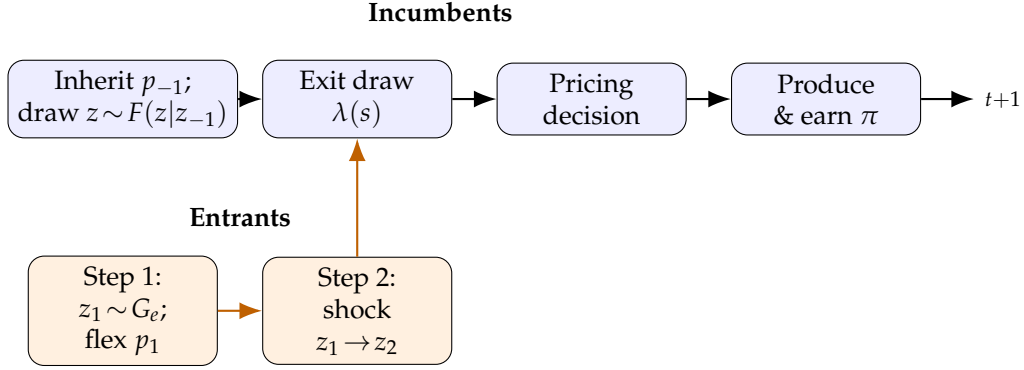
### 3.5 Equilibrium

Let  $\mathbf{S}$  denote the firm-level state space of triples  $s = (p_{-1}, z, \mathcal{S})$ , equipped with the Borel  $\sigma$ -algebra  $\Sigma_{\mathbf{S}}$ , and let  $\mu$  be a probability measure on  $(\mathbf{S}, \Sigma_{\mathbf{S}})$ . The incumbent transition kernel  $Q: \mathbf{S} \times \Sigma_{\mathbf{S}} \rightarrow [0, 1]$  describes how a surviving incumbent moves through the state space within a period: conditional on survival,  $Q(s, \cdot)$  applies the price policy  $p(s)$ , draws a new productivity  $z'$  from  $F(z' \mid z)$ , and updates the aggregate state from  $\mathcal{S}$  to  $\mathcal{S}'$  according to the perceived law of motion  $G$ . The entrant distribution  $\nu_e: \Sigma_{\mathbf{S}} \rightarrow [0, 1]$  is the image measure on  $\mathbf{S}$  induced by the two-step entry block: an entrant draws  $z_1 \sim G_e$  from (6), sets the flex-pricing reset  $p_1 = \arg \max_p V_A(z_1, \mathcal{S})$ , and then receives the within-period idiosyncratic shock that yields end-of-period productivity  $z_2$ .

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selective exit rather than on aggregate market size.

Figure 6: Timing within a period.



Notes: The figure shows the within-period sequence of events. Incumbents (top lane) inherit the previous price and draw productivity, face the exit hazard, choose whether to adjust, then produce. Entrants (bottom lane) follow a two-step block (draw an initial productivity, set a flexible price at no menu cost, then receive a within-period idiosyncratic shock) before joining the incumbent timeline at the exit-draw step.

**Definition.** A stationary recursive competitive equilibrium consists of value functions  $V_N(s)$ ,  $V_A(s)$ , a price-adjustment policy  $p(s)$ , an exit hazard  $\lambda(s)$ , an entrant distribution  $v_e$  and entrant mass  $M$ , an invariant cross-sectional measure  $\mu$ , aggregate quantities  $\{C, W/P\}$ , and a perceived law of motion  $G$  for the aggregate state, such that:

1. household optimization gives  $C = S/P$  and  $W/P = \chi C$ ;
2. given  $G$ , the wage  $W/P$ , and the hazard  $\lambda$ , the policy  $p(s)$  solves the incumbent's Bellman equation (4);
3. the exit hazard satisfies (5), with the cross-sectional integral in the denominator computed against  $\mu$ ;
4. the entrant distribution  $v_e$  is the image measure of the two-step entry block applied to  $G_e$  in (6), and the entrant mass equals total exit,  $M = \int \lambda(s) d\mu(s)$ ;
5. the cross-sectional measure  $\mu$  is invariant under the law of motion (7);
6. goods and labor markets clear at the implied  $(P, Y)$ ;
7. the perceived law of motion  $G$  is consistent with the inflation path generated by the firm sector at every aggregate state.

### 3.6 Solution method

Due to heterogeneity, rational expectations, and aggregate risk, the firm's problem depends on the cross-sectional distribution  $\mu_t$ , an infinite-dimensional object. Following [Krusell and Smith \(1998\)](#) and [Nakamura and Steinsson \(2010\)](#), I assume that firms forecast future aggregates with a finite set of its moments. The summary is a single aggregate state  $\mathcal{S} := S/P_{-1}$ , nominal

demand deflated by last period's price index, and firms forecast inflation as a function of it:

$$\frac{P_t}{P_{t-1}} = G\left(\frac{S_t}{P_{t-1}}\right)$$

Firms solve their dynamic programs taking  $G$  as given; I then update  $G$  from the model-implied inflation at each grid point and iterate to convergence. Appendix B.1 reports the algorithm in full. The forecasting rule is sufficient for the stationary equilibrium, and is used both to solve for the stationary  $\mu$  and to compute the long simulations behind the unconditional output variance reported in Section 4. For the headline impulse-response analysis, however, I solve a deterministic perfect-foresight transition path that tracks the cross-section  $\{\mu_t, V_t, \lambda_t, \pi_t\}$  explicitly along the IRF horizon.

## 4 Results

This section uses the model to study how selective exit affects monetary transmission. Section 4.1 describes the calibration. Section 4.2 reports the headline aggregate result: selective exit raises the unconditional variance of output and the cumulative real response to a one-time monetary shock. Section 4.3 unfolds the underlying mechanism into two channels, an option-value channel and a censoring channel, using cross-sectional comparisons and a partial-equilibrium counterfactual exercise. Section 4.4 quantifies the relative weight of the two channels by decomposing the CIR itself through counterfactual economies that activate one channel at a time.

Selective exit amplifies the effects of an aggregate shock: it raises the one-year cumulative output response to a nominal shock by about fifteen percent relative to random exit at the same exit rate, and raises the unconditional variance of output by a similar margin. Selection operates through two margins: through the cross-sectional distribution of firms, and through the policy functions. The key idea is the following: the real effects of a nominal shock depend on the stock of misaligned prices in the cross-section. Every exit corrects a price because the exiting firm hands its slot to an entrant that prices on arrival.<sup>14</sup> What turnover does to monetary non-neutrality therefore depends on where exit lands. Exit that removes a firm whose wrong price had time left deletes standing mispricing; exit that removes a firm about to adjust deletes basically nothing, since the slot gets a corrected price either way. Selection concentrates exit on firms about to adjust in the tail of the distribution (firms who received a large negative productivity shock and would like to increase their price) and at a fixed turnover rate, every exit spent on these firms is one exit fewer available for the interior of the cross-section, where wrong prices persist. Section 6 characterizes analytically the conditions under which selection

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<sup>14</sup>Strictly speaking, a new entrant is not distinguishable from an incumbent due to the two-step entry process, hence it might carry a mispriced price too. However, since its price goes through only one productivity shock, it carries a smaller pricing gap than an incumbent.

amplifies the effects of an aggregate shock in a reduced version of the model. The selective hazard also works through anticipation (the other channel): it discounts continuation values, widens the inaction bands of the most exposed firms, and lets their wrong prices stand longer. Section 4.4 decomposes the fifteen percent between the two channels and finds the largest piece in their interaction.

## 4.1 Calibration

The model period is one month. Five parameters are internally calibrated to match five empirical moments from Dutch retail scanner data (Section 2); the remaining structural parameters are externally calibrated. The internally calibrated parameters and targets are (i) the menu cost  $f$ , targeting the monthly frequency of price changes among incumbents; (ii) the average exit rate  $\delta$ , targeting the monthly UPC exit rate; (iii) the exit selectivity  $\zeta$ , targeting the relative size of exiting products (computed as the average revenue of exiting products over the average revenue of incumbents); (iv) the entry selectivity  $\tau$ , targeting the relative size of entrants (computed as the average revenue of new products over the average revenue of incumbents); (v) the productivity volatility  $\sigma_\epsilon$ , targeting the mean log price change. The exit and entry rate moments together pin down  $\delta$  and the entrant mass;<sup>15</sup> the relative-size targets pin down  $\zeta$  and  $\tau$  separately, since they discipline the amount of selection conditional on the level of turnover. Externally, I follow Nakamura and Steinsson (2010) for the trend nominal-demand drift  $\bar{\pi}$  and the volatility of the nominal-demand innovation  $\sigma_S$ , take the productivity autocorrelation  $\rho = 0.83$  from the autocovariances of prices in the same panel (Appendix A.5), and fix the standard preference and demand parameters at  $\beta = 0.96$ ,  $\theta = 4$ , and  $\chi = 1$ . Table 3 reports the full parameter set together with model and data values for the targeted moments and several untargeted moments.

The model matches the targeted moments closely: the calibrated economy has frequency 0.09, exit rate 0.02, relative size of exiters 0.65 (data: 0.66) and of entrants 0.69 (data: 0.67). Higher moments of the price-change distribution are not targeted: the mean absolute price change in the model is about twice as large as in the data, and the standard deviation is similarly inflated.<sup>16</sup> The fraction of upward changes is matched without targeting.

The model does not target directly the distributions behind the three empirical facts from Section 2.2 (smaller products exit more often, adjust less often, and adjust by larger amounts when they do). Here I show that the model manages to match the slope of these untargeted moments reasonably well.

To do the comparison I compute the same objects used in the data: product revenues, exit

<sup>15</sup>The number of products in the data is not strictly stationary: the entry rate is slightly above exit, meaning that over time the sample of products grows. I solve the model with an invariant distribution of firms, where entry and exit are balanced.

<sup>16</sup>This is a standard feature of menu-cost models with a single Gaussian productivity shock and no leptokurtic component (Midrigan, 2011; Aruoba et al., 2024): with one shock, the model cannot generate both small adjustments (from firms close to the inaction band) and large adjustments (from firms in the tails). Since the goal of this paper is to isolate the role of selective exit, I do not extend the model along this dimension.

**Table 3: Benchmark calibration.**

| Parameter                    | Description                  | Value  | Target                        | Data | Model |
|------------------------------|------------------------------|--------|-------------------------------|------|-------|
| $f$                          | Menu cost                    | 0.155  | Frequency of price changes    | 0.09 | 0.09  |
| $\delta$                     | Average exit rate            | 0.020  | Monthly exit rate             | 0.02 | 0.02  |
| $\xi$                        | Exit selectivity             | 20     | Rel. size of exiting firms    | 0.66 | 0.64  |
| $\tau$                       | Entry selectivity            | 5      | Rel. size of entrants         | 0.67 | 0.69  |
| $\sigma_\epsilon$            | St. dev. productivity        | 0.086  | Mean $\Delta \log p$          | 0.04 | 0.04  |
| <i>Untargeted moments</i>    |                              |        |                               |      |       |
| —                            | —                            | —      | Mean $ \Delta \log p $        | 0.10 | 0.21  |
| —                            | —                            | —      | St. dev. $\Delta \log p$      | 0.11 | 0.21  |
| —                            | —                            | —      | Fraction upward changes       | 0.63 | 0.62  |
| <i>Externally calibrated</i> |                              |        |                               |      |       |
| $\bar{\pi}$                  | Trend nominal demand         | 0.0028 | Nakamura and Steinsson (2010) |      |       |
| $\sigma_S$                   | St. dev. nominal demand      | 0.0065 | Nakamura and Steinsson (2010) |      |       |
| $\rho$                       | Productivity autocorrelation | 0.83   | Estimated, Appendix A.5       |      |       |
| $\beta$                      | Discount factor              | 0.96   | Externally calibrated         |      |       |
| $\theta$                     | Demand elasticity            | 4      | Externally calibrated         |      |       |
| $\chi$                       | Labor disutility             | 1      | Externally calibrated         |      |       |

*Notes:* The table reports calibrated parameter values, targeted moments (data vs. model), and selected untargeted moments. Five parameters are internally calibrated jointly to match the five targeted moments. The model matches the targeted first-moment pricing and turnover moments closely. The mean absolute price change and the standard deviation of price changes are not targeted and are larger in the model than in the data, a known feature of single-shock menu-cost models that abstract from leptokurtic shocks or random adjustment opportunities. The fraction of upward changes is essentially matched.

probability, frequency and magnitude of price changes. A product with productivity  $z$  and inherited relative price  $p/P$  supplies  $y = (p/P)^{-\theta}Y$  and earns revenue  $py \propto (p/P)^{1-\theta}Y$ , up to aggregate-state factors common to all products. I rank products in the stationary cross-section by log revenue, sort them into ten equal-mass deciles, and compute three within-decile moments: the average six-month exit probability  $1 - (1 - \lambda(s))^6$ , to match the empirical exit horizon; the average monthly adjustment probability; and the mean absolute log price change conditional on adjustment. Figures 7–9 report the three gradients alongside their OLS fits.

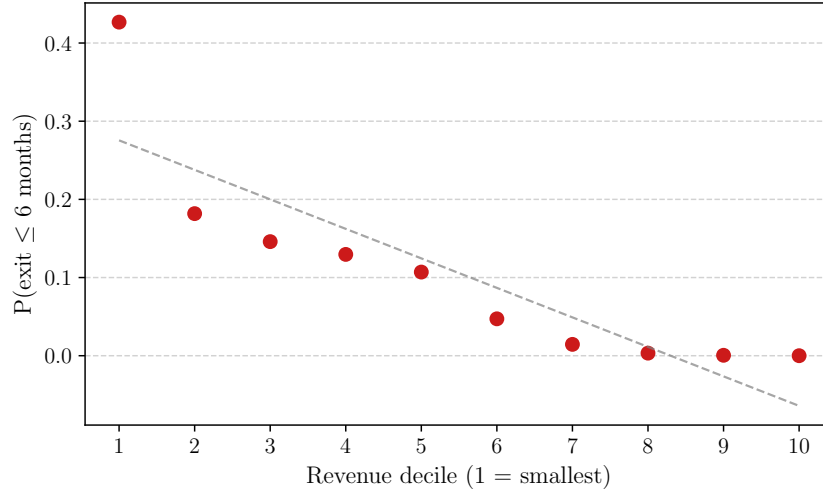
The signs of the three slopes match the data on all three margins, and the rank ordering across deciles is preserved. The match is closest on the exit and adjustment-frequency gradients: smaller-revenue firms exit more and adjust less in the model, and the gradient is concentrated in the bottom half of the distribution in both cases. The size-of-price-changes gradient is matched in shape over roughly the first two-thirds of the distribution (adjustments are larger at the bottom and shrink as revenue rises) but increases at the upper end, where the data continue to decline. The shortfall in the upper deciles is symptomatic of the difficulty of single-shock menu-cost models in matching the full shape of the price-change distribution (Midrigan, 2011).

## 4.2 Aggregate effect of selective exit

I compare two economies, both sharing the calibrated structural parameters:

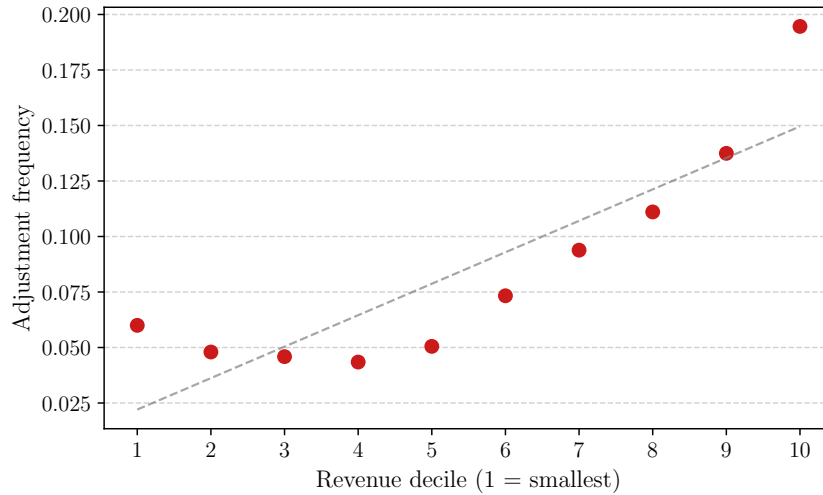
1. **Benchmark.** The fully calibrated economy: selective exit ( $\xi = 20$ ) and selective entry ( $\tau = 5$ ), at  $\delta = 0.02$ .

**Figure 7: Six-month exit probability by revenue decile, benchmark.**



Notes: For each decile of log revenue in the benchmark model's stationary cross-section, the figure plots the within-decile average six-month exit probability  $1 - (1 - \lambda(s))^6$ . The dashed line is the OLS fit through the ten bin means.

**Figure 8: Adjustment frequency by revenue decile, benchmark.**

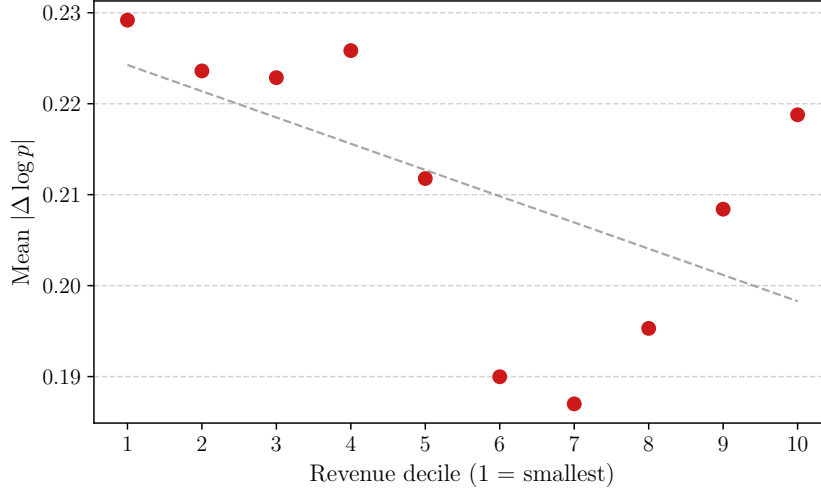


Notes: For each decile of log revenue in the benchmark model's stationary cross-section, the figure plots the within-decile average monthly adjustment probability. The dashed line is the OLS fit through the ten bin means.

2. **No selection.** Same average exit rate ( $\delta = 0.02$ ) but no selectivity ( $\xi = 0, \tau = 0$ ). Total exit equals the benchmark, but turnover is i.i.d. random across firms and entrants are statistically identical to the incumbent pool.

I also report an economy without entry or exit (**No churn.**): no exit at all ( $\delta = 0$ ), and consequently no entry. The comparison between Benchmark and No selection isolates the role of which firms leave: the two economies have the same total exit, the same menu cost, and similar pricing moments, but they differ in exit composition. Comparing No selection and No churn isolates the role of having turnover at all, holding selection at zero. Together, the three economies decompose the contribution of turnover into a churn component (random replace-

**Figure 9: Mean absolute log price change by revenue decile, benchmark.**



Notes: For each decile of log revenue in the benchmark model's stationary cross-section, the figure plots the average  $|\Delta \log p|$  across price-change events in the decile. The dashed line is the OLS fit through the ten bin means.

ment) and a selection component (composition).

I summarize aggregate dynamics with two objects. The first is the unconditional variance of output  $\text{Var}(Y)$ , obtained from a long simulation of the model with purely nominal aggregate shocks. The simulation draws monetary innovations  $\eta_t \sim N(0, \sigma_\eta^2)$ , accumulates nominal demand  $S_t$  as a random walk with drift, applies the forecasting rule  $P_t/P_{t-1} = G(S_t/P_{t-1})$  to obtain the inflation path, and tracks real value-added output as  $Y_t = S_t - P_{t+1}$ . I report  $\text{Var}(Y)$  over a 25,000-period simulation after a burn-in period. The second object is the impulse response of output to a one-time nominal shock and its cumulative counterpart, computed along a deterministic perfect-foresight transition path.

Additionally, [Alvarez et al. \(2016\)](#) show that in a wide class of menu-cost and time-dependent pricing models, the cumulative impulse response of output to a one-time nominal shock is captured to first order by a simple sufficient statistic: the ratio of the kurtosis of the stationary distribution of price changes to the frequency of price changes,<sup>17</sup>

$$\text{CIR}(Y) \propto \frac{\text{Kurt}(\Delta \log p)}{\text{freq}}.$$

Table 4 reports stationary moments and the unconditional output variance for the three economies. The frequency of price changes drives in part this result. Selective exit lowers it from 0.088 in No selection to 0.086 in Benchmark, a 0.2 percentage-point drop. Appendix D decomposes the frequency effect into channels.

To compute the short-run response to a one-time nominal shock, I solve for the impulse

<sup>17</sup>Section F5 in the appendix proves that the same sufficient statistic approach holds in a reduced form of this model.

**Table 4: Pricing moments and unconditional output variance.**

| Model        | $\delta$ | $\xi$ | $\tau$ | Frequency | Kurtosis | $\text{Var}(Y) \times 10^5$ |
|--------------|----------|-------|--------|-----------|----------|-----------------------------|
| Benchmark    | 0.02     | 20    | 5      | 0.086     | 1.41     | 2.36                        |
| No selection | 0.02     | 0     | 0      | 0.088     | 1.32     | 2.05                        |
| No churn     | 0.00     | 0     | 0      | 0.092     | 1.31     | 1.97                        |

*Notes:* Benchmark is the calibrated economy with selective exit and tilted entry. No selection has the same average exit rate at  $\xi = \tau = 0$  (random iid exit, untilted entry). No churn shuts down exit and entry entirely. Frequency is the monthly frequency of price changes; kurtosis is the kurtosis of  $\Delta \log p$  conditional on a price change.  $\text{Var}(Y) \times 10^5$  is the variance of real value-added output across a 25,000-period stationary simulation of the model with nominal aggregate shocks.

response along a deterministic perfect-foresight transition path that tracks  $\{V_t, \mu_t, \lambda_t, \pi_t\}_{t \geq 0}$ . In practice, a one-time log-nominal innovation  $\Delta$  hits at  $t = 0$  and the economy is assumed to have perfect foresight thereafter.<sup>18</sup> Table 5 reports the one-year cumulative impulse response for the three economies.

First, the Benchmark accumulates about fifteen percent more cumulated output response than No selection, in line with the unconditional-variance gap of Table 4. The two economies share the same total exit rate, the same menu cost, and very similar standard pricing moments, so the gap reflects the differential response of the cross-section to the shock under selective versus random exit; Section 4.3 traces it to two underlying channels. Second, the No-churn economy sits within half a percent of No selection: at the calibrated parameters, uniform exit decreases monetary non-neutrality and slightly increases the flexibility of the overall economy. The no churn margin is itself the net of two opposing forces that the rest of the paper separates. On one hand, random turnover subsidizes aggregate price flexibility, because each exit replaces a mispriced incumbent with a (relatively) correctly priced entrant; Section 6 prices this subsidy at one to two percent of the CIR at the calibrated volatility, small because idiosyncratic shocks

**Table 5: One-year cumulative impulse response of output.**

| Model        | CIR (1 year) | vs. No selection |
|--------------|--------------|------------------|
| Benchmark    | 0.178        | +14.6%           |
| No selection | 0.155        | –                |
| No churn     | 0.156        | +0.5%            |

*Notes:* One-year cumulative impulse response  $\text{CIR} = \frac{1}{12} \sum_{s \leq 12} \psi_s$ , in annualized units, where  $\psi_t = (\log Y_t - \log Y_t^0) / \Delta$  is the output response to a one-time log-nominal innovation  $\Delta$  at  $t = 0$ , computed along the perfect-foresight transition path and differenced against the contemporaneous no-shock path  $Y_t^0$  (Appendix B.2). Firms set their date-0 prices before observing the shock, so  $\psi_0 = 1$  in all economies. The last column is the percent difference from No selection, the economy with the Benchmark's exit flow and no selectivity.

<sup>18</sup>The algorithm is backward-forward shooting on the inflation path  $\{\pi_t\}_{t=0}^{T-1}$ . The shock enters through the aggregate state:  $Y_0 = Y^{ss} + \Delta$ , with the cross-section initialized at its stationary distribution  $\mu_0 = \mu^{ss}$ . Given a guess of the inflation path, I solve the firm Bellman backwards from the terminal condition  $V_T = V^{ss}$  to obtain  $\{V_t, p_t\}$ , push the distribution forward under the deterministic transition operator to obtain  $\{\mu_t\}$ , recompute the exit hazard  $\lambda_t$  from the contemporaneous  $(V_t, \mu_t)$ , and back out a new inflation path from the CES condition. Appendix B.2 describes the algorithm in full.

are large, so most mispricing at any moment is new rather than standing. On the other hand, against it, the exit hazard widens the inaction bands of incumbents through the option-value channel of Section 4.3, which pushes the CIR up by a comparable amount.

### 4.3 Mechanisms

The real effect of a nominal shock is governed by the amount of misaligned prices in the economy, and by how fast these prices catch up with the shock. Two features of the menu-cost equilibrium affect that speed: which firms find it worthwhile to pay the menu cost in any given period, and which firms are present in the cross-section to do so. Selective exit affects both. It changes the pricing policy by lowering the option value of paying the menu cost for the firms most exposed to exit (the option-value channel), and it changes the composition of the cross-section by removing a non-random subset of would-be price adjusters (the censoring channel). The key is how the removal differs between selective and random exit: in the latter, exit removes firms who, on average, did not want to change their price, and replaces them with fresh entrants. In the former, the exiting firm's slot passes within the period to an entrant that prices on arrival. This means that selection removes this sort of flexibility subsidy that uniform exit was granting, and hence withholds new price changes from the rest of the cross-section. The two channels operate on overlapping but distinct sets of firms; Section 4.4 quantifies their contributions in units of monetary non-neutrality.

#### Option-value channel

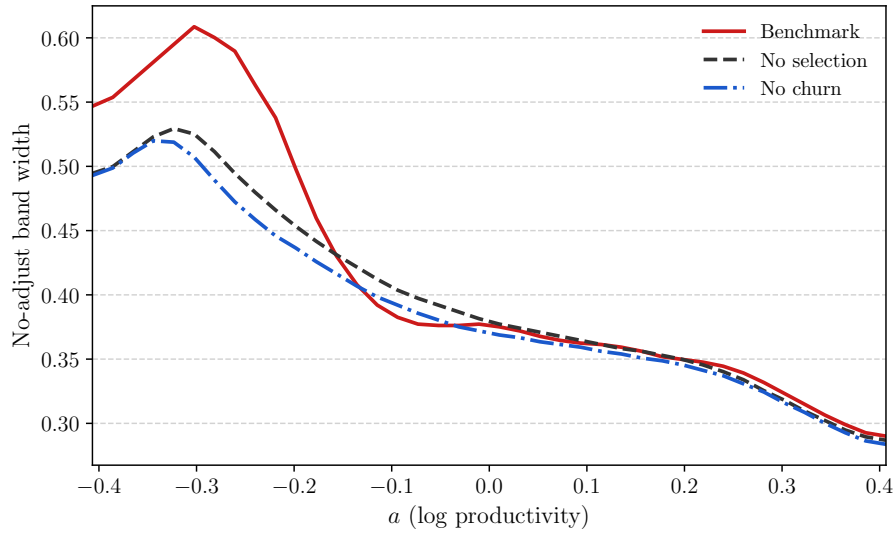
Higher exit hazards lower the present value of paying the menu cost, especially for low-productivity firms whose hazard is largest. The result is that the inaction region  $V_N(s) \geq V_A(s)$  widens for those firms relative to an economy with random exit. Figure 10 plots the implied  $S$ - $s$  band (the no-adjustment region of  $\log p_{-1}/P$  as a function of productivity) across the three economies.

The inaction band is widest at low productivity in all three economies: less productive firms have lower flow profits, and the gain from a price reset is small relative to the menu cost. Selective exit amplifies this pattern. In the Benchmark, low-productivity firms face a disproportionately higher exit hazard, which acts as an additional discount on the future returns to any current price reset, and they therefore find it less worthwhile to pay the menu cost today.<sup>19</sup> For more productive firms the hazard is closer to the cross-sectional average, the discount effect is small, and the band is essentially unchanged across economies. The selective hazard therefore reshapes the pricing policy at low productivity before any firm has actually exited. In a similar fashion, the No churn economy has the smallest inaction band as this additional discounting due to exit is absent.

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<sup>19</sup>Whether this policy widening translates into a lower aggregate adjustment frequency depends on where the stationary cross-section actually places mass. Appendix D returns to this point in the moments-based decomposition.

**Figure 10: Inaction region by productivity, across models.**



*Notes:* The figure shows the boundaries of the inaction region in the  $(\log z, \log p_{-1}/P)$  plane for the three economies. Inside the band, firms keep their inherited price; outside, they pay the menu cost and reset. The band is wider for low- $z$  firms in the Benchmark relative to the no-selection and no-churn economies, reflecting the option-value effect of a higher exit hazard at low  $V$ .

### Censoring channel

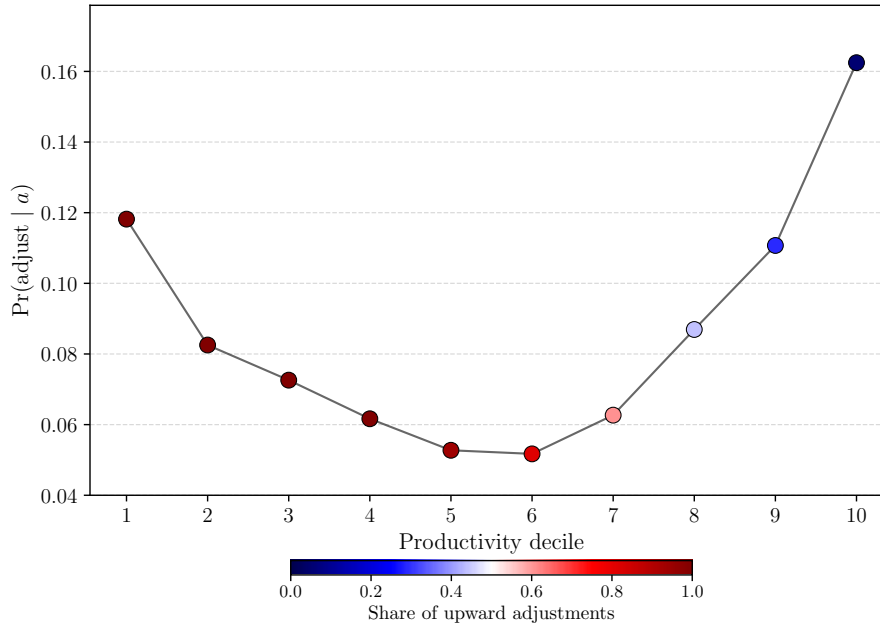
The second channel works through what would have happened to the firms that selective exit removes. Figure 11 reports the conditional frequency of price changes by productivity decile in the No selection economy.

Adjustment is concentrated in the tails of the productivity distribution: low- $z$  firms reset upward (their inherited price is too low relative to a higher marginal cost), high- $z$  firms reset downward, and firms in the middle stay put.<sup>20</sup> The firms that selective exit removes, those at the bottom of the productivity distribution, therefore come from the high-adjustment tail. Once they exit, the price changes they would have made never appear in the data. Crucially, those changes are not random. A firm that survives a sharp negative productivity draw typically resets its price upward in the period of the shock and then gradually trims that price downward as productivity mean-reverts. Selective exit removes both legs of this trajectory: the firm is gone before either the upward reset or the subsequent downward corrections can take place. The two sides have different fates. The upward reset is replaced on the spot: the exiting firm's slot goes, within the period, to an entrant that prices (relatively more) flexibly on arrival,<sup>21</sup> so the correction the incumbent was about to make happens anyway, recorded as an exit and an entry

<sup>20</sup>This U-shape is a feature of menu-cost models with idiosyncratic shocks and is observed also in Cooper and Öztürk (2022).

<sup>21</sup>Strictly speaking, a new entrant is not distinguishable from an incumbent due to the two-step entry process, hence it might carry a mispriced price too. However, since its price goes through only one productivity shock, it will carry a smaller pricing gap than an incumbent.

**Figure 11: Frequency of price adjustment by productivity decile.**



*Notes:* For each decile of productivity  $z$ , the figure reports the fraction of firms in that decile that adjust their price in a given month. Adjustment is concentrated in the tails: very low- and very high-productivity firms adjust relatively most often, while firms in the middle of the distribution adjust the least.

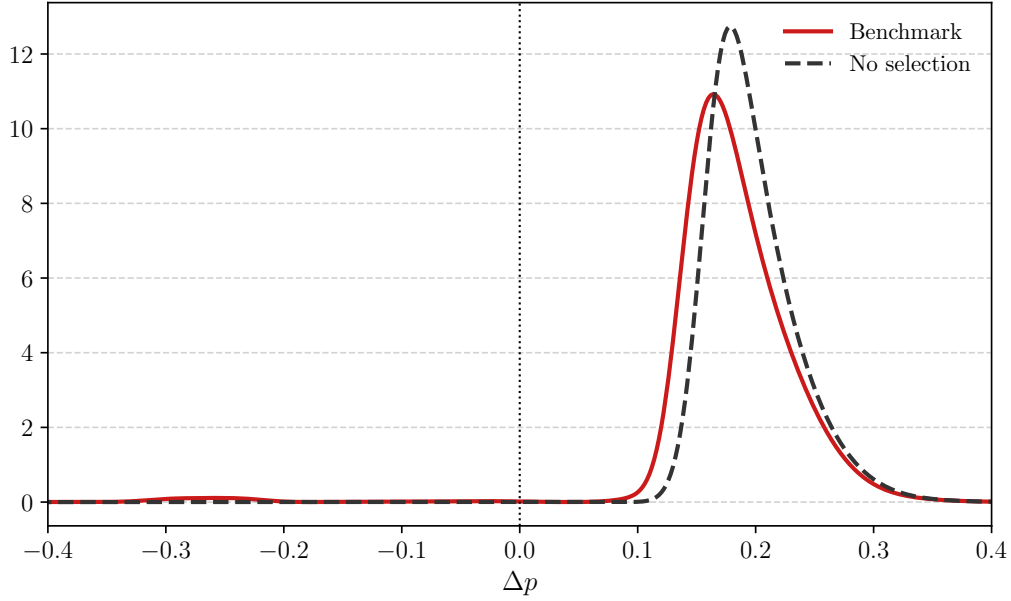
rather than as a price change. The downward corrections are not replaced, because the entrant draws a fresh productivity and does not inherit the exiter's recovery path. The cross-section of price adjustments is therefore systematically different under selection, and this difference is visible directly in the conditional distribution of price changes by productivity quartile.

This is where entry and exit already matter. When selection removes firms from the left tail, it simply replaces one price change with another; this is exactly what selection does. Uniform exit, by contrast, removes more firms, on average, from the interior of the distribution. Firms in the middle are the ones that would not have changed their price (and so would have remained mispriced); uniform exit replaces them with new entrants, who tend to be more flexible. Uniform turnover therefore effectively subsidizes flexibility. Removing it, i.e. moving from uniform exit to selection, eliminates this subsidy and so amplifies the real effects of a monetary shock.

Figure 12 shows the lowest quartile, Q1, which captures firms that have just received a sharp negative productivity draw. Their inherited price is too low relative to a higher marginal cost, and most of their adjustment activity is upward resets. Selective exit removes some of these firms, and the missing mass shows up directly in the right tail of the Q1 distribution.

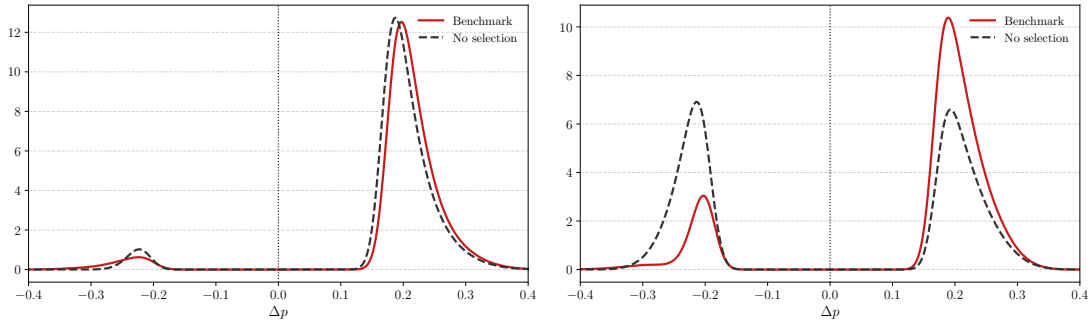
Figure 13 shows the next two quartiles. Relative to No selection, the Benchmark has visibly less mass in the moderate-downward-adjustment region. These are firms that, in the No-selection economy, survived an initial low-productivity episode and are now mean-reverting upward; as productivity recovers, their inherited price becomes too high relative to current conditions, and they generate a sequence of moderate downward corrections. Selective exit

**Figure 12: Distribution of price changes within productivity quartile Q1.**



*Notes:* Kernel density of  $\Delta \log p$  conditional on adjustment, for firms in the lowest productivity quartile (Q1), across the two economies. Quartile boundaries are defined using the Benchmark productivity distribution, so the same productivity range is compared across models.

**Figure 13: Distribution of price changes within productivity quartiles Q2 and Q3.**



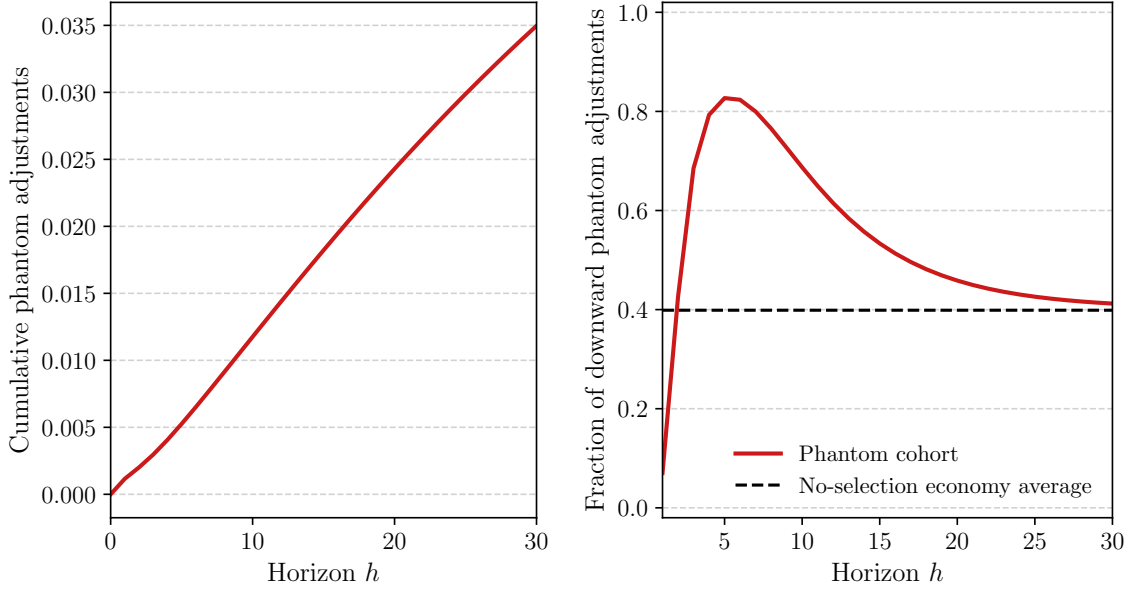
*Notes:* Kernel density of  $\Delta \log p$  conditional on adjustment, for firms in productivity quartiles Q2 (left) and Q3 (right), across the two economies. Quartile boundaries are defined using the Benchmark mass.

removes a fraction of these firms before they reach the migration stage. The missing mass therefore shows up not only at the moment of exit (Q1) but also one step later in the productivity trajectory (Q2 and Q3), where mean reversion has begun.

To quantify the price adjustments that go missing along with the missing mass, I propagate the cohort of selective exiters forward under the No-selection dynamics and compute the price changes they would have made. I call these missing partial-equilibrium corrections phantom adjustments. Let  $\mu_b$  and  $\lambda_b$  denote the stationary distribution and exit hazard of the Benchmark economy, and define the exit-source measure

$$de_0(s) = \lambda_b(s) d\mu_b(s),$$

**Figure 14: Phantom price adjustments by horizon.**



*Notes:* The left panel reports the cumulative mass of phantom adjustments at different horizons, computed as the integral of the no-selection adjustment policy against the cumulative phantom measure  $\Phi_T$ . The right panel reports the share of these adjustments that are downward. At short horizons, the phantom cohort consists of immediate upward resetters from the low-productivity tail; the downward share rises with the horizon as productivity mean-reverts and the surviving cohort migrates into states that generate downward corrections.

which records the mass that exits under selection. I propagate this cohort forward under the No-selection transition kernel  $Q_{ns}$  and survival rate  $1 - \delta$ :

$$v_{t+1}(s') = (1 - \delta) \int Q_{ns}(s', s) dv_t(s), \quad v_0 := de_0, \quad (8)$$

and define the horizon- $T$  and lifetime cumulative phantom measures

$$\Phi_T = \sum_{t=1}^T v_t, \quad \Phi_\infty = \sum_{t=1}^{\infty} v_t.$$

At each horizon, I evaluate the No-selection price-adjustment policy  $p_{ns}(s)$  against the phantom measure to compute the cumulative mass of adjustments and their sign composition.  $\Phi_\infty$  measures the entire missing trajectory of price changes that the selective-exit cohort would have made over its remaining lifetime in the absence of selection.

Figure 14 reports the cumulative mass of phantom adjustments (left panel) and the share of those adjustments that are downward (right panel). The downward share moves with the horizon: at short horizons the missing cohort generates mostly upward resets (the immediate response to a bad productivity draw), but as the cohort propagates forward and productivity mean-reverts, the share of downward corrections rises. Selective exit therefore censors a price-adjustment trajectory that is upward at impact and downward thereafter, and different from what a purely random exit would remove.

The phantom exercise delivers two facts. First, it shows that the removed cohort was

about to reprice. The removed cohort's counterfactual adjustments arrive almost entirely at short horizons, so selection removes firms whose correction was imminent. If exit instead removed stale prices unlikely to change any time soon, the amplification effect would probably be smaller, and possibly even reverse.<sup>22</sup> Second, it shows that the mispricing selection eliminates matters beyond impact as well: these firms would have kept adjusting their price for several periods, as their productivity mean-reverts only gradually. The phantom mass therefore captures not just how much mispricing selection removes, but how long that mispricing would otherwise have persisted, evidence that the temporal dimension of idiosyncratic states, and not just their cross-sectional distribution, shapes the transmission of monetary shocks.

#### 4.4 Decomposing the CIR

The exit hazard works through two channels in the model. The first job is anticipation. In the Bellman equation (4), survival discounts every continuation value,

$$V_N(p_{-1}, z, \mathcal{S}) = \pi(p_{-1}, z, \mathcal{S}) + \mathbb{E}[\mathbb{E}(1 - \lambda(s')) V(p_{-1}, z', \mathcal{S}')],$$

so a selective hazard reshapes pricing incentives before any firm has actually exited: this is where the option-value channel operates. The second job is realization. In the law of motion (7), the hazard determines which firms are deleted from the cross-section,

$$\mu'(s') = \int (1 - \lambda(s)) Q(s, s') d\mu(s) + M v_e(s'),$$

and this is where the censoring channel operates. In equilibrium a single hazard does both jobs at once. To quantify each channel in units of monetary non-neutrality, I separate the two mechanisms: firms solve their pricing problem under a perceived hazard  $\lambda_B$ , while the cross-section evolves under a realized hazard  $\lambda_F$ . Each takes the form (5) with its own selectivity parameter,

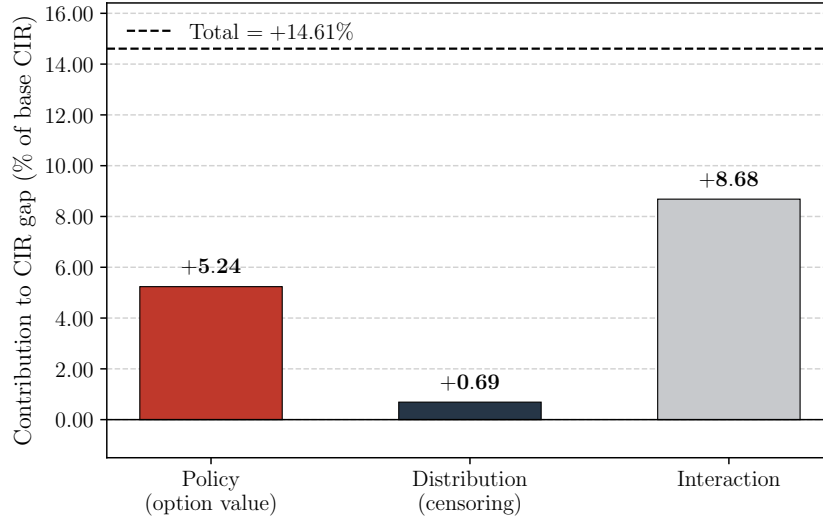
$$\lambda_j(s) = \delta \cdot \frac{\exp(-\xi_j V(s))}{\mathbb{E}_\mu[\exp(-\xi_j V)]}, \quad j \in \{B, F\},$$

so both carry the same exit flow  $\delta$  and all economies below share the same total exit rate. The entry tilt  $\tau$  of (6) enters the model only through the entrant distribution  $v_e$  in the law of motion, so it is assigned to the realization side. A configuration  $(\xi_B, \xi_F, \tau)$  then defines a complete economy: I solve its stationary equilibrium, hit it with the same one-time nominal shock, and compute the one-year CIR under the protocol of Section 4.2.<sup>23</sup> The mixed configurations embody deliberately inconsistent beliefs, since firms perceive an exit process different from the realized one, which is the point of the exercise: switching off one transmission channel at a

<sup>22</sup>Theorem A.1 in Section F.4 states a sufficient condition for amplification.

<sup>23</sup>Both hazards are evaluated at the counterfactual economy's own equilibrium value function and cross-section, so each remains an equilibrium object of that economy; only the assignment of the selectivity tilt differs. The computation is identical to Appendix B.2, with  $\lambda_B$  entering the backward pass and  $\lambda_F$  the forward pass.

**Figure 15: CIR decomposition: No selection to Benchmark.**



*Notes:* Contribution of each channel to the one-year CIR gap between the No-selection economy  $(\xi_B, \xi_F, \tau) = (0, 0, 0)$  and the Benchmark  $(20, 20, 5)$ , in percent of the No-selection CIR. The policy bar is the CIR of the economy  $(20, 0, 0)$ , the distribution bar that of  $(0, 20, 5)$ , each in deviation from the base; the interaction is the residual to the total gap (dashed line). All CIRs are computed along the perfect-foresight transition path of Section 4.2.

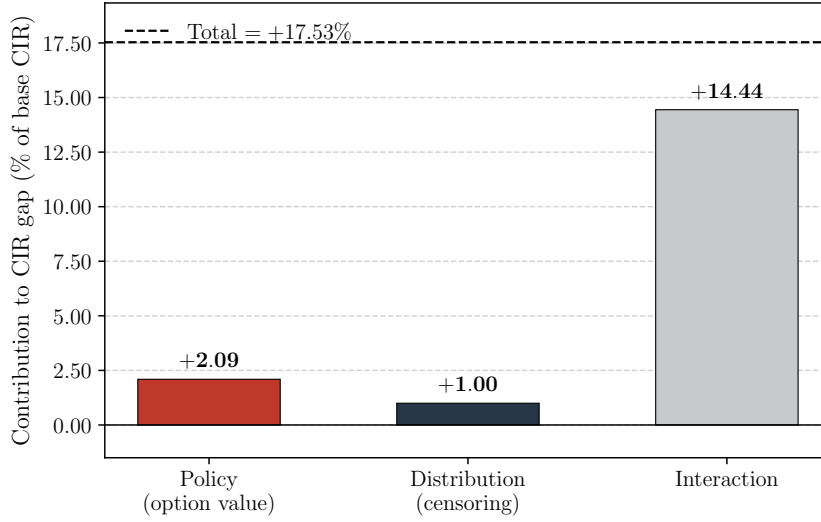
time.

Turning on the selection only in the Bellman equation quantifies the option-value channel: firms price as if exit were selective, but actual exit falls at random. Turning it on only in the law of motion prices the censoring channel: exit falls selectively, but firms price as if it were random. Each counterfactual CIR, in deviation from the economy with no selection anywhere, gives one channel's standalone contribution; the two contributions sum to the total gap up to an interaction term, which measures the part of the amplification that requires both jobs to be active at once.

Figure 15 reports the decomposition, from the No-selection economy to the Benchmark, whose one-year CIRs of 0.155 and 0.178 imply a gap of about fifteen percent. The option-value bar alone carries about a third of the gap: wider inaction bands slow the catch-up of the price level even when exit itself is random. The censoring bar alone is small: with pricing policies undistorted, the firms that selective exit removes sit close to their adjustment margin, and the entrants replacing them restore most of the price flexibility that their departure takes away. The largest entry is the interaction. Each channel supplies what the other lacks: the widening builds standing mispricing, and the withdrawal of exit from the interior protects it. Under the selective discount, exposed firms linger inside widened bands holding wrong prices. Random exit would trim part of that build-up away at rate  $\delta$ ; realized selective exit instead sits at and beyond the adjustment margin, on corrections that entrants replace anyway, so the lingering mispricing survives whole. The interaction bar is the stock that the widening creates and the withdrawal spares.

Figure 16 repeats the decomposition for selectivity routed through entry versus through exit. The base is the entry-only economy  $(\xi = 0, \tau = 5)$ , the target the exit-only economy  $(\xi =$

**Figure 16: CIR decomposition: entry-only to exit-only selection.**



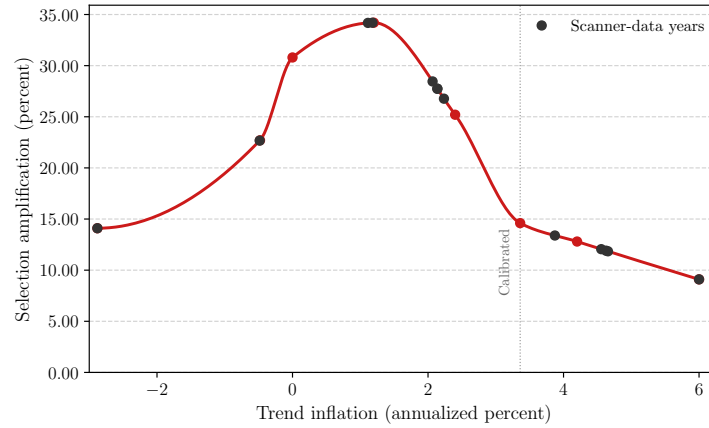
*Notes:* Contribution of each channel to the one-year CIR gap between the entry-only economy  $(\xi_B, \xi_F, \tau) = (0, 0, 5)$  and the exit-only economy  $(20, 20, 0)$ , in percent of the entry-only CIR. The policy bar is the CIR of the economy  $(20, 0, 5)$ , the distribution bar that of  $(0, 20, 0)$ , each in deviation from the base; the interaction is the residual to the total gap (dashed line). All CIRs are computed along the perfect-foresight transition path of Section 4.2.

$20, \tau = 0$ ); their one-year CIRs of 0.157 and 0.184 imply a gap of about eighteen percent. Here the standalone bars nearly vanish and the interaction is essentially the whole effect. Perceived selective exit alone widens bands, but random exit still deletes wrong prices at the average rate, so little standing mispricing accumulates; realized selective exit alone censors imminent adjusters, but with undistorted bands the affected firms adjust quickly and the queue it can censor is short. Selection on exit generates large real effects only as a package of the two.

A wrong price survives until one of two events occurs: the firm corrects it, or the firm exits and hands its slot to a correctly priced entrant. With perceived selection alone, correction is delayed, but exit still falls at random, so while a firm sits longer with a wrong price, that same randomness means most of the resulting mispricing still gets deleted by chance. With realized selection alone, exit is withdrawn from the interior of the distribution, sparing firms there from random exit, but since bands are undistorted, those firms tend to correct their prices quickly anyway, so there is little mispricing left to protect. Only when both margins are active does a wrong price both wait long and face no exit: mispricing is delayed by perceived selection and then, because it is delayed, falls into the interior that realized selection protects. The extra months a wrong price survives beyond what either margin alone would deliver are the interaction bar.

Section 6 works out the distribution side of this decomposition in closed form. In a reduced version of the model the CIR equals the dispersion of price gaps in units of the idiosyncratic variance, the effect of one unit of exit at each point of the cross-section has an explicit formula, and a sufficient condition signs the equal-flow comparison between a selective and a uniform hazard; evaluated on the hazard measured from the solved benchmark, the condition delivers the amplification at the calibrated band widths.

**Figure 17: Selection amplification across trend inflation.**



*Notes:* The amplification is the percent increase in the one-year CIR of output under selective relative to random exit, at a common total exit rate. The amplification across trend inflation including deflation, with each Dutch scanner-data year (2008–2020) placed at its realized annual price drift; the dotted line marks the calibrated rate.

## 5 Low and high inflation

The results so far fix trend inflation at its calibrated value  $\bar{\pi}$ , but how much amplification selection delivers should itself depend on the inflation regime: since selection operates through the cross-section of pending price adjustments, its strength depends on how much repricing the trend generates in the first place, and hence on  $\bar{\pi}$ . This raises a natural question: does selection amplify monetary shocks by the same amount regardless of the inflation environment, or does its effect vary with different inflation regimes? This section answers it by studying how the amplification of the one-year cumulative impulse response under selective relative to random exit varies as trend inflation changes, holding the average exit rate  $\delta$  and all other parameters fixed at their calibrated values. Amplification is far from constant: it exceeds thirty percent when inflation is low (near zero, including mild deflation), then falls as inflation rises, dropping off markedly for inflation above six percent. At high inflation, selection appears closer to neutral, and may even dampen rather than amplify the real effects of monetary shocks for extremely high values. Figure 17 shows the amplification against annualized trend inflation. It is hump-shaped: for small values, or below zero inflation, amplification reaches around 35% of the uniform exit benchmark, declining steadily as inflation increases further.

Random exit buys price flexibility. It mostly removes a firm sitting on a stale price that no adjustment was about to fix, and the entrant taking its slot posts a correct price instead. Selective exit mostly removes a firm that was about to raise its price anyway, so the entrant only carries out a correction the incumbent had already queued up. The amplification measures exactly the flexibility that substitution fails to buy. So when amplification vanishes at high trend inflation, the two rules have stopped differing. Only two things can make that happen: selection stops landing on firms whose correction was imminent, or random exit stops finding stale prices worth deleting.

Selection keeps landing on imminent adjusters as trend inflation rises, and if anything lands

on them harder. The firms it removes are low-value, underpriced firms whose marginal cost has risen, and a faster-eroding price level pushes them toward the adjustment margin sooner, not later. What changes is the average firm. At higher trend inflation every price erodes faster, so much more of the cross-section has already queued up a price increase, and a randomly placed exit increasingly lands on a firm that was going to reprice anyway. Random exit buys less flexibility, the two rules converge, and the amplification goes with them.

I compare what an exiting firm is about to do with what the average firm is about to do. For a firm in state  $s = (p_{-1}, z, \mathcal{S})$ , let  $p^*(z, \mathcal{S})$  be the reset price, the maximizer in  $V_A(z, \mathcal{S})$ , and let

$$g(s) = \mathbb{1}\{V_A(s) > V_N(s)\} [\log p^*(z, \mathcal{S}) - \log p_{-1}] \quad (9)$$

be the price increase the firm is about to post: zero if it keeps its price, the distance from its inherited price to its reset if it adjusts. Two averages of  $g$  follow. One runs over the firms exit removes, the other over all firms,

$$\mathbb{E}_\lambda[g] = \frac{1}{\delta} \int \lambda(s) g(s) d\mu(s), \quad \mathbb{E}[g] = \int g(s) d\mu(s). \quad (10)$$

$\mathbb{E}_\lambda[g]$  weights each firm by its chance of dying, which (5) normalizes to average  $\delta$ : it is the pending price increase of the firm exit takes.  $\mathbb{E}[g]$  weights every firm alike: it is the pending increase of a firm drawn at random, equal to trend inflation up to the resets entrants perform in the slots firms leave behind. These are the two lines Figure 18 plots against trend inflation: the pending increase of the firm exit takes, and the pending increase of the average firm.<sup>24</sup>

At the calibrated trend, the firm exit takes is about to raise its price by 5.4 percent, the average firm by 0.3 percent. That gap separates selective exit from random exit. Selective exit removes the first kind of firm, so the entrant only carries out a raise the incumbent had already queued up. Random exit lands on the average firm instead, and the entrant then supplies a raise nobody was going to make.

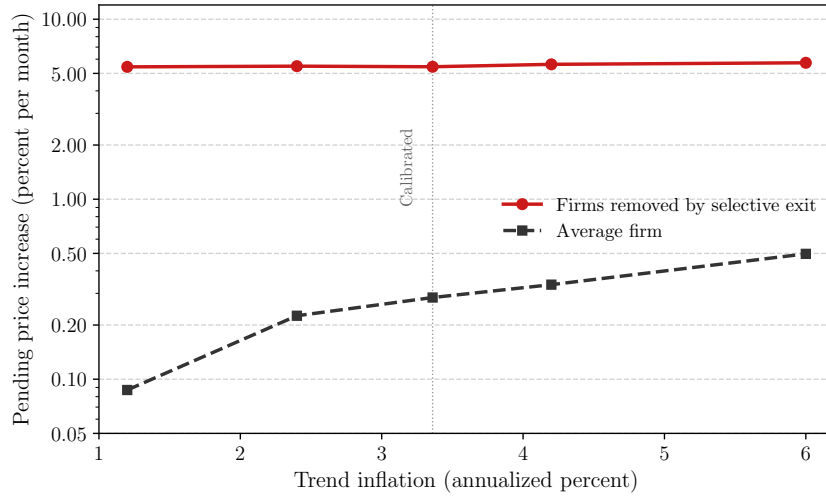
Through the same mechanism, amplification is largest where trend inflation is smallest, because near zero nothing pushes a firm out of its inaction band except its own idiosyncratic shocks, so far fewer firms have a price change pending. A random exit almost always lands on a firm parked inside its band, holding a stale price that nothing was about to fix; with no drift that price would have stayed wrong for a long time, so deleting it and handing the slot to a correctly priced entrant buys more flexibility.

The size of the shock acts on the amplification the way trend inflation does. I hold every parameter at its calibrated value, vary the size of the one-time nominal shock from half a percent to five percent of the price level, and recompute the one-year CIR under both exit rules at each trend inflation level. Trend inflation and shock size both raise the repricing that happens for

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<sup>24</sup>In the computation,  $g$  and the adjustment policy come from the No-selection economy, and the hazard and the measure from the Benchmark, so both averages count the corrections the removed firms would have recorded absent selection, matching the design of the phantom cohort in (8).

**Figure 18: Pending price increase of the firms selective exit removes and of the average firm.**



*Notes:* The monthly log price increase a firm records if it survives to the pricing stage,  $g(s)$  of equation (9), averaged over the firms exit removes with weight  $\lambda(s) d\mu(s)/\delta$  and over the whole cross-section with weight  $d\mu(s)$ .  $g$  and the adjustment policy come from the No-selection economy and the hazard and the measure from the Benchmark. The vertical axis is logarithmic, so the gap between the two series reads as the ratio  $\mathbb{E}_\lambda[g]/\mathbb{E}[g]$ . The dotted line marks the calibrated rate.

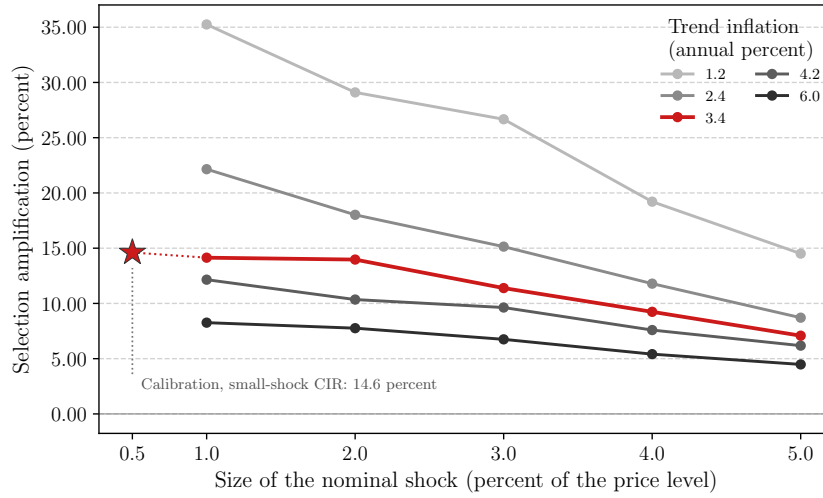
reasons other than exit. The trend does it every month; a large shock does it once, by pushing firms out of their inaction band on impact. Selection removes firms that were going to reprice anyway, so the extra adjusters a large shock recruits are firms random exit takes, and the two rules converge again.

Figure 19 reports the result. Amplification falls with the size of the shock at every trend inflation level, and the ranking across levels holds at every size. A five percent shock cuts the amplification by half or more: from 14.6 to 7.1 percent at the calibrated trend, from 34.2 to 14.5 at a 1.2 percent trend, from 9.1 to 4.5 at six percent. Shock size and trend inflation substitute: a five percent shock at 1.2 percent trend inflation gives 14.5 percent amplification, the same amplification a small shock gives at the calibrated 3.36 percent trend. The fall comes from the selective side. At the calibrated trend the No-selection CIR moves from 0.155 to 0.151 as the shock grows from half a percent to five percent, while the Benchmark CIR falls from 0.178 to 0.161. What selection builds is a stock of mispriced firms it keeps from correcting, and a large shock releases that stock at once, the way a higher trend releases it month by month.

## 6 Understanding the results

The results so far quantify the amplifying effect of selection on the real effects of nominal shocks and decompose it into channels. To understand what drives these findings, I consider a reduced

**Figure 19: Selection amplification across shock sizes.**



*Notes:* Selection amplification of the one-year CIR (the percent increase under selective relative to random exit, at a common total exit rate) against the size of the one-time nominal shock, one line per annualized trend-inflation level. Shocks are expressed as a percent of the price level. The star marks the calibrated economy at the small-shock limit, the headline amplification of Section 4.2. Zero trend inflation is omitted: its net repricing flow is near zero, so the ratio is numerically unstable.

representation of the model in which the mechanism can be solved in closed form.<sup>25</sup> I collapse the firm's state to a single dimension, its price gap, hold the inaction band fixed, and shut down general equilibrium feedback. What survives is the distribution channel of Section 4.4 in isolation: everything selection does through the composition of the cross-section, holding the pricing policy of every firm fixed. First, in line with the literature, monetary non-neutrality equals the dispersion of price gaps in the cross-section, so what exit does to money is decided by where in the gap distribution it lands. Second, the two extreme landing spots carry opposite prices: exit spread randomly over the cross-section removes standing mispricing and makes money more neutral, while exit at the adjustment margin removes nothing, because the prices it censors were about to be corrected anyway. Third, holding the total exit flow fixed turns the second result into an amplification theorem, confirming the findings of Section 4. At a fixed flow, concentrating exit at the adjustment margin necessarily withdraws it from the interior of the band, so the mispricing that random exit would have removed now survives, and money stays non-neutral longer. Selection amplifies exactly when it re-targets exit in this direction. All derivations and proofs are in Appendix F.

I use a continuous time version of the model and the tools coming from the more general literature on lumpy adjustments (Baley and Blanco, 2021): the aggregation result below extends formulas in Alvarez and Lippi (2014), Alvarez et al. (2016), Baley and Blanco (2021), and Al-

<sup>25</sup>The choice of continuous time follows the analytical menu-cost literature: with the gap following a diffusion, expected spell values and the stationary density solve ordinary differential equations, which is what makes the CIR computable in closed form.

varez et al. (2022) to an economy with entry and state-dependent exit, while the exit layer, the pricing of exit placement and the amplification theorem, is new to the best of my knowledge. The reading of turnover as a source of aggregate price flexibility that emerges below connects to Bilbiie (2021), who shows that with sticky prices the entry margin can substitute for price flexibility and even restore monetary neutrality when entry is frictionless; here the substitution operates through the composition of the surviving cross-section in a state-dependent pricing model, and selection is precisely what switches it off.

## 6.1 The reduced model

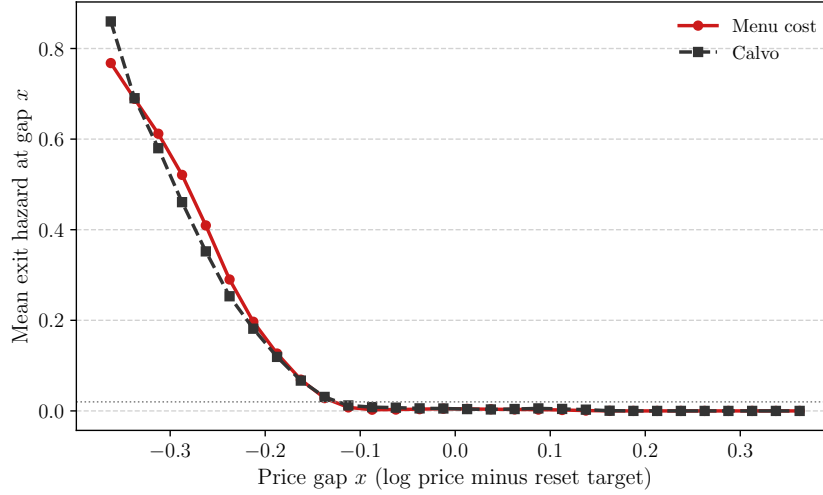
A firm's state is its price gap  $x$ : the log difference between the price it charges and the price it would set if adjustment were free. Trend inflation erodes every gap downward, and idiosyncratic shocks move the target price around, so between adjustments the gap follows  $dx = -\bar{\pi} dt + \sigma dW$ , with  $\sigma$  the standard deviation of gap innovations. A firm whose gap reaches the edge of the inaction band  $(-a, a)$  pays the menu cost and resets its gap to zero. Exit arrives at a hazard  $\lambda(x)$  that may depend on the gap, and the exiting firm is replaced by an entrant born at gap zero: the entrant prices optimally on entry, so it starts correctly priced.<sup>26</sup> Adjustment and exit therefore both place a correct price at that slot of the economy; what distinguishes them is which firms they hit. The monetary shock is a small once-and-for-all increase in nominal demand, which lowers every gap by  $\Delta$  on impact, and the object of interest is the cumulative impulse response of output per unit of shock, the continuous time counterpart of the CIR of Section 4.2. Every comparison holds the exit flow fixed: the average hazard over the cross-section equals  $\delta$  in each economy, mirroring the normalization of the structural hazard (5). The no-selection benchmark is the uniform hazard  $\lambda_u := \delta$ ; a selective economy is any hazard  $\lambda_s(x)$  with the same flow.<sup>27</sup>

Furthermore, exit in the full model selects on the firm's value, which is mostly driven by productivity, not by mispricing. While this is true, it is irrelevant for the mechanism I study

<sup>26</sup>In the quantitative model an entrant sets its flexible price and then receives one within-period productivity shock before joining the cross-section (Section 3), so it starts the next period with a gap of one idiosyncratic innovation rather than exactly zero. The wedge is one period of noise and vanishes in the limit as the period length shrinks; birth at gap zero is the continuous time limit of the model's entry block.

<sup>27</sup>Two remarks on the reduced model. First, a negative gap can have two sources in the full model: a bad productivity draw, which raises the firm's target price above its posted price at once, and trend inflation, which erodes every posted price slowly and can accumulate price gaps. The reduced model carries both, as the diffusion term and the drift term, and treats two firms with the same gap identically regardless of the source. For everything that happens on impact this is not an approximation: repricing incentives, and the event probabilities outside the inaction band, depend on the gap alone, however the firm got there. Yet, the zero-drift results of this section focus on the idiosyncratic sources of a price gap: at  $\bar{\pi} = 0$  there is no erosion at all and the negative gaps are possible only because of a productivity draw, so erosion does not drive the amplification. Second, what the reduced model does give up is the continuation and the persistence. In the full model productivity mean-reverts, so a firm whose gap came from a bad draw would correct its price now and then walk it back down as productivity recovers, while an erosion-driven gap carries no such trajectory. The gap process here has no mean reversion, so every statement in this section concerns the impact aspect of the counterfactual; the delayed element belongs to the persistence margin the reduced model deliberately omits, and it is exactly what the phantom-adjustment exercise of Section 4.3 measures in the full model.

**Figure 20: The exit hazard in the space of price gaps.**



*Notes:* Mean monthly exit hazard conditional on the price gap,  $\mathbb{E}[\lambda(s) \mid \text{gap} = x]$ , over the stationary cross-section: benchmark menu-cost economy (red circles) and the Calvo economy of Section 7 (black squares). The dotted line marks the average exit rate  $\delta = 0.02$ . The gap is the log distance between a firm's price and its reset target, so the left side collects underpriced firms who would raise their price but are most likely to exit.

here, because the results below depend on the hazard only through where exit lands in gap space, and that object can be simply measured. Figure 20 plots the mean exit hazard conditional on the price gap,  $\mathbb{E}[\lambda(s) \mid \text{gap} = x]$ , over the stationary cross-section of the benchmark economy. The hazard is flat and negligible over the interior of the gap distribution and rises steeply on the negative side, the side populated by firms that have just received bad productivity draws, and thus find themselves underpriced and in need of raising their prices (hence a negative gap). Therefore, even though it is true that productivity is the main driver of selection, it is also the main predictor of price gaps. Exit selects on value, the lowest values belong to the least productive firms, and those firms' inherited prices sit furthest below the prices they would like to charge. They are the firms at, or past, the margin where prices get raised. In gap space, selection on value is selection on imminent repricing.<sup>28</sup>

<sup>28</sup>The projection matches the exit flow out of every gap by construction, since it is a conditional expectation, but it collapses the dispersion of exit risk within a gap. A low-productivity firm that resets re-enters at gap zero and inherits the low hazard of the firms sitting there, while in the full model it keeps its productivity and its exit risk. The reduced model therefore lets a firm shed its exposure by adjusting, which the full model does not. This is again the continuation margin set aside above, and it leaves untouched where exit lands, which is what the results below price and what Figure 20 measures on the solved model.

## 6.2 Monetary non-neutrality and exit

**Proposition 1** (Non-neutrality is gap dispersion). *In the reduced model with zero trend inflation, for any exit hazard and any inaction band,*

$$\text{CIR} = \frac{\text{Var}_f(x)}{\sigma^2},$$

where  $f$  is the stationary distribution of price gaps. Trend inflation adds a correction term of order  $\bar{\pi}$ .

Money is non-neutral because, at any moment, the economy carries a spread of wrong prices, and the shock survives in that spread:  $\text{Var}_f/\sigma^2$  measures how much standing mispricing the economy tolerates per unit of new mispricing injected by idiosyncratic shocks each period. The formula is the price-gap version of the sufficient statistics of [Alvarez et al. \(2016\)](#) and [Baley and Blanco \(2021\)](#): the aggregation logic survives entry and exit exactly, for any hazard and any placement of exit. Every reset, whether an adjustment or an exit and replacement, moves one firm from wherever it stands to a correct price, so what exit does to money depends only on where in the cross-section it lands. Exit that removes a firm sitting in the middle of the distribution deletes mispricing that would otherwise have persisted, compresses the spread, and makes money more neutral. Exit that removes a firm about to adjust deletes nothing, because that firm's price was about to be corrected anyway.

Since the exit flow is fixed, every unit of exit that selection spends on a firm about to adjust, the harmless kind, is a unit it does not spend in the middle of the distribution, the kind that would have deleted persistent mispricing. Censoring the imminent adjusters is therefore equivalent to sparing the mispriced firms in the interior of the inaction band, where more persistent price gaps would have survived, and hence works as the source of the amplification.

Consider first the no-selection model, where exit falls uniformly on the cross-section. A random draw from the gap distribution typically lands on a firm in the middle of the band, holding a wrong price that would have persisted; exit eliminates that mispricing now and hands the slot to a correctly priced entrant. Random churn therefore works like a subsidy to aggregate price flexibility: it continuously removes standing mispricing that the pricing friction would have let stand. Proposition [A.2](#) in the appendix shows that with a uniform hazard  $\lambda$  the CIR admits a closed form, strictly decreasing in  $\lambda$ ,

$$\text{CIR}(\lambda) = \frac{a^2}{6\sigma^2} - \frac{a^4}{60\sigma^4}\lambda + O(\lambda^2),$$

The no-selection economy is thus not a neutral yardstick: it is an economy in which turnover continuously buys price flexibility, and whatever amplification selection delivers must come from reverting this effect. The opposite case is exit that lands on a firm outside the inaction band, one that would adjust immediately (Proposition [A.3](#)). Outside the band an event happens with probability one; the exit hazard only decides whether the event is recorded as a price adjustment by the incumbent or as an exit followed by a correctly priced entrant, and the slot carries a correct price either way. Between the two cases, the marginal effect of exit at each gap is a function of the gap.

**Proposition 2** (The price of exit placement). *At zero trend inflation, let  $\lambda$  be any baseline hazard with stationary distribution  $f$  and mean gap  $\bar{x}$ . Adding exit in any direction  $h$ ,  $\lambda \rightarrow \lambda + \theta h$ , changes the CIR at rate*

$$\left. \frac{d \text{CIR}}{d\theta} \right|_{\theta=0} = \int_{-a}^a \omega(x) h(x) dx, \quad \omega(x) = \frac{f(x)}{\sigma^2} \mathbb{E}_x \left[ \int_0^\tau (\text{Var}_f - (x_t - \bar{x})^2) dt \right], \quad (11)$$

where  $\mathbb{E}_x$  follows one firm from gap  $x$  until its next reset at time  $\tau$ , by adjustment or exit. At the no-exit baseline the kernel has the closed form

$$\omega_0(x) = - \frac{x^2 (a - |x|)^2 (a + |x|)}{6 \sigma^4 a^2} \leq 0.$$

The kernel<sup>29</sup> compares the firms removed by exit with the average firm. Follow a firm from gap  $x$  over its remaining spell, and at each instant compare the squared mispricing it carries,  $(x_t - \bar{x})^2$ , with the economy's per-firm squared deviation,  $\text{Var}_f$ ; the kernel accumulates the difference. A unit of exit at  $x$  raises the CIR if the firm it kills was going to carry less mispricing than average over its remaining life, so that the cross-section could spare it, and lowers the CIR if the firm was going to carry more, because exit then deletes more standing mispricing than the entrant that replaces it restores. The density  $f(x)$  counts how many firms the extra hazard actually hits at  $x$ , and  $1/\sigma^2$  converts dispersion into CIR units through Proposition 1.

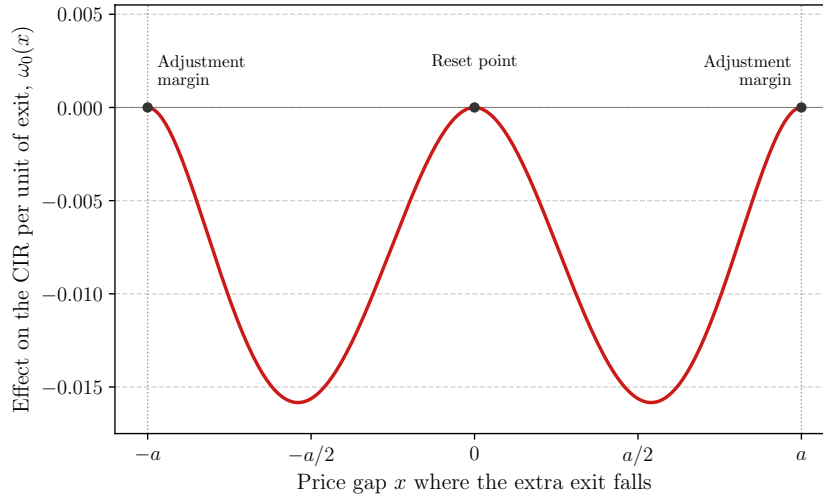
Figure 21 plots the closed form of the kernel at the no-exit baseline. Exit generates no change in the CIR at the reset point and at the adjustment margins.<sup>30</sup> At the reset point the effect is null because the killed firm carries a correct price, exactly like the entrant that replaces it. At the adjustment margin the same happens for the opposite reason: the killed firm's spell has nothing left in it, its correction was imminent, so exit merely relabels that correction as turnover. Between the two zeros the kernel is strictly negative. A firm there holds a wrong price that no imminent event would fix, so killing it trades mispricing for a correct price and makes the aggregate price level effectively more flexible. The trough sits at intermediate gaps because the loss is the product of two forces that move against each other: how wrong the price is, which grows toward the margin, and how long it would have stayed wrong, which shrinks to zero at the margin.

The kernel is the analytical counterpart of the phantom-adjustment exercise of Section 4.3. Both computations start from the same firms, the ones exit removes: the exit-source measure  $\lambda_b d\mu_b$  in (8) there, the extra hazard times the density here. Both then ask the same counterfactual question: what would this firm have done had it not been removed? The phantom exercise answers by counting events. It follows the removed cohort under the no-selection dynamics and records the price changes that never happened: upward at impact, downward

<sup>29</sup>Kernel here means the weight function of an integral, as in the Green's function of an ODE: (11) computes the change in the CIR by adding up the extra exit placed at every gap, each unit weighted by  $\omega(x)$ . The kernel is that weight, the price of one unit of exit at gap  $x$ .

<sup>30</sup>The zeros are robust to any baseline hazard and any trend inflation, see Lemma A.3

**Figure 21: The price of exit at each point of the cross-section.**



*Notes:* The placement kernel  $\omega_0$  of Proposition 2 at the no-exit baseline, in normalized units ( $a = \sigma = 1$ ): the marginal effect on the CIR of one unit of exit placed at gap  $x$ . The dots mark the exact zeros. Exit at the reset point replaces a correctly priced firm with a correctly priced entrant; exit at either adjustment margin removes a firm whose price was about to be corrected anyway.

later as productivity mean-reverts. The kernel answers by valuing states. It follows the same counterfactual firm and adds up, instant by instant until the next reset, the mispricing the firm would have carried in excess of the cross-sectional average; since by Proposition 1 the CIR is exactly the economy's standing mispricing, that sum is the exit's effect on the CIR. So the exercise measures what disappears from the price-change data, and the kernel measures what the disappearance is worth for money non-neutrality.

### 6.3 When selection amplifies

Random exit dampens because it lands mostly in the interior of the band, where the kernel is negative; exit at the margin is free. At a fixed total exit flow the two placements compete: every unit of exit that selection concentrates on a firm about to adjust is a unit withdrawn from the interior of the band. Amplification is that withdrawal. Selection does remove mispricing; in fact it removes the largest gaps in the economy. But the CIR does not price a gap by its size at the moment of exit. By Proposition 1 the CIR is the standing stock of mispricing, a unit of exit removes everything the killed firm would have still contributed: how wrong its price was, times how long the price would have stayed wrong. That product is the kernel's spell integral, and the two factors move in opposite directions.

For a firm in the interior of the band the product is large: the gap is moderate but nothing was about to correct it. For a firm at the margin the product is zero: the gap is the largest in the economy, but its remaining life is not, because the firm was about to reset on its own. Selection concentrates exit on the second kind of firm, so at an equal flow it deletes less of the standing stock than random exit does, and the extra stock that survives is what the nominal shock moves. For the amplification to take place, selection must re-target in the right direction, and I provide

one sufficient condition below under which selection increases monetary non-neutrality.

**Condition M (margin-concentrated exit).** *Relative to the uniform hazard with the same exit flow,*

$$\int_{-a}^a \omega_0(x) (\lambda_s(x) - \lambda_u(x)) dx > 0.$$

Condition M states in one integral what the numerical censoring channel requires: relative to spreading the same exit flow evenly, selection must on net remove firms that would have changed their price soon anyway, the firms at or near the edges of the band, and spare the firms in the interior, whose wrong prices had time left and whose removal would therefore delete standing mispricing. Because  $\omega_0 \leq 0$  vanishes only at the reset point and at the band edges, extra hazard placed near the edges costs the integral almost nothing, while hazard withdrawn from the interior, where the kernel is strictly negative, pushes it up. The condition holds for any hazard with this profile: it does not require the hazard to vanish anywhere, to be monotone, or to take any particular functional form. It fails when the targeting runs the other way, exit concentrated at intermediate gaps, on firms mispriced enough to matter but whose correction was not imminent. For a similar reason, selection must target firms close to the reset point at zero. Finally, the condition compares placements, not levels: multiplying both hazards by a common factor leaves its sign unchanged, so one check on the measured hazard settles it at every exit intensity the theorem considers.

**Theorem 1** (Selection amplifies). *Let  $h \geq 0$  be a bounded hazard direction,  $c = \mathbb{E}_{f_0}[h]$  its flow-matched uniform level, and scale both by an intensity  $\theta > 0$ :  $\lambda_s = \theta h$ ,  $\lambda_u = \theta c$ . Then*

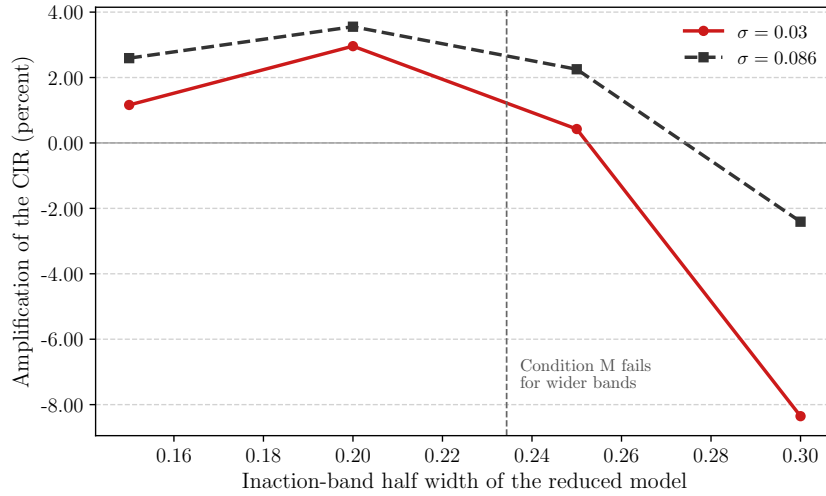
$$\text{CIR}_s - \text{CIR}_u = \theta \int_{-a}^a \omega_0(x) (h(x) - c) dx + O(\theta^2),$$

*so under Condition M there is  $\bar{\theta}(h) > 0$  such that  $\text{CIR}_s > \text{CIR}_u$  for every  $\theta \in (0, \bar{\theta})$ : at an equal exit flow, the selective economy has the larger real effects of nominal shocks.*

Theorem A.1 in the appendix extends the statement beyond small exit flows: the difference between any two equal-flow economies equals the kernel-weighted difference of their hazards, averaged along the path connecting them, and the sign survives whenever the extra exit stays confined near the adjustment margin.

Condition M is only sufficient, hence selection might still be amplifying even when it fails. However, it can be used to construct an educated guess of when selection would instead dampen the effects of a monetary shock. Feeding the measured  $\lambda(x)$  into the reduced model at an equal exit flow and comparing the two CIRs runs the theorem's experiment quantitatively (Figure 22). At band half widths between 0.15 and 0.25, the selective economy's CIR exceeds the uniform one by one to four percent. Stretching the band while holding the hazard fixed moves the same exit into the kernel's trough: gaps that used to trigger an immediate adjustment now sit well inside the band, so the hazard concentrated at those gaps starts deleting standing mispricing instead of relabeling imminent corrections. The figure shows both boundaries of this reversal. Condition M's integral, evaluated on the measured hazard, turns negative

**Figure 22: The theorem on the measured hazard, and its boundary.**



*Notes:* Equal-flow amplification in the reduced model when the exit hazard is the measured  $\lambda(x)$  of Figure 20 instead of the uniform  $\delta$ , in percent of the uniform-exit CIR, as a function of the reduced model's band half width, for two values of the innovation volatility. The vertical dashed line marks the half width 0.23 at which the Condition M integral, evaluated on the measured hazard, changes sign. At bands in the calibrated range the amplification is positive; widening the band moves the same hazard into the interior of the band, where the kernel is negative, and the amplification turns into dampening.

at a half width of 0.23, the dashed line; the amplification itself turns negative between 0.25 and 0.30. Again, the two lines do not need to coincide as the condition is only sufficient rather than necessary: amplification survives slightly past the point where the condition stops guaranteeing it, but no further. Amplification is therefore a property of exit selecting firms whose price correction it censors, and the effect comes from where the hazard sits relative to the adjustment margin.

Two final remarks. First, on magnitudes. The theorem signs the distribution channel at a fixed inaction band, and the exercise of Figure 22 sizes it at a few percent, against the fifteen percent of Section 4.2. The difference is the option-value channel and its interaction with composition: in the full model the hazard also changes the inaction bands of exposed firms (Section 4.3), exposed firms then spend longer in exactly the queue that selective exit censors, and Section 4.4 prices the split. The theorem explains why the composition of exit matters and when its sign flips; the full model quantifies the magnitude. Second, on the literature. The mechanism runs opposite to the classical selection effect of Golosov and Lucas (2007). In Golosov and Lucas, selection concerns who adjusts. Holding the frequency of price changes fixed, the menu-cost economy directs its adjustments to the most mispriced firms, each price change deletes more mispricing than a randomly assigned one would, and the real effects of nominal shocks are a fraction of those of the Calvo economy, whose adjusters are drawn at random (Alvarez et al., 2016). Here, selection concerns who exits. Holding the exit flow fixed, the benchmark directs its exits to firms about to adjust, each exit deletes less mispricing than a randomly assigned one would, because the removed firm's correction was happening anyway, and the real effects are larger than in the no-selection economy. Selection on adjustment concen-

trates correction events where they delete the most mispricing, selection on exit concentrates them where they delete the least. In both cases what decides the aggregate consequence of a repricing technology is not how many correction events it generates but which firms receive them.

## 7 Calvo model

In this section I replace the menu cost with a Calvo lottery, holding the household, the productivity process, the exit hazard, and the entry block fixed. The comparison matters for two reasons. First, there is a long-standing debate in the literature about whether state- and time-dependent pricing models are interchangeable or not (Alvarez et al., 2016, 2017, 2022; Auclert et al., 2024). Selective exit works in a slightly different way under the two pricing technologies, adding to the differences, but it still works as an amplifying force. Second, the reason the net effects differ locates which parts of the mechanism belong to the technology.

Selective exit increases the CIR only by around two percent, and the reason is that now the two channels, policy and censoring, work against each other. The option-value channel survives in rerouted form: a firm that faces a very high exit probability stops treating the pricing decision as a dynamic one, and will charge a price closer to the static optimum. The censoring channel however ceases to exist, and works against the policy one: since the timing of a price change is exogenous now, any exit, whether it is selective or not, ends up picking firms at random, and behaving like the uniform exit from the previous section, hence subsidizing flexibility through entry.

### 7.1 Setup

The household block, the final-good sector, the productivity process, the exit hazard, and the entry block are unchanged. The only change is to the pricing margin. There is no menu cost in flow profits,

$$\pi(p_{it}, z_{it}, \mathcal{S}_t) = \frac{p_{it}}{P_t} y_{it} - \frac{W_t}{P_t} l_{it},$$

and in each period a firm receives an independent adjustment opportunity with probability  $\alpha$ . With probability  $1 - \alpha$  it keeps its inherited price; with probability  $\alpha$  it resets the price.

Let  $V_N(p, z, \mathcal{S})$  denote the firm's value if it cannot adjust this period and stays at inherited price  $p$ , and let  $V_A(z, \mathcal{S})$  denote its value if it can. The two satisfy

$$\begin{aligned} V_N(p, z, \mathcal{S}) &= \pi(p, z, \mathcal{S}) + \mathbb{E}[\Xi(1 - \lambda(s'))((1 - \alpha)V_N(p, z', \mathcal{S}') + \alpha V_A(z', \mathcal{S}')], \\ V_A(z, \mathcal{S}) &= \max_{p'} V_N(p', z, \mathcal{S}). \end{aligned}$$

The first equation says that a non-adjusting firm earns flow profit at its inherited price and then enters next period with the same  $p$ , where it either draws an adjustment opportunity (continuation value  $V_A$ , probability  $\alpha$ ) or does not (continuation value  $V_N$ , probability  $1 - \alpha$ ). The second equation says that an adjusting firm picks the price  $p'$  that maximizes its continuation; the cho-

**Table 6: Calvo model: pricing moments under selective exit, baseline calibration.**

|              | $\xi$ | $\tau$ | Frequency | $\sigma(\Delta \log p)$ |
|--------------|-------|--------|-----------|-------------------------|
| No selection | 0     | 0      | 0.088     | 0.081                   |
| Benchmark    | 20    | 5      | 0.088     | 0.084                   |

*Notes:* Moments are computed on the stationary distribution of  $\Delta \log p$  conditional on a price change, in the Calvo model with  $\alpha = 0.088$  at the baseline calibration ( $\rho = 0.85$ ,  $\sigma_\varepsilon = 0.086$ ). The selective-exit benchmark has  $\xi = 20$ ,  $\tau = 5$ ; the no-selection economy has  $\xi = \tau = 0$ .

sen  $p'$  then becomes the inherited price going forward, and the value the firm attaches to that inherited price is  $V_N(p', z, \mathcal{S})$ . Iterating the  $V_N$  recursion forward gives the discounted-sum form,

$$V_N(p, z, \mathcal{S}) = \mathbb{E} \left[ \sum_{t \geq 0} (1 - \alpha)^t D_t \pi(p, z_t, \mathcal{S}_t) \right] + \alpha \mathbb{E} \left[ \sum_{t \geq 0} (1 - \alpha)^t D_{t+1} V_A(z_{t+1}, \mathcal{S}_{t+1}) \right],$$

where  $D_t = \prod_{j=0}^{t-1} \Xi_{j,j+1}(1 - \lambda(s_{j+1}))$  is the cumulative discount-survival factor up to date  $t$ , with  $D_0 = 1$ . The first sum is the expected discounted profit over the random spell of non-adjustment, weighted by the probability  $(1 - \alpha)^t$  that the firm has not yet drawn an adjustment opportunity by horizon  $t$ . The second is the present value once an adjustment opportunity arrives. The optimal reset price  $p^*(z, \mathcal{S}) = \arg \max_p V_N(p, z, \mathcal{S})$  is the maximizer of  $V_N$  in its first argument.

For implementation, the two recursions collapse into the unconditional value the firm enters the period with, before the Calvo coin is flipped:

$$V(s) = \alpha V_A(s) + (1 - \alpha) V_N(s). \quad (12)$$

Adjustment timing is independent of the firm's state. In the timing of Figure 6, step 3 is replaced by an exogenous Bernoulli draw on whether each surviving incumbent is allowed to adjust. The exit hazard (5) is unchanged and still depends on  $V(s)$ . I fix  $\alpha = 0.088$  across both Calvo economies, matching the monthly adjustment frequency of the calibrated menu-cost model and mirroring the menu-cost exercise of Section 4 in which the menu cost was held at its calibrated value across Benchmark and No-selection. The underlying nominal friction is therefore held constant within each comparison, so any change in the conditional distribution of  $\Delta \log p$  reflects the response of the equilibrium to the exit margin rather than a change in the friction. All other parameters remain at their calibrated values.

## 7.2 Selective exit in the Calvo economy

Table 6 reports stationary moments of the conditional distribution of  $\Delta \log p$  for the two Calvo economies at the baseline calibration: the no-selection case ( $\xi = 0$ ,  $\tau = 0$ ) and the selective-exit benchmark ( $\xi = 20$ ,  $\tau = 5$ ). Adjustment frequency is identical by construction (both economies face the same Calvo lottery rate  $\alpha$ ) and the standard deviation of  $\Delta \log p$  is also similar.

The decomposition of Section 4.4 applies to the Calvo economy without modification: firms

discount survival with a perceived hazard  $\lambda_B$ , the law of motion deletes mass with a realized hazard  $\lambda_F$ , each of the form (5) with its own tilt  $\xi_j$  and the common flow  $\delta$ , and the Calvo lottery rate  $\alpha$  is the same in every economy. The one-year CIR of the no-selection economy is 0.552 and that of the benchmark 0.565: selection amplifies Calvo non-neutrality by about two percent, against the fifteen percent of the menu-cost model.

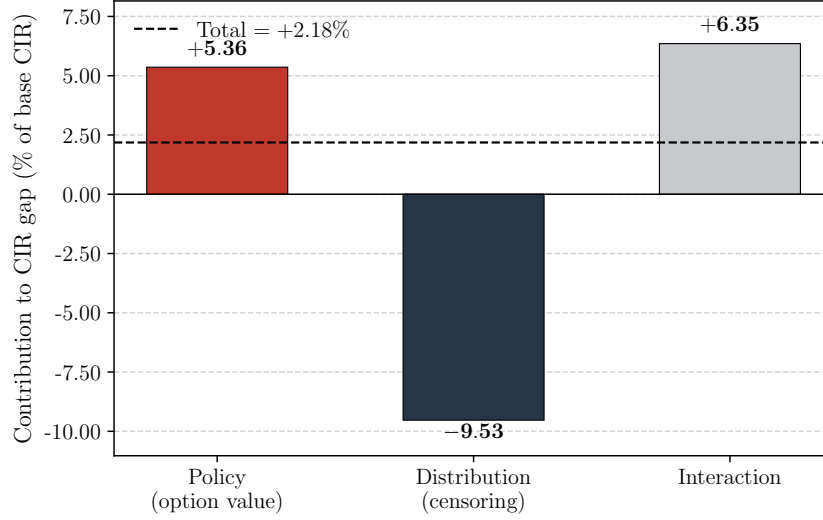
Figure 23 reports the decomposition. Selective exit raises the CIR by two percent, against fifteen percent under the menu cost. The censoring channel explains the gap: it amplified the response under the menu cost and dampens it under Calvo, at a distribution-only CIR of 0.500 against 0.552 with no selection. An exit's worth is the mispricing it deletes: how wrong the price is, times how long it would have stayed wrong. Under the menu cost, selection kills firms sitting at the adjustment margin, whose price was about to move on its own. Their remaining wrong life is zero, so killing them is free, and at a fixed exit flow that exit comes out of the interior of the cross-section instead, where the mispricing sits. The Calvo lottery has no memory and no state dependence: every firm, doomed or not, waits an expected  $1/\alpha$  months for its next opportunity to reprice, so no firm's remaining wrong life is zero and no deletion is free. Whether selection beats uniform exit then turns on where the hazard lands in the space of price gaps. Figure 20 measures the hazard, projected on price gaps: it sits near zero over the interior and the positive-gap side and rises to about 0.86 per month at the most negative gaps, against a uniform rate of 0.02. Selection lands on the firms carrying the largest standing price gaps, prices that would otherwise have waited out the lottery: a firm in the tail is far more likely to die, at a hazard of 0.86, than to reprice on its own, at  $\alpha = 0.088$ . Each selective exit therefore deletes more mispricing than a uniform one would, a larger flexibility subsidy per unit of exit flow, and the bar goes negative.

In the calibrated economy, selective exit under Calvo recreates the selection effect of Golosov and Lucas (2007) through the turnover margin. The exit margin is the only state-dependent correction technology left under Calvo: the lottery moves prices at random, blind to the price gap. A state-dependent hazard turns exit and replacement into a correction device instead: it aims at the firms with the largest gaps, and on exit an entrant with a freshly set price takes their place. Each of those exits deletes more mispricing than one assigned at random, the force that pushes toward monetary neutrality, and it is why the censoring bar carries the dampening sign. The menu cost escapes this logic because the pricing technology already corrects the largest gaps on its own, so a hazard aimed at the tail catches corrections that were about to happen anyway, and selection on exit has nothing left to delete.

Nonetheless, amplification survives to some extent through the policy channel. Figure 24 reports the optimal reset price  $p^*(a)$  as a function of productivity, both for the Calvo benchmark and the Calvo no-selection economy, alongside the static flex-price optimum at the benchmark aggregate state.

The static optimum (dotted line) is a constant markup over marginal cost and is therefore linear in productivity with slope  $-1$ . The Calvo reset price differs from this contemporaneous static optimum because the firm sets a price it expects to hold across a random spell of non-adjustment. With mean-reverting productivity, future productivity is expected to differ

**Figure 23: CIR decomposition in the Calvo model: No selection to Benchmark.**



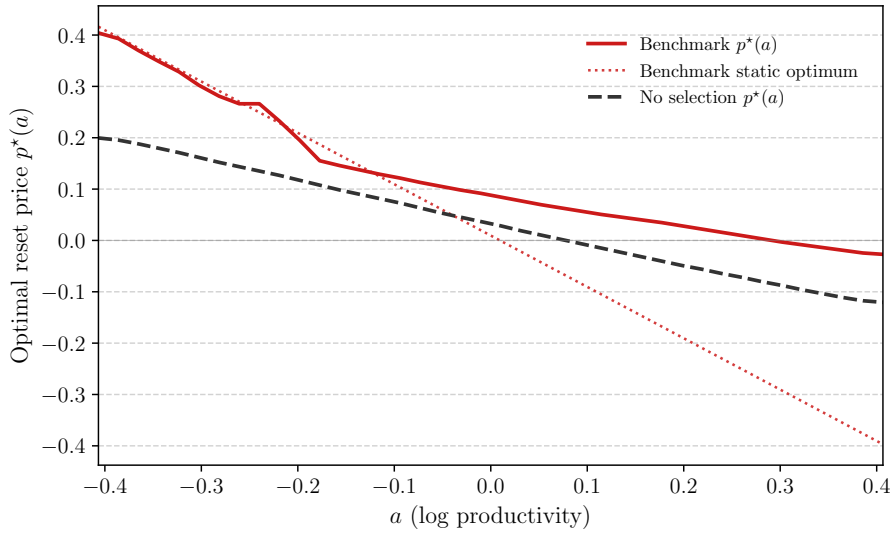
*Notes:* Contribution of each channel to the one-year CIR gap between the Calvo no-selection economy and the Calvo benchmark, in percent of the no-selection CIR. The policy bar is the CIR of the economy with the tilt only in the perceived hazard,  $(\xi_B, \xi_F) = (20, 0)$ , the distribution bar that of the economy with the tilt only in the realized hazard,  $(0, 20)$ , each in deviation from the base; the interaction is the residual to the total gap (dashed line). Both counterfactual economies retain the benchmark entrant distribution, so the policy bar here also carries the small entry-tilt offset that Figure 15 assigns to the distribution side. One-year CIRs: no selection 0.552, policy only 0.582, distribution only 0.500, benchmark 0.565.

from current productivity: a firm at low  $z$  expects productivity to recover during the spell, the forward-looking optimum is therefore below the contemporaneous static optimum (future marginal cost will be lower), and the firm resets below the static line; a firm at high  $z$  faces the reverse and resets above it. Mean reversion thus flattens the reset-price function in productivity relative to the static optimum. The No-selection curve in the figure (dashed) is the textbook Calvo shape that emerges from this logic.

The figure shows that at very low productivities in the benchmark model, the Calvo reset price overlaps with the static optimum. A low-productivity firm in the Benchmark faces an extremely high exit hazard, so the cumulative survival-discount factor  $D_t = \prod_{j=0}^{t-1} \Xi_{j,j+1} (1 - \lambda(s_{j+1}))$  inside the discounted-sum representation of  $V_N$  collapses within one or two periods: the firm expects to die before mean reversion can bring its productivity back up. With effectively no horizon to optimize over, the firm picks the price that maximizes contemporaneous profit, which is the static markup over current marginal cost. Because marginal cost is high at low productivity, the static optimum is high. The same firm under no selection has a much longer effective horizon, internalizes future mean reversion, and trims its reset price toward the mean-reverted future, which lies well below the current static optimum. The Benchmark–No-selection gap at low  $z$  is therefore the response of the reset price to a shorter effective horizon, and it is largest in the region where the hazard binds.<sup>31</sup>

<sup>31</sup>The level shift is a general-equilibrium effect of selection. Selective exit removes mass from the low-

**Figure 24: Optimal reset price by productivity in the Calvo model.**



*Notes:* Optimal reset price  $p^*(a)$  as a function of log productivity, at the stationary aggregate state. The solid line is the calibrated benchmark with selective exit and tilted entry; the dashed line is the no-selection economy. The dotted line is the static flex-price optimum at the benchmark aggregate state, with slope  $-1$  in log productivity. The benchmark curve attains the static optimum at the low-productivity end of the support and is uniformly above the no-selection curve over the rest of the support; the gap is the Calvo analog of the inaction-band widening of Section 4.3 in the menu-cost model.

Figure 24 is the Calvo analog of the inaction-band figure 10 of Section 4.3. The underlying economic force is the same in both: a higher exit hazard shortens the firm's effective horizon and shifts weight from future to present payoffs. Under the menu cost the firm responds by widening its inaction band: the option-value response acts on the timing of adjustment. Under Calvo the firm cannot change timing because the lottery is exogenous, so the same economic force is forced into the level of the reset price.

In both models the interaction term is positive: the marginal effect of realized selection on the CIR is larger when firms price under the selective hazard, but it works in opposite ways in the two models. Under the menu cost, censoring amplifies, and anticipation raises what its protection is worth when compared to uniform exit. A wrong price dies by adjustment or by exit, and the two hazards compete for the same event: the option-value channel lowers the first for exposed firms, realized selection withdraws the second from the interior of the band, and each extension is worth more when the other is in place. The reduced model prices this in one derivative: uniform exit buys flexibility worth  $(a^4/60\sigma^4)\lambda$  of CIR, so the subsidy censoring reclaims grows with the fourth power of the band width while the CIR grows with the second; widening the bands raises the price of the exit that selection withdraws. Under

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productivity tail of the cross-section, so the surviving distribution has higher average productivity, aggregate output is higher, labor demand is higher, and the equilibrium real wage is higher. Marginal cost is proportional to the real wage divided by productivity, so a higher wage shifts the static optimum (and with it the reset price) up uniformly in log space.

Calvo, censoring dampens, and anticipation shrinks the dampening. With a memoryless lottery the remaining life of a wrong price is the same at every gap, so the value of a deletion is its squared gap, and selective exit, targeting the largest gaps, deletes more standing mispricing than uniform exit: its standalone contribution is then minus nine percent. Anticipation pulls the doomed firms onto the static optimum, so the firms selection kills carry little mispricing left to delete, and censoring's contribution falls to minus three percent. The interaction bar is that difference: one channel disarming the other.

## 8 Conclusion

This paper studies how the selectivity of firm exit affects the transmission of aggregate monetary shocks. I started from the simple observation that firms removed by exit are not random: using scanner data, I find that products with lower sales are disproportionately more likely to exit and change their prices less often. To quantify the role of selection in the transmission of nominal aggregate shocks, I build a general-equilibrium menu-cost economy with state-dependent exit and entry, calibrated to the rate and composition of turnover in the scanner data, and compare it to a counterfactual with the same total exit rate but no selectivity. Holding total turnover fixed, selective exit raises the real effects of a monetary shock by fifteen percent, and by up to thirty-five percent in periods of low inflation.

The amplification runs through two channels that the exit hazard operates at once. Through the policy functions, the hazard discounts continuation values, widens the inaction bands of the most exposed firms, and lets their wrong prices stand longer. In the cross-section, selection concentrates exit on firms at the adjustment margin, whose corrections entrants replace on the spot, and at a fixed exit flow this withdraws exit from the interior of the cross-section, where random turnover would have deleted standing mispricing. I use a reduced continuous-time model to characterize the distributional channel in closed form: monetary non-neutrality equals the dispersion of price gaps, and I find a sufficient condition for selection to amplify the effects of an aggregate shock in comparison with uniform exit at the same flow. The condition asks only where exit lands relative to the adjustment margin, and it holds on the hazard measured from the calibrated model. The Calvo economy shows a similar amplification, albeit of smaller magnitude. With exogenous adjustment timing there is no margin to censor: selective exit lands instead on the largest price gaps, whose remaining life is as long as anyone's, so the censoring channel flips from amplifying to dampening, and the two percent of amplification that survives runs through the policy channel alone, as firms facing near-certain exit price close to the static optimum.

These results carry three implications. For measurement, the real effects of nominal shocks depend not only on how often prices change but on which products disappear: two economies with the same pricing moments and the same turnover rate differ by fifteen percent in monetary non-neutrality, so calibrations that treat product exit as random, or drop exiting products from the sample, absorb part of the transmission mechanism into measurement error. For policy, the amplification is regime-dependent: it exceeds thirty percent near zero inflation and dies

out above six percent, so the turnover margin makes monetary policy least neutral precisely in low-inflation environments, and changes in transmission across inflation regimes may partly reflect the composition of churn rather than the pricing friction itself. For firm dynamics, entry and exit act as a repricing technology: turnover continuously replaces standing mispricing with fresh prices, and selection is what switches this substitute for price flexibility off. Anything that reshapes the composition of churn, entry costs, retail concentration, the secular decline in product dynamism, therefore moves the potency of monetary policy without touching a single pricing moment. The model holds the entrant mass fixed by replacement and the nominal-demand rule exogenous; letting entry respond to the aggregate state, letting the central bank respond to the inflation that selection reshapes, and confronting the cyclicity of the exit hazard itself with data on churn over the business cycle are the natural next steps.

The cross-sectional channel rests on one idiosyncratic shock doing two jobs. A bad productivity draw lowers a firm's value, exposing it to exit, and raises its desired price, putting it at the repricing margin. The mechanism needs only the second half: exit must land on firms whose price change was imminent. Idiosyncratic demand does the same job once the demand system lets scale move the desired price: outside CES, where a multiplicative shifter would drop out of the pricing condition, a firm losing demand wants a lower price, and the direction of the censored correction flips. Exit still intercepts an imminent price change and hands the slot to a freshly priced entrant; the censoring channel goes through. The sign flip changes the interaction with trend inflation. Erosion opposes the desired correction instead of reinforcing it, and the inflation-regime results would shift. With productivity and demand together, the channel's strength becomes a quantitative question about their joint distribution: which shock drives exit, and how exit correlates with price gaps.

The model treats each product as an independent price setter and each exit as an independent death. In the data, most products are sold by multi-product firms. Under [Midrigan \(2011\)](#)'s model of price changes with economies of scope, the discussion of imminence moves to the firm level. In a similar way, a richer entry and exit process of both products and firms would affect the transmission of aggregate shocks differently. Finally, the exit hazard is reduced-form and flow-normalized by design. If, on one hand, it nests classic exit processes such as [Hopenhayn \(1992\)](#) when the exit elasticity is taken to the limit, on the other hand, the average exit rate is constant by construction, and hence the amount of turnover cannot respond to changes in the aggregate economy. An exit margin microfounded with operating costs would add a level response on top of the composition effect measured here. Both extensions point to the same next step: data that follow pricing and exit inside the firm.

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## A Data

This appendix describes the data cleaning and construction of the estimation samples used in Section 2. Section A.1 shows how to go from the raw transaction records to the sample of Table 1. Section A.2 shows that a chain posts one price per product nationwide, and that price changes of the same product are synchronized across chains. Section A.3 collects the robustness of the three cross-sectional patterns of Section 2.2, and Section A.4 collects the exit diagnostics. Section A.5 estimates the persistence of idiosyncratic productivity that the calibration of Section 4.1 uses.

### A.1 Sample construction

The raw data are transaction records: a household buys a barcode in a store of a retail chain in a province, on a date, in some units, for some value. The price of a transaction is the unit value, transaction value over units. I keep non-promotion transactions (12.8% of the records carry a promotion flag), drop the transactions with a missing province, round prices to the euro cent, and drop transactions with a price below €0.20 or above €25. The upper cut removes 0.05% of the remaining transactions, misplaced decimals and bulk-entry records. The lower cut removes 3.2% of transactions but 1.6% of value: items sold per piece for a few cents, half of them small bread rolls, where a one-cent move on the rounding grid is already a five percent price change.

The unit of observation is the cell, a UPC in a chain in a province. Within cell  $i$  and month  $t$ , let  $n_{it}(p)$  count transactions at price  $p$ . Since I do not observe the physical store location, but only the first two digits of a zip code, in principle there might be multiple stores belonging to the same chain in the same province posting different prices. To make sure I capture genuine price changes, and not some store-idiosyncratic temporary pricing policy, I use the following definition of posted price as the mode

$$p_{it} = \arg \max_p n_{it}(p), \quad s_{it} = \frac{n_{it}(p_{it})}{\sum_p n_{it}(p)}, \quad (13)$$

where  $s_{it}$  is the modal share. The mode returns a price that was actually posted, while a mean or a median of unit values moves with basket composition even when no posted price changes. A cell-month is retained when the mode rests on enough transactions and describes the month:  $\sum_p n_{it}(p) \geq \underline{n}$  and  $s_{it} \geq \underline{s}$ .

The retained months of a cell form its lifecycle. Writing  $\underline{t}_i$  and  $\bar{t}_i$  for the first and last retained month, the span is  $\bar{t}_i - \underline{t}_i + 1$ , the density is the number of retained months over the span, and the gap is the longest run of consecutive missing months inside the span. A lifecycle is valid when its span is at least  $\underline{T}$ , its density at least  $\underline{d}$ , and its longest gap at most  $\bar{g}$ . Lifecycles that begin in the first sample month are left-censored and lifecycles that end in the last are right-censored: their start and end are not observed events, and they are excluded from the entry and exit calculations respectively.

I use the following thresholds:  $\underline{n} = 20$ ,  $\underline{s} = 0.50$ ,  $\underline{T} = 12$ ,  $\underline{d} = 1/2$ ,  $\bar{g} = 6$ . Two further steps separate a genuine price change from residual noise. The first removes temporary sales

that the data provider failed to flag: the price drops for a month or two and returns exactly to its pre-sale level. I replace these V-shaped drops with the surrounding price before classifying changes: whenever

$$p_{i,t+\ell} = p_{i,t-1} \text{ for some } \ell \leq 2, \quad p_{i\tau} \neq p_{i,t-1} \text{ for some } \tau \in \{t, \dots, t + \ell - 1\}, \quad (14)$$

the filtered price is  $\tilde{p}_{i\tau} = p_{i,t-1}$  for  $\tau \in \{t, \dots, t + \ell - 1\}$ , and  $\tilde{p}_{i\tau} = p_{i\tau}$  otherwise. A sale that ends with a return to the old price is then no longer two price changes.

The second step restricts where a change can be observed at all. A month-to-month comparison is meaningful only when both months are retained, so I classify changes inside uninterrupted runs of consecutive retained months, and only in runs of at least twelve months. Within such a run, month  $t$  is an adjustment when

$$0.001 \leq |\Delta \log \tilde{p}_{it}| \leq \log(10/3), \quad (15)$$

where the lower bound removes rounding artifacts and the upper bound removes barcode reuse and misplaced decimal points, and both follow from the literature, e.g. [Alvarez et al. \(2016\)](#). The frequency of price changes is the count of adjustments over the count of consecutive month pairs inside such runs; the pricing rows of Table 1 report the resulting moments.

Entry in month  $t$  is a valid, non-left-censored lifecycle with  $\underline{t}_i = t$ ; exit is a valid, non-right-censored lifecycle with  $\bar{t}_i = t$ . The monthly exit rate averages, over the interior of the sample,

$$x_t = \frac{\#\{i : \bar{t}_i = t, \text{ not right-censored}\}}{\#\{i : \underline{t}_i \leq t \leq \bar{t}_i\}}, \quad (16)$$

where the interior drops the first and last twelve months, in which censoring mechanically depresses measured turnover; the entry rate is symmetric. The relative revenue of exiters is the average revenue of the products exiting in month  $t$  over the average revenue of incumbents in the same month, averaged over interior months, and likewise for entrants.

## A.2 Geography of price setting

A chain sets one price per product for the whole country. Comparing the modal posted price across the provinces in which a chain sells a product, in 98% of product-chain-months the price is the same in every one of them, and the standard deviation of the log price across provinces averages 0.001 (Table A.1). The same statistic computed across chains rather than provinces is 0.028, twenty-eight times larger, so prices do differ, but along the chain dimension and not the geographic one. The province is a sampling unit, not a market: I keep it in the cell definition because the panel records it and because it lets me hold the local market fixed in the tests below, but it carries no separate decision. A product that stops appearing in a chain has left that chain's national assortment.

When one chain changes the price of a product, a different chain selling the same product changes its price within one month in 63% of cases (Table A.1). Under independent timing the

estimated rate is 14%: shifting each cell's price-change months by a random offset of three to twenty-four months and recomputing the index yields 0.14. The matched changes also agree in direction: among cross-chain pairs of price changes within one month of each other, 94% have the same sign. Price changes of the same product are timed together across chains, four times as often as chance, and move the same way. Common movements in the product's cost generate this pattern; chain-specific pricing policies do not. I therefore read an observed price change as belonging to the product, which is what the competitive final-good sector of the model assumes.

The test matters because the data report retail prices and contain no wholesale costs, so an observed price change could in principle reflect a repricing decision by the chain rather than a change in the product's underlying cost; DellaVigna and Gentzkow (2019) document that chains in the United States set prices uniformly across their stores, so chain-level pricing policies are a real possibility. Synchronization across chains discriminates between the two: a chain policy moves many products within one chain, while a producer-side shock moves one product across many chains.

A cross-chain comparison requires a product sold in at least two chains; these are branded products by construction, since private labels are sold in a single chain. For product  $u$  in month  $t$ , an opportunity is an ordered pair of cells  $(i, j)$  in different chains such that cell  $i$  changes the price in  $t$  and cell  $j$  has a defined change indicator in  $t$ ; a match occurs if cell  $j$  changes the price in  $t - 1$ ,  $t$ , or  $t + 1$ . The synchronization index is the share of matches over opportunities, pooled across products and months:

$$\text{Sync} = \frac{\sum_{u,t} \sum_{(i,j) \in \mathcal{O}_{u,t}} \mathbb{1} \left\{ \text{chg}_{j,t-1} + \text{chg}_{j,t} + \text{chg}_{j,t+1} > 0 \right\}}{\sum_{u,t} |\mathcal{O}_{u,t}|}, \quad (17)$$

where  $\mathcal{O}_{u,t}$  is the set of opportunities and  $\text{chg}_{j,t}$  indicates a price change of cell  $j$  in month  $t$ . Table A.1 reports two variants that tighten the comparison: restricting the pair to cells in the same province, which holds the local market fixed, and aggregating cells to the product-chain-month before pairing, so that a chain selling the product in many provinces counts once. Both leave the conclusion unchanged. The same-month rate, with no window, is 0.46, six times the unconditional monthly frequency of price changes of 0.08 in this sample, so the co-movement is not an artifact of the one-month tolerance.

**Table A.1: Geography of price setting.**

|                                            | Rate | Observations |
|--------------------------------------------|------|--------------|
| Same price in every province, within chain | 0.98 | 51,907       |
| Synchronization, $\pm 1$ month             | 0.63 | 4,778        |
| Same province only                         | 0.62 | 799          |
| Chain level                                | 0.55 | 2,067        |
| Same month only                            | 0.46 | 4,778        |
| Placebo: random time shifts                | 0.14 | 4,778        |
| Same sign among matched changes            | 0.94 | 3,446        |

Notes: The first row covers the 51,907 product-chain-months of the pricing sample observed in two or more provinces, and reports the share in which the modal posted price is identical in every one of them; the mean standard deviation of the log price across provinces is 0.001, against 0.028 across chains in the same product-month (4,242 product-months). The remaining rows use the 185 products of the pricing sample sold in two or more chains (22,165 cell-months with a defined change indicator). An opportunity is a pair (product, month) in which one chain changes the product's price and a different chain sells the product with a defined change indicator in that month; a match occurs if the second chain changes the price within one month. Same province restricts the pair to cells in the same province. Chain level aggregates cells to the product-chain-month before pairing. The placebo shifts each cell's change months by a random offset of between three and 24 months (20 draws, mean reported). The last row is the share of matched cross-chain change pairs in which the two changes have the same sign.

### A.3 Robustness of the three patterns

Table A.2 tests the three patterns of Section 2.2 along four margins. First, the exit definition: the within-category exit gradient survives restricting the outcome to national exits, in which the product leaves every chain and province within six months, and further to exits after which the brand leaves the category entirely; relative to its base rate, the brand-leaves-category gradient is the steepest of the three. Second, the sample period: all three patterns hold separately in the two halves of the sample. Third, the outcome margin: the exit gradient loads on quantity rather than price, so products exit because nobody buys them, not because they are cheap. Fourth, brand structure: the exit gradient holds within multi-product brands and among each brand's highest-revenue product, so it is not an artifact of small brands carrying small products.

**Table A.2: Robustness of the size gradients.**

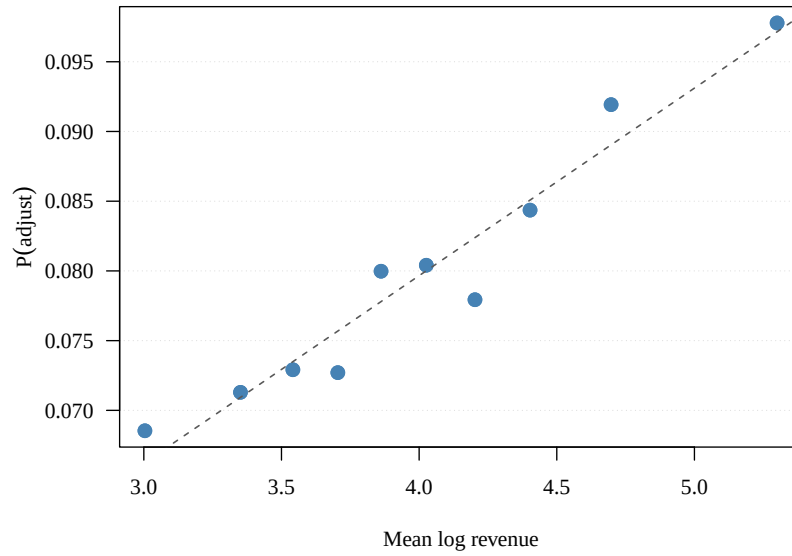
|                    | Outcome and sample                            | Pooled         | Cat. + month FE | N       |
|--------------------|-----------------------------------------------|----------------|-----------------|---------|
| Exit definition    | Cell exit within 6 months (baseline)          | −0.014 (0.005) | −0.045 (0.005)  | 250,744 |
|                    | National product exit                         | −0.005 (0.005) | −0.021 (0.004)  | 250,744 |
|                    | National exit, brand leaves category          | −0.002 (0.001) | −0.006 (0.001)  | 250,744 |
| Subperiods         | Exit, 2008–2013                               | −0.014 (0.005) | −0.036 (0.005)  | 98,207  |
|                    | Exit, 2014–2020                               | −0.018 (0.007) | −0.050 (0.008)  | 152,537 |
|                    | Frequency, 2008–2013                          | +0.017 (0.007) | −0.000 (0.002)  | 65,450  |
|                    | Frequency, 2014–2020                          | +0.011 (0.003) | −0.005 (0.002)  | 103,629 |
|                    | Mean $ \Delta \log p $ , 2008–2013            | −0.012 (0.004) | +0.004 (0.003)  | 5,225   |
|                    | Mean $ \Delta \log p $ , 2014–2020            | −0.012 (0.002) | −0.005 (0.002)  | 8,344   |
|                    |                                               |                |                 |         |
| Price vs. quantity | Exit on lagged log quantity                   | −0.027 (0.004) | −0.049 (0.004)  | 250,744 |
|                    | Exit on lagged log price                      | +0.004 (0.008) | −0.032 (0.014)  | 250,744 |
|                    | Mean $ \Delta \log p $ on lagged log quantity | −0.003 (0.002) | +0.006 (0.001)  | 13,569  |
|                    | Mean $ \Delta \log p $ on lagged log price    | −0.030 (0.005) | −0.039 (0.005)  | 13,569  |
| Brand structure    | Exit, multi-product brands                    | −0.016 (0.006) | −0.045 (0.006)  | 244,968 |
|                    | Exit, star products                           | −0.031 (0.011) | −0.038 (0.013)  | 17,838  |

Notes: Each row reports the slope from a linear regression of the stated outcome on lagged log revenue (or, in the price-quantity panel, on lagged log quantity and lagged log price jointly), on the pricing sample of Table 1, with standard errors clustered by category in parentheses. National product exit restricts the exit indicator to cell exits in which the UPC leaves the entire panel within six months; brand-leaves-category further requires that the brand has no other product active in the category afterwards. Star products are each brand’s highest-revenue UPC.

The three patterns documented in Section 2.2 use the previous month’s log revenue at the product-chain-province cell as the size proxy. As an additional robustness check, I substitute the lagged trailing twelve-month mean of log revenue. This proxy is dominated by persistent product size and is less affected by single-month sales spikes; the cost is a slightly smaller sample, since it requires at least twelve months of revenue history per cell.

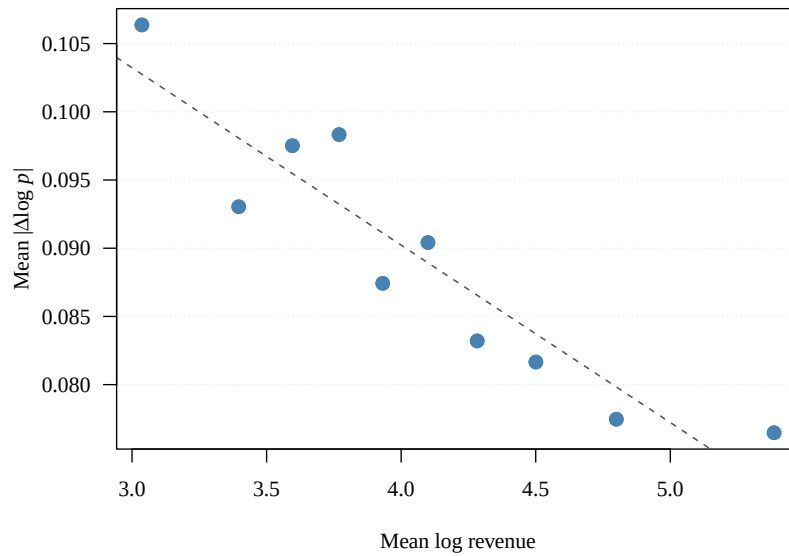
Figures A.1–A.3 reproduce the three bin-scatters with the rolling-window proxy. The qualitative pattern is unchanged: adjustment frequency rises with the size proxy, the size of adjustments falls with it, and the six-month exit probability falls with it. Slopes are within sampling error of the lagged-revenue versions in the main text.

**Figure A.1: Adjustment frequency by trailing twelve-month size decile.**



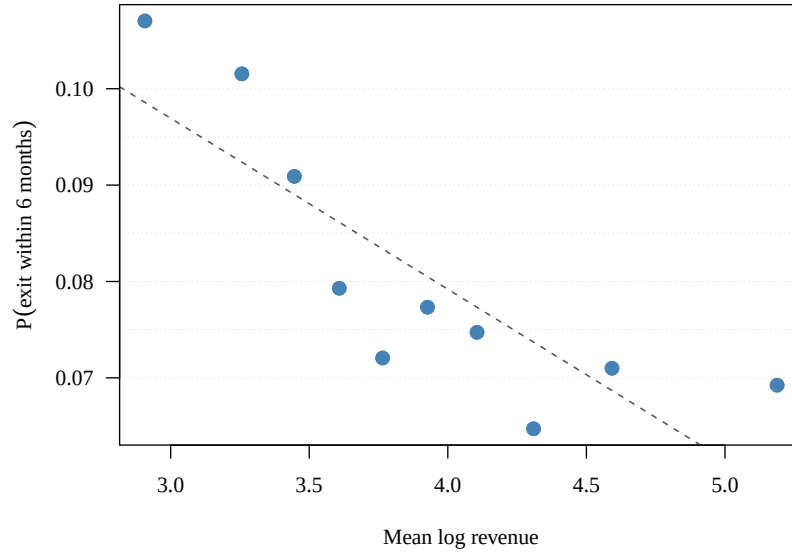
Notes: For each decile of the trailing twelve-month log-revenue mean, the figure plots the mean indicator that the cell adjusted its price in the current month. Each decile contains 14,153 product-months. The dashed line is the OLS fit through the ten bin means; no fixed effects.

**Figure A.2: Mean absolute log price change by trailing twelve-month size decile.**



Notes: For each decile of the trailing twelve-month log-revenue mean, the figure plots the mean of the absolute log price change among product-months that adjust. Each decile contains 1,129 adjustment events on average. The dashed line is the OLS fit through the ten bin means; no fixed effects.

**Figure A.3: Six-month exit probability by trailing twelve-month size decile.**



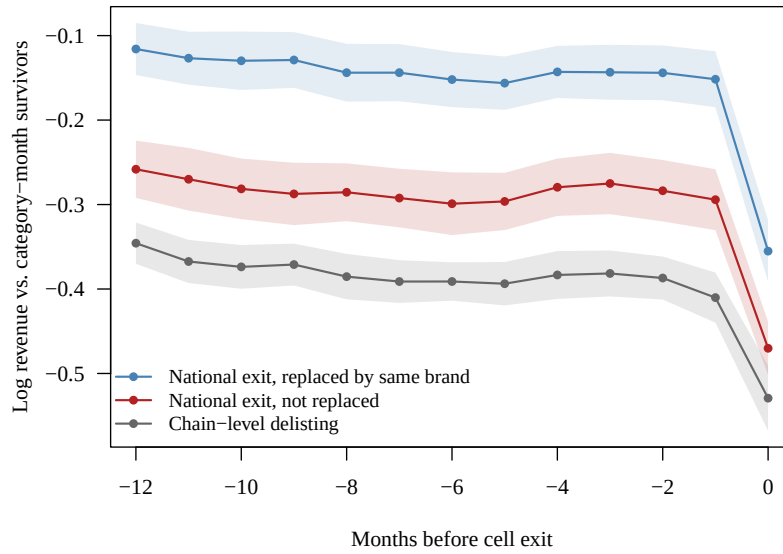
Notes: For each decile of the trailing twelve-month log-revenue mean, the figure plots the share of product-months for which the cell exits within the next six months. Each decile contains 18,789 product-months. The dashed line is the OLS fit through the ten bin means; no fixed effects.

#### A.4 Exit diagnostics

Section 2.2 claims that measured exit is the exit of a product from the shelf. The first diagnostic follows every exiting product across all chains and provinces: an exit is national when the product disappears from the entire panel within six months of the cell's end. Weighted by revenue, 0.82 of all exit is national, and 0.70 among products sold in two or more chains, for which a within-panel substitution would be visible (Table A.3). Chain-level delisting, in which the product survives elsewhere, accounts for the remainder.

The second diagnostic studies barcode relabeling: a brand that changes a product's flavor or pack size retires one barcode and issues another, and the data would record an exit and an entry where no product actually leaves the market. For each national exit I look for a same-brand entrant in the same category within a window from three months before to six months after the exit; one appears in 0.62 of cases. However, the benchmark is not zero: these are large, active brands that launch products continuously. At a random interior month of the same brand and category, the same window contains an entry in 0.50 of cases, so the excess replacement tied to the exit itself is twelve percentage points. The pack sizes of the closest same-brand entrants split about evenly between smaller, identical, and larger than the exiting product, which is not the signature of systematic relabeling or shrinkflation.

**Figure A.4: Revenue before exit, by type of exit.**



Notes: Same construction as Figure 4, with exiting products split into three groups: exits followed by a same-brand entrant in the same category within the  $[-3, +6]$ -month window, national exits without such a replacement, and chain-level delistings in which the product survives elsewhere. The shaded bands are ninety-five percent confidence intervals, standard errors clustered by category.

**Table A.3: What barcode exit measures.**

|                                                         | Share              |
|---------------------------------------------------------|--------------------|
| National product exit within 6 months, revenue-weighted | 0.82               |
| products sold in 2+ chains                              | 0.70               |
| Same-brand entrant in same category, $[-3, +6]$ months  | 0.62               |
| at a random month of the same brand-category (placebo)  | 0.50               |
| excess over placebo                                     | 0.12               |
| closest entrant: same / smaller / larger pack           | 0.34 / 0.30 / 0.34 |

Notes: Sample: valid lifecycles of the turnover sample (Table 1), interior months; 96,915 cell exits, of which 27,263 national product exits with an identified manufacturer brand. A national product exit occurs when the UPC leaves every chain and province in the panel within six months of the cell exit; shares are weighted by lifecycle revenue. The replacement rows report the share of national product exits followed by a new barcode of the same brand in the same category within three months before to six months after the exit; the placebo evaluates the same window at a random interior month for the same brand-category, averaged over ten draws. Pack size compares volume per unit of the exiting product and its closest replacement, when both share a measurement unit.

Figure A.4 splits the pre-exit revenue path of Figure 4 by type of exit: national exits followed by a same-brand replacement, national exits without one, and chain-level delistings. All three groups sell below their category-month peers for at least a year before exiting, and the replaced exits decline as clearly as the unreplaced ones. A healthy product continuing under a new barcode would show a flat path up to the switch, so the replacement pattern reads as brands retiring dying variants: an exit and an entry in the model's sense.

The last diagnostic asks whether products change their pricing behavior on the way out. Table A.4 reports the monthly adjustment frequency by distance to exit: frequency in the final three months equals the sample average, and the months just before show slightly more adjust-

ment, not less. Exit does not announce itself in the time dimension; what distinguishes exiting products is their level, small and declining revenue, which is the cross-sectional selection the model encodes.

**Table A.4: Adjustment frequency by months before exit.**

| Months before exit          | Frequency | <i>N</i> |
|-----------------------------|-----------|----------|
| 1–3                         | 0.079     | 5,825    |
| 4–6                         | 0.100     | 4,896    |
| 7–12                        | 0.088     | 10,823   |
| 13 or more                  | 0.079     | 76,701   |
| Censored (no observed exit) | 0.079     | 70,834   |

Notes: Mean monthly adjustment probability on the incumbent panel of the pricing sample, by months between the observation and the product’s exit from the sample. The censored row collects products whose exit is never observed.

### A.5 Persistence of idiosyncratic productivity

I estimate the persistence of idiosyncratic productivity from the autocovariances of prices at lags of one to three months rather than from a regression on the first lag, because a monthly price computed from transactions carries measurement error, and the error moves the variance but not the autocovariances.

In the model the desired price of a firm is a constant markup over  $W_t/z_{it}$ , so  $\log z_{it}$  enters the log relative price one for one with the opposite sign and the persistence of the relative price is the persistence of productivity. A relative price in the data also moves with the product’s own demand, which  $z$  does not represent. I therefore strip out the part of the price that moves with the product’s market share and keep the rest. Within a category and month, the log relative price is the log unit value minus the median log unit value and the share is the product’s revenue over category revenue; I regress the first on the second with product and month fixed effects and call the residual  $x_{it}$ . The residual is the cost state only up to the demand movements that leave a trace in the share, which is the assumption this step rests on.

The unit of observation is the product-month, pooling chains and provinces, since Section A.2 shows that a price change is determined mainly by the product, and not by the chain. I keep months with at least three transactions and a unit value between €0.20 and €25, and products observed for at least twelve consecutive months, splitting a product into separate spells at each interruption. This leaves 3,550,549 pairs of consecutive months.

A monthly price is an average over the transactions of the month, so  $x_{it}$  measures the state with error. Write  $x_{it} = x_{it}^* + u_{it}$ , with  $x^*$  an AR(1) of persistence  $\rho$  and  $u$  uncorrelated over time and with  $x^*$ . The error then sits in the variance alone,

$$C_0 = \text{Var}(x^*) + \text{Var}(u), \quad C_k = \rho^k \text{Var}(x^*) \quad (k \geq 1), \quad (18)$$

so any two successive autocovariances recover the persistence and the variance of the state,

whatever the size of the error:

$$\rho = \frac{C_2}{C_1} = \frac{C_3}{C_2}, \quad \text{Var}(x^*) = \frac{C_1^2}{C_2}. \quad (19)$$

I compute the  $C_k$  after demeaning by product and by month, and treat a lag as missing whenever the two months are not exactly  $k$  apart on the calendar, so a hole in a product's panel never enters as an adjacent pair.

The two ratios in (19) give 0.830 and 0.833, and I set  $\rho = 0.83$ . Regressing on the first lag would have given 0.512: the variance of the state is 0.008 against a measured variance of 0.014, so 38 percent of the observed movement in relative prices is error, and the regression reads that error as mean reversion.

## B Computational details

This appendix collects the two solution algorithms used in the paper: the Krusell–Smith stationary solver (Section B.1) and the perfect-foresight transition path used to compute the impulse responses (Section B.2).

### B.1 Krusell–Smith solution algorithm

The forecasting rule for inflation is

$$G(\mathcal{S}) = \bar{\pi} + b_1 (\mathcal{S} - s^*),$$

parametrized by the aggregate-state fixed point  $s^*$  (such that  $G(s^*) = \bar{\pi}$ ) and the slope  $b_1$ . The algorithm operates on a grid  $\{\mathcal{S}_k\}_{k=1}^K$  in the aggregate-state dimension and iterates jointly on  $(s^*, b_1)$  until convergence:

1. **Guess.** Start from  $(s_{\text{old}}^*, b_{1,\text{old}})$ , defining  $G_{\text{old}}$  pointwise on the grid.
2. **Policies.** Iterate the firm Bellman (4) given  $G_{\text{old}}$  and the implied wage and discount factor. The exit hazard  $\lambda$  is recomputed from the contemporaneous  $(V, \mu)$  via (5) after each pass, ensuring that  $V$ ,  $\mu$ , and  $\lambda$  are mutually consistent.
3. **Distribution.** Given the price policy  $p(s)$ , the hazard  $\lambda(s)$ , and the entry block, compute the stationary measure  $\mu$  by iterating the law of motion (7) to convergence at  $G_{\text{old}}$ .
4. **Pointwise CES residual.** Forward-push  $\mu$  through pricing, the productivity transition, and the aggregate-state transition. For each grid point  $\mathcal{S}_k$ , the implied one-step-ahead inflation residual is

$$G_{\text{res}}(\mathcal{S}_k) = \log \left[ \sum_s e^{(1-\theta)p(s)} \mu(s \mid \mathcal{S}_k) \right]^{1/(1-\theta)}.$$

5. **Project.** Set  $G_{\text{target}}(\mathcal{S}_k) = G_{\text{old}}(\mathcal{S}_k) + G_{\text{res}}(\mathcal{S}_k)$  and fit

$$G_{\text{target}} \approx b_0 + b_1 (\mathcal{S} - \bar{\mathcal{S}})$$

by weighted least squares with weights  $\propto \mu(\mathcal{S}_k)$ . Recover the new fixed point as  $s_{\text{new}}^* = \bar{\mathcal{S}} + (\bar{\pi} - b_0)/b_1$ .

6. **Update.** Replace  $(s_{\text{old}}^*, b_{1,\text{old}}) \leftarrow (s_{\text{new}}^*, b_{1,\text{new}})$  with damping. Return to step 2 until  $\|G_{\text{new}} - G_{\text{old}}\|_{\infty} < \epsilon$ .

Two remarks. First, the projection in step 5 is the population counterpart of the [Krusell and Smith \(1998\)](#) simulation regression: the stationary distribution in step 3 delivers the cross-section at every aggregate-state grid point, so the weighted fit computes exactly what a regression on a simulated path estimates, with the ergodic mass as weights and no sampling error ([Nakamura and Steinsson, 2010](#)). Second, although the exit hazard (5) contains the cross-sectional functional  $\mathbb{E}_{\mu}[\exp(-\zeta V)]$ , this is not a separate aggregate state to be forecast: once  $V$  is parametrized by  $\mathcal{S}$  via  $G$ , the integral is determined contemporaneously by integration (i.e. the same approximation that summarizes  $\mu$  by  $\mathcal{S}$  in the Krusell–Smith logic). The accuracy of the forecast rule is reported in Section C.

## B.2 Perfect-foresight transition path

The IRFs in Section 4.2 are computed along a deterministic transition of length  $T$  following a one-off nominal innovation  $\Delta$  at  $t = 0$ , similar to [Clementi and Palazzo \(2016\)](#). Aggregate accounting under perfect foresight implies  $Y_{t+1} = Y_t + \bar{\pi} - \pi_t$ , deterministic given the inflation path  $\{\pi_t\}_{t=0}^{T-1}$ . The firm Bellman is solved on a time-varying  $Y_t$ :

$$\begin{aligned} V_t(s) &= \max\{V_t^{\text{stay}}(s), V_t^{\text{adj}}(s)\}, \\ V_t^{\text{stay}}(s) &= \pi(s; Y_t) + \beta (1 - \lambda_t(s)) \rho_t \mathbb{E}_z[V_{t+1}(s')], \\ V_t^{\text{adj}}(s) &= -K(Y_t) + \max_p \left[ \pi(p, z; Y_t) + \beta (1 - \lambda_t(p, z, \cdot)) \rho_t \mathbb{E}_{z'}[V_{t+1}(p, z', \cdot)] \right], \end{aligned}$$

where  $\rho_t = \exp(\gamma(\pi_t - \bar{\pi}))$  is the perfect-foresight stochastic discount factor,  $K(Y_t)$  is the menu cost expressed at the date- $t$  wage, and the terminal condition  $V_T = V^{\text{ss}}$  is the deterministic stationary value. The exit hazard is recomputed each period from the contemporaneous  $(V_t, \mu_t)$  via (5).

The cross-section evolves under a deterministic operator

$$\mu_{t+1} = \mathcal{T}_t[\mu_t; \lambda_t, M_t, p_t, \pi_t],$$

which sequentially applies (i) exit at  $\lambda_t$ ; (ii) injection of entrants of mass  $M_t = \int \lambda_t d\mu_t$  from  $v_e$  at the prevailing  $Y_t$ ; (iii) the date- $t$  price policy  $p_t$ ; (iv) the productivity transition  $F(z' | z)$ ; and (v) deflation of relative prices by  $\pi_t$ . The CES condition pins down  $\pi_t$  at every  $t$ :

$$1 = \mathbb{E}_{\mu_t^{\text{post-}p,F}}[\exp((1 - \theta) p_{t+1}(z', \tilde{p} - \pi_t, S'))],$$

with initial conditions  $\mu_0 = \mu^{ss}$  and  $Y_0 = Y^{ss} + \Delta$ .

The unknown is the inflation path  $\{\pi_t\}_{t=0}^{T-1}$ , with  $T$  chosen large enough that the IRF returns to zero. The algorithm iterates as follows:

1. **Guess** a path  $\{\pi_t\}_{t=0}^{T-1}$  and compute the implied  $\{Y_t\}_{t=0}^T$  from  $Y_{t+1} = Y_t + \bar{\pi} - \pi_t$ .
2. **Backward pass.** Set  $V_T = V^{ss}$ . For  $t = T - 1, \dots, 0$ , solve the Bellman to obtain  $(V_t, p_t)$  at  $(Y_t, \pi_t, \lambda_t)$ .
3. **Forward pass.** Set  $\mu_0 = \mu^{ss}$ . For  $t = 0, \dots, T - 1$ , evolve  $\mu_{t+1} = \mathcal{T}_t[\mu_t]$ .
4. **Update hazards.** Recompute  $\lambda_t$  from  $(V_t, \mu_t)$  at every  $t$ .
5. **CES update.** At each  $t$ , compute the new inflation  $\pi_t^{\text{new}}$  from the CES condition above.
6. **Damped update.**  $\pi_t \leftarrow (1 - \omega)\pi_t + \omega \pi_t^{\text{new}}$ . Return to step 1 until  $\max_t |\pi_t^{\text{new}} - \pi_t| < \text{tol}$ .

The terminal and initial conditions  $V_T = V^{ss}$  and  $\mu_0 = \mu^{ss}$  must be the stationary equilibrium of the economy without aggregate noise: at  $\sigma_\eta = 0$  the aggregate state rests at  $s^*$ , inflation equals  $\bar{\pi}$  in every period, and the equilibrium is a fixed point of the deterministic operator  $\mathcal{T}$ , so that the no-shock path reproduces stationarity exactly. This object differs from the Krusell–Smith equilibrium evaluated at  $s^*$  at the order of the aggregate noise: the stochastic value function prices future aggregate shocks, and the conditional cross-section at  $s^*$  averages over the stochastic histories passing through that point. Feeding the Krusell–Smith slice into  $\mathcal{T}$  accordingly leaves a one-period CES residual of order  $10^{-4}$ . I therefore solve the deterministic stationary equilibrium, using the slice as a warm start: an inner solver iterates  $\mathcal{T}$  at  $\pi = \bar{\pi}$  until  $(V, \mu, \lambda)$  are mutually consistent, and bisection on  $Y^{ss}$  until the stationary CES residual is zero.

The reported impulse response is the deviation of output from the contemporaneous no-shock transition path,  $\psi_t = (\log Y_t - \log Y_t^0)/\Delta$ , where  $Y_t^0$  is obtained by running the same operator  $\mathcal{T}$  with  $\Delta = 0$ . The cumulative response is  $\text{CIR}_t = \frac{1}{12} \sum_{s \leq t} \psi_s$  in annualized units, reported at the one-year horizon in Table 5.

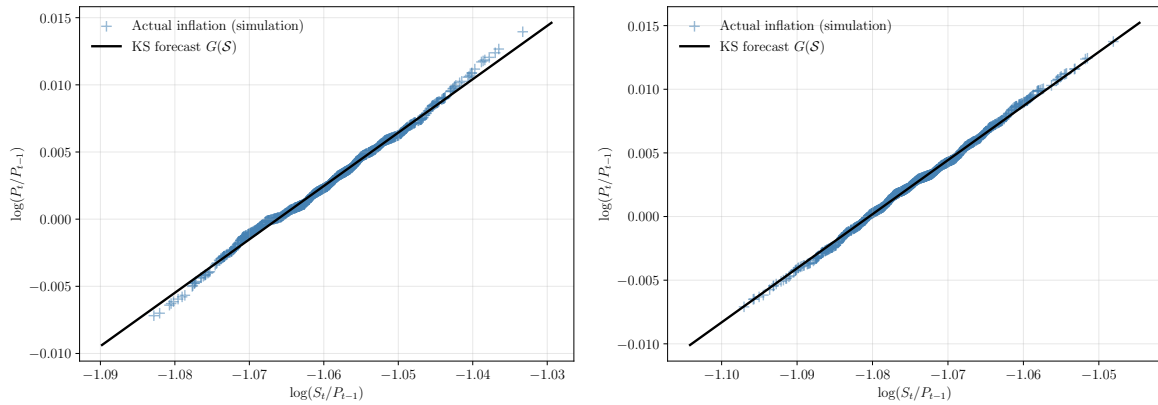
## C Accuracy and robustness of the forecasting rule

The stationary equilibrium relies on a parametric forecasting rule  $G(\mathcal{S})$  for inflation as a function of the aggregate state  $\mathcal{S} = S/P_{-1}$ . In the main text I use a linear specification,  $G(\mathcal{S}) = \bar{\pi} + b_1(\mathcal{S} - s^*)$ , which is the standard [Krusell and Smith \(1998\)](#) choice and is fitted analytically by weighted least squares on the aggregate-state grid ([Nakamura and Steinsson, 2010](#)). This appendix assesses the accuracy of that specification and shows that the headline results are robust to a more flexible cubic alternative.

### C.1 Diagnostics under the linear rule

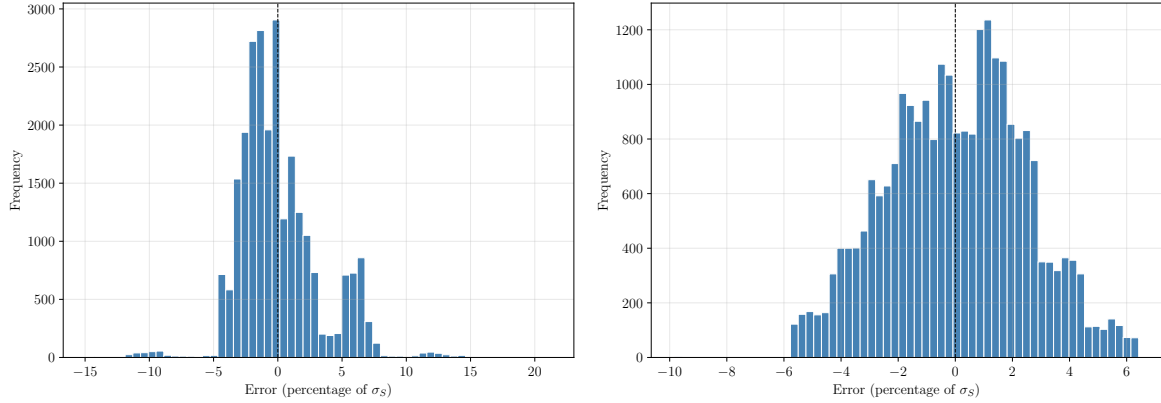
I evaluate the rule by simulating a long path of the aggregate state  $\mathcal{S}_t$  under the converged equilibrium and recording realized inflation  $\pi_t$  at each date. I then plot  $(\mathcal{S}_t, \pi_t)$  pairs against the linear rule  $G$  (Figure A.5) and the distribution of forecast errors  $\pi_t - G(\mathcal{S}_t)$ , expressed as a percentage of  $\sigma_S$ , the standard deviation of the nominal-demand innovation (Figure A.6).

**Figure A.5: Forecasting rule: perceived vs. realized inflation, linear rule.**



*Notes:* For each economy, the figure plots realized log inflation along the simulated path against the aggregate state  $\log(S_t/P_{t-1})$ , with the linear forecasting rule  $G$  superimposed in black. The Benchmark (left) displays a visible bow-shape: at low values of the aggregate state realized inflation lies below the rule, and at high values it lies above. The No-selection economy (right) shows essentially no such pattern.

**Figure A.6: Distribution of one-step forecast errors, linear rule.**



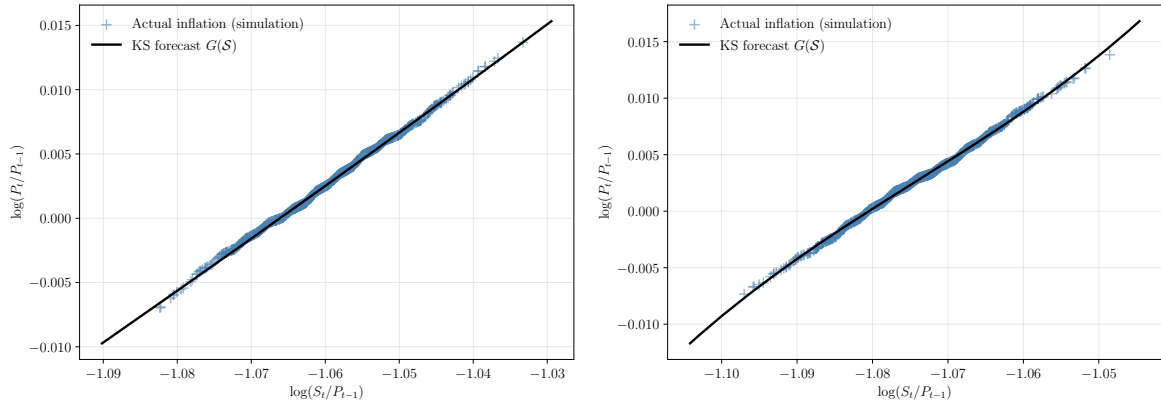
*Notes:* For each economy, the histogram reports the distribution of  $\pi_t - G(\mathcal{S}_t)$  as a percentage of  $\sigma_S$ . The Benchmark exhibits a few isolated tail masses around  $\pm 10$ – $15\%$  of  $\sigma_S$ , corresponding to rare visits of the simulated aggregate state to its support boundaries; under No-selection, errors are smaller and more uniformly distributed.

Under No-selection, model-implied inflation is essentially linear in  $\mathcal{S}$  and the linear rule fits well throughout the support. Under the Benchmark, selective exit reshapes the cross-section in a state-dependent way, and the model-implied inflation is mildly nonlinear in  $\mathcal{S}$ . The mass-weighted least-squares fit absorbs that nonlinearity well in the center of the simulated support, where most of the stationary mass lies, but allows residuals to grow at the boundaries, producing the S-shape in the scatter and the tail humps in the histogram. Therefore, I fit a cubic forecasting rule next.

## C.2 Cubic forecasting rule

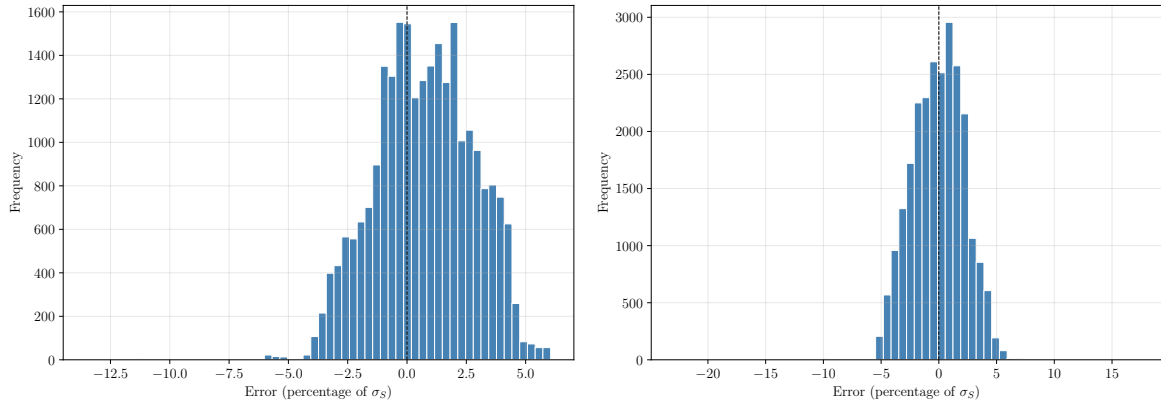
To assess whether these residuals matter quantitatively, I re-solve all three economies under a cubic specification,  $G(\mathcal{S}) = \sum_{k=0}^3 b_k \mathcal{S}^k$ , fitted by the same weighted least-squares procedure on the aggregate-state grid. Under the cubic rule the diagnostics shown in Figures A.7 and A.8 look cleaner: the tail humps are absorbed into a smooth, roughly bell-shaped error distribution, with the standard deviation of the forecast error dropping by approximately one-third in the Benchmark.

**Figure A.7: Forecasting rule: perceived vs. realized inflation, cubic rule.**



Notes: Same construction as Figure A.5, with the rule replaced by a cubic polynomial. The bow-shape that was present in the Benchmark scatter under the linear rule is no longer visible.

**Figure A.8: Distribution of one-step forecast errors, cubic rule.**



Notes: Same construction as Figure A.6. The Benchmark histogram is now smooth and unimodal; the isolated tail masses observed under the linear rule have disappeared.

I now report the stationary moments and the impulse-response analysis under the cubic rule. Table A.5 compares them to the linear-rule values from Table 4.

**Table A.5: Stationary moments under linear vs. cubic forecasting rule.**

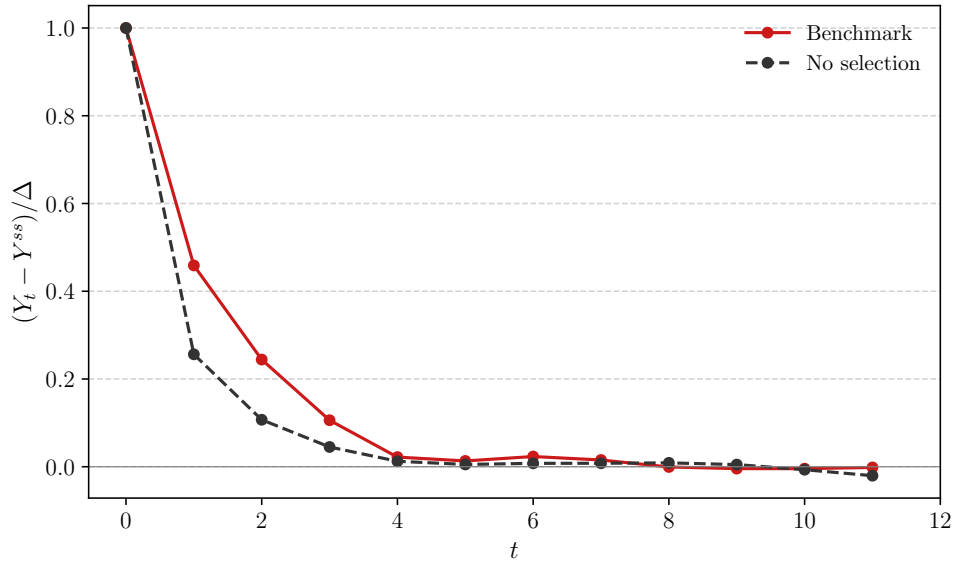
| Model        | Linear rule |          |                             | Cubic rule |          |                             |
|--------------|-------------|----------|-----------------------------|------------|----------|-----------------------------|
|              | Frequency   | Kurtosis | $\text{Var}(Y) \times 10^5$ | Frequency  | Kurtosis | $\text{Var}(Y) \times 10^5$ |
| Benchmark    | 0.086       | 1.41     | 2.362                       | 0.086      | 1.39     | 2.190                       |
| No selection | 0.088       | 1.32     | 2.051                       | 0.088      | 1.32     | 1.988                       |
| No churn     | 0.092       | 1.31     | 1.967                       | 0.092      | 1.31     | 1.988                       |

Notes: Frequency is the monthly frequency of price adjustment. Kurtosis refers to the conditional distribution of  $\Delta \log p$  given a price change.  $\text{Var}(Y) \times 10^5$  is the unconditional variance of output around its trend, computed as in the main text.

The frequency and kurtosis of price changes are unchanged across rules. The unconditional variance of output is somewhat smaller under the cubic rule, with the largest movement

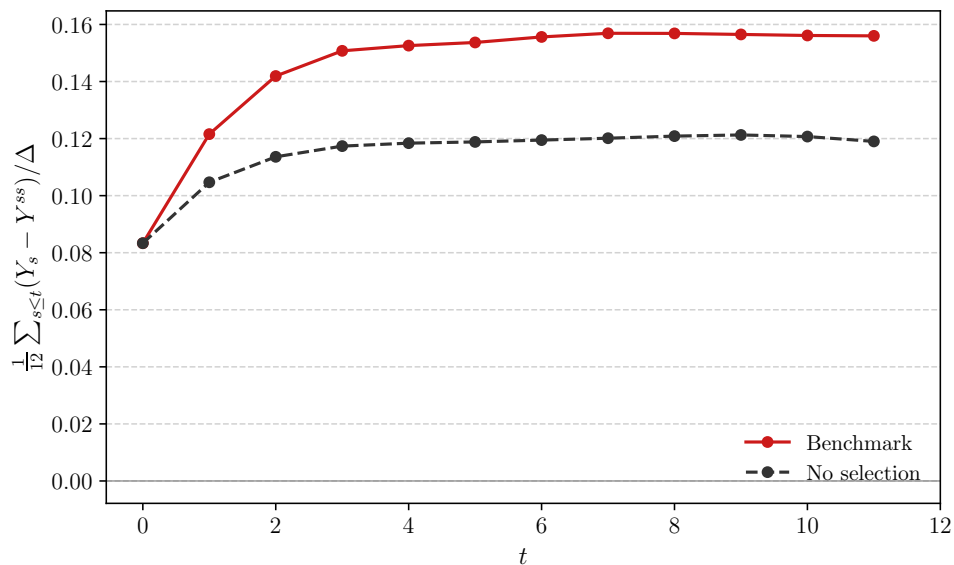
in the Benchmark; the Benchmark/No-selection variance gap shrinks from +15.4% to +10.3%. Figures A.9 and A.10 report the perfect-foresight impulse and cumulative responses for the Benchmark and No-selection economies under the cubic rule. Their shape is essentially indistinguishable from the linear-rule responses underlying Table 5: the one-year cumulative responses differ by less than one percent of their values, and the relative gap between the two economies is preserved.

**Figure A.9: Impulse response of output, cubic rule.**



Notes: Output impulse response  $\psi_t = (\log Y_t - \log Y^{ss})/\Delta$  along the perfect-foresight transition path, computed under the cubic forecasting rule, truncated at twelve months.

**Figure A.10: Cumulative impulse response of output, cubic rule.**



Notes: Cumulative impulse response of log output,  $CIR_t = \frac{1}{12} \sum_{s \leq t} \psi_s$ , in annualized units, along the perfect-foresight transition path under the cubic forecasting rule, truncated at twelve months.

The exercise confirms that the linear-rule approximation is adequate for the headline results: pricing moments are unchanged, the unconditional output variance shifts modestly with the qualitative ranking preserved, and the one-year cumulative impulse responses are essentially identical across rules.

## D Moments-based decompositions and sensitivity to mean reversion

This appendix applies an additive policy-distribution decomposition to stationary pricing moments in both models, and uses it to examine the sensitivity of the Calvo results to the speed of mean reversion. The main text prices the same two channels directly in units of monetary non-neutrality (Sections 4.4 and 7); the moments-based decomposition is retained here because it shows how the two channels load on the micro-price moments, and because the mean-reversion exercise is built on it.

### D.1 Menu-cost model

Section 4.3 described two channels through which selective exit alters pricing: one on the firms' policy function, the option-value response of the firm's policy, and one on the composition of the cross-sectional distribution, the censoring mechanism. Because both channels act on overlapping firms, I decompose the individual effects into the two contributions. I use a decomposition that expresses the change as the sum of a counterfactual policy, a counterfactual distribution, and an interaction term.

Let  $M(p, \mu)$  denote any moment of the conditional distribution of  $\Delta \log p$ , where  $p$  is the price-adjustment policy and  $\mu$  is the stationary measure. For any pair of stationary equilibria  $(p_0, \mu_0)$  and  $(p_1, \mu_1)$ , the identity

$$M(p_1, \mu_1) - M(p_0, \mu_0) = \underbrace{[M(p_1, \mu_0) - M(p_0, \mu_0)]}_{\Delta_{\text{policy}}} + \underbrace{[M(p_0, \mu_1) - M(p_0, \mu_0)]}_{\Delta_{\text{dist}}} + \Delta_{\text{int}} \quad (20)$$

holds exactly.<sup>32</sup> The policy channel  $\Delta_{\text{policy}}$  holds the cross-section at  $\mu_0$  and switches the policy from  $p_0$  to  $p_1$ : it isolates the option-value response of pricing behavior. The distribution channel  $\Delta_{\text{dist}}$  holds the policy at  $p_0$  and switches the cross-section to  $\mu_1$ : it isolates the censoring response of the cross-section. The interaction term collects the part of the change shared by the two channels.

I apply (20) in two steps. First, I use it to split the headline difference between the No-selection and Benchmark economies of Table 4 into the two channels. Second, holding the average exit rate fixed at  $\delta = 0.02$ , I apply it to a pair that differs only in whether the selectivity is routed through entry or through exit (entry-only versus exit-only). Table A.6 reports both for the conditional mean of  $\Delta \log p$ , its kurtosis, and the frequency of price changes.

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<sup>32</sup>The construction parallels the Foster-Haltiwanger-Krizan within/between/covariance accounting for aggregate productivity. Take frequency,  $M(p, \mu) = \mathbb{E}_\mu[p_{\text{adj}}(s)]$ , as an example. The full change can be decomposed in three steps: (i) switch the policy from  $p_0$  to  $p_1$  while holding the cross-section at  $\mu_0$ , giving  $\Delta_{\text{policy}} = \mathbb{E}_{\mu_0}[p_{1,\text{adj}} - p_{0,\text{adj}}]$ , a within-cell change; (ii) starting from  $(p_0, \mu_0)$ , switch the cross-section from  $\mu_0$  to  $\mu_1$  while holding the policy at  $p_0$ , giving  $\Delta_{\text{dist}} = \mathbb{E}_{\mu_1 - \mu_0}[p_{0,\text{adj}}]$ , a between-cell shift; (iii) the residual  $\Delta_{\text{int}} = \mathbb{E}_{\mu_1 - \mu_0}[p_{1,\text{adj}} - p_{0,\text{adj}}]$  is the covariance between the policy change and the mass redistribution, and would be zero if the two channels acted on disjoint cells. For higher moments the same three-step construction applies; the residual is the non-linear analog of the covariance.

**Table A.6: Decomposition of selection effects, menu-cost model.**

| Moment                                                                                                          | $M_0$  | $M_1$  | $M_1 - M_0$ | $\Delta_{\text{policy}}$ | $\Delta_{\text{dist}}$ | $\Delta_{\text{int}}$ |
|-----------------------------------------------------------------------------------------------------------------|--------|--------|-------------|--------------------------|------------------------|-----------------------|
| <i>Pair 1: No-selection (<math>\zeta = 0, \tau = 0</math>) vs Benchmark (<math>\zeta = 20, \tau = 5</math>)</i> |        |        |             |                          |                        |                       |
| mean $\Delta \log p$                                                                                            | +0.032 | +0.042 | +0.010      | -0.008                   | +0.015                 | +0.003                |
| kurtosis $\Delta \log p$                                                                                        | 1.318  | 1.415  | +0.097      | -0.022                   | +0.124                 | -0.006                |
| frequency (pp)                                                                                                  | 8.785  | 8.581  | -0.204      | +0.050                   | -0.193                 | -0.061                |
| <i>Pair 2: entry-only (<math>\zeta = 0, \tau = 5</math>) vs exit-only (<math>\zeta = 20, \tau = 0</math>)</i>   |        |        |             |                          |                        |                       |
| mean $\Delta \log p$                                                                                            | +0.012 | +0.052 | +0.040      | -0.030                   | +0.047                 | +0.023                |
| kurtosis $\Delta \log p$                                                                                        | 1.203  | 1.508  | +0.305      | +0.016                   | +0.366                 | -0.076                |
| frequency (pp)                                                                                                  | 8.746  | 8.496  | -0.250      | -0.963                   | -0.140                 | +0.853                |

Notes: The table reports the additive decomposition (20) applied to two pairs of stationary equilibria. Pair 1 compares No-selection ( $\zeta = 0, \tau = 0$ ) to the Benchmark ( $\zeta = 20, \tau = 5$ ) and quantifies the headline difference of Table 4. Pair 2 holds the total exit rate at  $\delta = 0.02$  and changes only the source of selectivity, from entry-only ( $\zeta = 0, \tau = 5$ ) to exit-only ( $\zeta = 20, \tau = 0$ ). Conditional moments are computed under weights  $p \cdot Q$ , where  $Q$  is the stationary mass. Frequency is reported in percentage points; all entries to three decimals.

The difference between No-selection and Benchmark (Pair 1) is split almost entirely by the censoring channel. The frequency drop of  $-0.204$  pp is accounted for by  $\Delta_{\text{dist}} = -0.193$ , with a negligible policy contribution of  $+0.050$  and an interaction of  $-0.061$ . The kurtosis rise of  $+0.097$  is similarly carried by  $\Delta_{\text{dist}} = +0.124$  against a partly offsetting policy term of  $-0.022$ . The mean of  $\Delta \log p$  shifts by only  $+0.010$ , again predominantly through  $\Delta_{\text{dist}} = +0.015$ . Selection therefore raises the kurtosis of price changes, the moment that maps into monetary non-neutrality through standard sufficient-statistic formulae, almost entirely by removing mass from the right tail of the No-selection price-change distribution, while leaving pricing behavior at any given state nearly unchanged.

When comparing selection on exit versus selection on entry (Pair 2), the option-value channel is instead the predominant one. Frequency now responds primarily through the policy channel:  $\Delta_{\text{policy}} = -0.963$  pp accounts for most of the drop, alongside a  $\Delta_{\text{dist}} = -0.140$  pp distribution contribution and a large positive interaction of  $+0.853$  pp. The interaction is large precisely because the two channels overlap heavily: the firms whose inaction band widens under selective exit are also the firms whose mass is moved by selective exit. Kurtosis, by contrast, continues to be driven by the distribution channel:  $\Delta_{\text{dist}} = +0.366$  against a small policy term of  $+0.016$ . Routing the selectivity through exit rather than entry concentrates more mass at low productivity, where the option-value widening of inaction is largest; the policy channel becomes informative precisely because the reference cross-section places enough mass at low productivity to amplify it.

Reading the two pairs together delivers a clean picture. The censoring channel is robust across pairs: it lowers frequency, raises kurtosis, and accounts for essentially all of the headline difference between No-selection and Benchmark. The option-value channel is conditional on the reference cross-section: it can move frequency by nearly a full percentage point when low-productivity firms are well represented, but the contributions at low and high productivity offset each other when the cross-section is balanced across the productivity distribution, as in No-selection. For one of the two moments driving monetary non-neutrality, the kurtosis of

$\Delta \log p$ , the censoring channel is the load-bearing force in both pairs.

## D.2 Calvo model

The additive decomposition of Section D.1 applies also to the Calvo economy. As in the menu-cost case, the underlying nominal friction is held fixed across the comparison (here, the Calvo lottery rate  $\alpha$  is set to the same value in every cell, just as the menu cost was fixed across Benchmark and No-selection in Section D.1) so that any difference in the conditional distribution of  $\Delta \log p$  reflects the response of the equilibrium to the exit margin. Because  $\alpha$  is exogenous and identical, the adjustment frequency is the same in every Calvo equilibrium and the frequency channel is mechanically zero; the decomposition therefore speaks only to moments of the conditional distribution of  $\Delta \log p$ . Table A.7 reports the mean, skewness, and kurtosis decompositions for the same two pairs of equilibria used in Table A.6: No-selection versus Benchmark (Pair 1), and entry-only versus exit-only (Pair 2).

**Table A.7: Decomposition of selection effects, Calvo model.**

| Moment                                                                                                      | $M_0$  | $M_1$  | $M_1 - M_0$ | $\Delta_{\text{policy}}$ | $\Delta_{\text{dist}}$ | $\Delta_{\text{int}}$ |
|-------------------------------------------------------------------------------------------------------------|--------|--------|-------------|--------------------------|------------------------|-----------------------|
| <i>Pair 1: No-selection (<math>\xi = 0, \tau = 0</math>) vs Benchmark (<math>\xi = 20, \tau = 5</math>)</i> |        |        |             |                          |                        |                       |
| mean $\Delta \log p$                                                                                        | +0.026 | +0.026 | −0.001      | +0.059                   | −0.054                 | −0.006                |
| skewness $\Delta \log p$                                                                                    | +0.377 | +0.205 | −0.173      | +0.450                   | −0.580                 | −0.044                |
| kurtosis $\Delta \log p$                                                                                    | 3.667  | 4.795  | +1.128      | +0.612                   | +0.384                 | +0.132                |
| <i>Pair 2: entry-only (<math>\xi = 0, \tau = 5</math>) vs exit-only (<math>\xi = 20, \tau = 0</math>)</i>   |        |        |             |                          |                        |                       |
| mean $\Delta \log p$                                                                                        | +0.018 | +0.035 | +0.016      | +0.026                   | −0.003                 | −0.007                |
| skewness $\Delta \log p$                                                                                    | +0.342 | +0.701 | +0.359      | +0.354                   | +0.161                 | −0.156                |
| kurtosis $\Delta \log p$                                                                                    | 3.692  | 4.626  | +0.934      | +0.825                   | +0.092                 | +0.018                |

*Notes:* The decomposition follows the construction of Table A.6. In Pair 1,  $(p_0, \mu_0)$  is the Calvo no-selection economy and  $(p_1, \mu_1)$  is the Calvo benchmark; in Pair 2,  $(p_0, \mu_0)$  is the entry-only Calvo economy and  $(p_1, \mu_1)$  is the exit-only Calvo economy. The Calvo lottery rate is  $\alpha = 0.088$  in all cells, so the adjustment frequency is identical across pairs and the frequency channel is identically zero.

In Pair 1, kurtosis rises by 1.128. The policy channel contributes +0.612, the distribution channel +0.384, and the interaction +0.132: a large increase to which both channels contribute, with the policy channel the larger of the two. In Pair 2, where the comparison flips the source of selectivity from entry to exit, the policy channel becomes overwhelming: +0.825 out of a total of +0.934, with the distribution channel small and the interaction negligible.

The mean and skewness rows tell the same story in two moments. Selection mainly affects low-productivity firms, and these firms make large upward resets when they adjust because their inherited price is too low for their current marginal cost; large upward resets populate the right tail of  $\Delta \log p$ . The policy channel makes those upward jumps even larger by raising the price these firms target (Figure 24), pushing the right tail of  $\Delta \log p$  outward; the distribution channel removes a fraction of these firms from the cross-section, thinning the same right tail. The two effects act on the same low-value firms in opposite directions and at similar magnitudes, and on the mean and on the skewness they almost cancel, leaving the small net moves of −0.001 and −0.173 in Pair 1.

The two channels reinforce on the kurtosis through different shape effects. The policy channel turns the residual low- $z$  firms into outliers: a few firms now adjust by an even larger amount than they would have under no selection, fattening the extreme right tail. The distribution channel thins the moderate- $\Delta \log p$  shoulder of the distribution by removing the low-value firms that produce intermediate-sized upward adjustments. The combined effect is fewer observations in the shoulder, with the remaining low- $z$  adjustments sitting further out: a distribution with a more peaked center and a relatively heavier extreme right tail, which is precisely a distribution with higher kurtosis. In Pair 2 the two channels stop offsetting each other on skewness and push it up jointly, mirroring the larger policy contribution to kurtosis.

There is a clear contrast with the menu-cost case. In the menu-cost model, the policy response of selective exit operates on the firm's choice of when to adjust (the inaction band) and shows up also in the frequency response (Section D.1, Pair 2). The conditional moments of  $\Delta \log p$  inherit only a small policy contribution, and the kurtosis increase is carried by the cross-sectional composition of the surviving adjusters. In the Calvo model, the firm cannot choose when to adjust because the lottery is exogenous; the policy response of selective exit therefore loads on the reset-price function of Figure 24, which directly enters the conditional distribution of  $\Delta \log p$ . Both channels are at work in both models, but the moment they show up on is determined by the firm's available margin of response. The same selective-exit mechanism therefore generates a positive kurtosis effect in both economies but routes it through different firm-level adjustments.

### D.3 Sensitivity to the speed of mean reversion

The Benchmark reset price coinciding with the static optimum at low productivity, as in Figure 24, relies on a comparison between how quickly productivity reverts and how often firms re-price. At the values  $\rho = 0.85$  and  $\alpha = 0.088$  productivity reverts faster than firms re-price: a firm observed at low productivity has typically arrived there from a recent draw rather than from a long spell at low  $z$ , so the gap between its inherited price and its current desired price is large. Selection removes mass whose pending Calvo adjustment would have been a large upward correction, and the low-productivity segment of  $p^*(a)$  jumps up to the static optimum because high-hazard firms expect to die before mean reversion brings their productivity back up.

If productivity were more persistent, Calvo adjustments would have had time to track productivity downward before a low- $z$  firm exits, the gap between inherited and desired price would shrink, and the effect of selection would weaken. I check this by redoing the comparison at  $\rho = 0.95$ , rescaling the conditional volatility  $\sigma_\epsilon$  so that the unconditional standard deviation  $\sigma_\epsilon / \sqrt{1 - \rho^2}$  of  $\log z$  stays at the calibrated value (isolating the role of persistence from changes in the dispersion of  $\log z$ ). Table A.8 reports the result. The kurtosis ranking reverses: kurtosis is now slightly lower under selection than under no selection (4.27 versus 4.46), and the skewness gap shrinks as well. At  $\rho = 0.95$  the No-selection reset price already lies close to the static optimum on its own: mean reversion is slow enough that the Calvo firm has little room to price below the static optimum, so the Benchmark low-productivity segment of  $p^*(a)$

no longer stands out from No-selection, the cross-section of reset prices is no longer more dispersed than under no selection, and what remains is the censoring effect of removing low- $V$  adjusters, which in this region of the parameter space slightly trims the right tail and lowers kurtosis on net.

**Table A.8: Calvo model: pricing moments by persistence and selection.**

| Parameterization                          | $\zeta$ | $\tau$ | Frequency | $\sigma(\Delta \log p)$ | skew( $\Delta \log p$ ) | Kurtosis |
|-------------------------------------------|---------|--------|-----------|-------------------------|-------------------------|----------|
| $\rho = 0.85, \sigma_\varepsilon = 0.086$ | 0       | 0      | 0.09      | 0.08                    | +0.38                   | 3.67     |
|                                           | 20      | 5      | 0.09      | 0.08                    | +0.20                   | 4.79     |
| $\rho = 0.95, \sigma_\varepsilon = 0.051$ | 0       | 0      | 0.09      | 0.09                    | +0.52                   | 4.46     |
|                                           | 20      | 5      | 0.09      | 0.08                    | +0.43                   | 4.28     |

*Notes:* Moments are computed on the stationary distribution of  $\Delta \log p$  conditional on a price change. The first block uses the baseline calibration; the second block raises persistence to  $\rho = 0.95$  and rescales the conditional shock standard deviation so that the unconditional standard deviation of  $\log z$  remains  $\sigma_\varepsilon / \sqrt{1 - \rho^2} = 0.163$ .

By contrast, the menu-cost economy is robust to this change. Table A.9 repeats the higher-persistence exercise for the calibrated menu-cost model: the baseline rows reproduce the moments of Table 4, and the  $\rho = 0.95$  rows apply the same rescaling of  $\sigma_\varepsilon$  used in the Calvo case. The kurtosis gap between Benchmark and No-selection remains positive at  $\rho = 0.95$  and is slightly larger than at the baseline (+0.17 versus +0.10), and the skewness gap widens.

**Table A.9: Menu-cost model: pricing moments by persistence and selection.**

| Parameterization                          | $\zeta$ | $\tau$ | Frequency | $\sigma(\Delta \log p)$ | skew( $\Delta \log p$ ) | Kurtosis |
|-------------------------------------------|---------|--------|-----------|-------------------------|-------------------------|----------|
| $\rho = 0.85, \sigma_\varepsilon = 0.086$ | 0       | 0      | 0.09      | 0.21                    | -0.41                   | 1.32     |
|                                           | 20      | 5      | 0.09      | 0.21                    | -0.45                   | 1.41     |
| $\rho = 0.95, \sigma_\varepsilon = 0.051$ | 0       | 0      | 0.06      | 0.17                    | -0.53                   | 1.39     |
|                                           | 20      | 5      | 0.06      | 0.16                    | -0.64                   | 1.56     |

*Notes:* Moments are computed on the stationary distribution of  $\Delta \log p$  conditional on a price change. The first block uses the baseline calibration; the second block raises persistence to  $\rho = 0.95$  and rescales the conditional shock standard deviation so that the unconditional standard deviation of  $\log z$  remains  $\sigma_\varepsilon / \sqrt{1 - \rho^2} = 0.163$ .

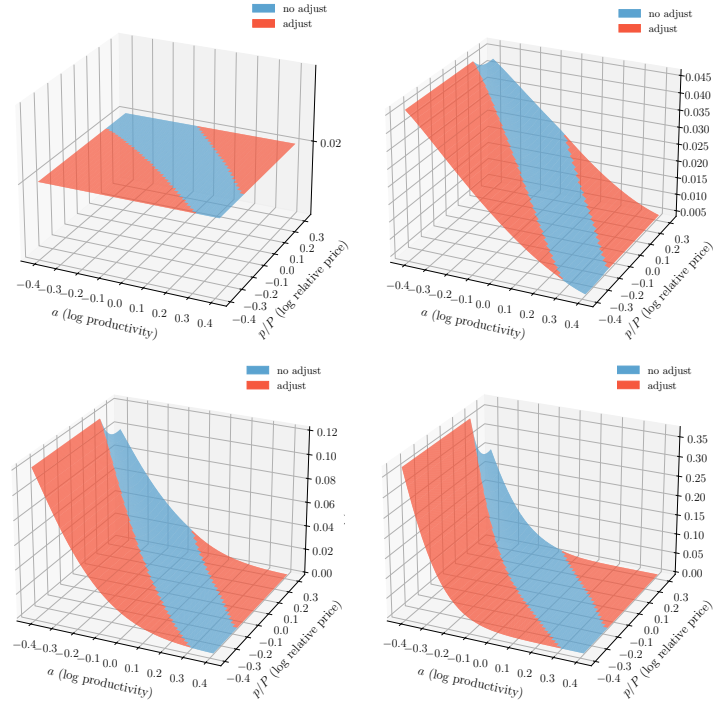
The contrasting persistence patterns map back to the channel decomposition of Sections D.1 and D.2. In the menu-cost model the kurtosis lift is carried by the cross-sectional composition of the surviving adjuster pool; neither the inaction band nor the cross-section depends tightly on  $\rho$ , and the  $\rho = 0.95$  rows of Table A.9 confirm the menu-cost lift is essentially unchanged. In the Calvo model the kurtosis lift is carried by the option-value response of the reset price, and this response exists at all because mean reversion is fast relative to the Calvo arrival rate: a low- $z$  firm cannot expect its inherited price to track productivity within its remaining lifetime, and the reset-price function turns sharply upward to meet the static optimum. Once  $\rho$  rises to 0.95 Calvo adjustments arrive fast enough relative to mean reversion that this segment flattens, the Benchmark–No-selection gap in reset prices collapses, and the Calvo kurtosis lift collapses with it.

The persistence exercise points to a subtle difference between time- and state-dependent pricing under selective exit. The two specifications need not generate the same response of micro-price moments to selection, and whether they do is ultimately a quantitative question,

one that depends on the underlying productivity process, beyond what the steady-state sufficient statistic of [Alvarez et al. \(2016\)](#) captures.

## E Additional figures

**Figure A.11: Exit hazard.**



*Notes:* The figure shows the exit hazard as a function of productivity and relative price, for values of  $\xi = 0, 2, 4, 10$ . The red region and blue region represent, respectively, the adjust region and no adjust regions of the value function. For positive selection ( $\xi > 0$ ), firms can decrease their exit probability by avoiding to adjust their price, i.e. saving on the menu cost.

## F Continuous time model

This appendix develops the reduced model of Section 6 and proves the results shown there. Section F.1 states the model. Section F.2 derives the three auxiliary objects that every proof uses: the expected value of a firm's remaining spell, the stationary density of gaps, and a stationarity identity linking the two. Section F.3 proves that the CIR equals the average slope of the spell value and, from it, Proposition 1: monetary non-neutrality equals the cross-sectional variance of price gaps divided by the innovation variance. Section F.4 proves the placement results and the amplification theorem. The representation results extend formulas of Alvarez et al. (2016), Bailey and Blanco (2021), and Alvarez et al. (2022); since the model here departs from that class through entry, exit, and an asymmetric gap-dependent hazard, I re-derive them in full, and the derivations double as the toolkit for the exit results, which are new.

Firms are the intermediate producers of Section 3, viewed through a single state variable, the price gap. A firm's flow loss increases with the distance between its price and the price it would set if adjustment were free, and paying the menu cost resets that distance to zero; with a constant hazard and a quadratic loss the optimal policy is an inaction band around the reset point, as in the standard menu-cost problem. The results below do not use the optimality of the band: they take the band as given, whatever objective generated it, and hold for any hazard and any band. Fixing the band removes the option-value response of the pricing policy and isolates the distributional channel quantified in Section 4.4. Throughout, the exit hazard  $\lambda$  is continuous on the closed band, hence bounded, with maximum  $\max \lambda$  (Weierstrass theorem).

### F.1 The reduced model

A firm's state is its price gap  $x$ : the log difference between the price it charges and the price it would set if adjustment were free. Trend inflation erodes every gap downward (the target drifts up while the posted price stands still); idiosyncratic shocks move the target around.

Start from a discrete monthly version, matching the timing of the quantitative model. A firm ends last month with gap  $x$ , draws an innovation  $\varepsilon \sim N(0, \sigma^2)$ , and arrives at  $y = x - \bar{\pi} + \varepsilon$ , where  $\bar{\pi}$  is the monthly erosion from trend inflation. On arrival: with probability  $\lambda(y)$  the firm exits and is replaced by an entrant born at gap 0; a surviving firm adjusts, resetting to 0, if  $y$  is outside the inaction band (in the Calvo variant, with constant probability  $\alpha$  instead). Both events place a gap of 0 at that slot in the economy: the adjuster corrects its own price, the exiter hands the slot to a correctly priced entrant. The probability that something fires is

$$e(y) = \lambda(y) + (1 - \lambda(y)) \mathbf{1}\{y \notin [b_\ell, b_u]\}. \quad (21)$$

Partition the month into short intervals of length  $dt$ : over one interval the gap moves by its share of the trend erosion,  $-\bar{\pi} dt$ , plus a Gaussian innovation with mean zero and variance  $\sigma^2 dt$ , independent across intervals. Denote the instantaneous change by  $dx = -\bar{\pi} dt + \sigma dW$ , then:

$$\mathbb{E}[dx] = -\bar{\pi} dt, \quad \mathbb{E}[(dx)^2] = \sigma^2 dt + O(dt^2), \quad \mathbb{E}[(dx)^k] = O(dt^2), \quad k \geq 3 :$$

the squared increment is of order  $dt$  and survives the limit, every higher power dies, which is why second derivatives appear in the equations below and higher ones never do. The band is  $(-a, a)$ ; touching a boundary triggers adjustment and reset to 0; exit fires at hazard  $\lambda(x)$ , meaning probability  $\lambda(x) dt$  per interval, with the entrant born at 0.  $W$  is a standard Wiener process.

The monetary shock is a permanent increase in nominal money of size  $\Delta$ : every target price rises by  $\Delta$ , so every gap drops by  $\Delta$  at  $t = 0$ . Output deviations are proportional to real balances, and the price level is the mean of prices, so the output deviation at  $t$  is proportional to minus the deviation of the mean gap from its stationary value. The object of interest is the cumulated impulse response per unit of shock,

$$\text{CIR} = \lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \int_0^\infty (\text{output deviation at } t) dt,$$

in months: the CIR divides the cumulated output deviation by the size of the shock, so a CIR of e.g. 5 means the shock generates the same cumulated output as output standing one full shock size above trend for five months. Amplification always means a higher CIR compared to some baseline.

The hazard of Section 3 is  $\lambda = \delta e^{-\xi V} / \mathbb{E}[e^{-\xi V}]$ , a function of the firm's value  $V$ ; in the reduced model it is a function of the gap. Writing the hazard on the gap does not contradict the model's finding that exit is driven mainly by productivity and not by mispricing: the projection concerns what the hazard looks like along the dimension that matters for pricing. The gap is informative about who exits precisely because the driver is productivity: a firm with a large negative gap is typically a firm whose target price just jumped above its posted price after a bad (productivity) draw, that is, a low-value firm with a high hazard. The projection  $\mathbb{E}[\lambda \mid \text{gap} = x]$  is simply a predictor of this feature of selection. The normalization in (5) fixes the average hazard over the cross-section at  $\delta$  every period: the exit flow is always  $\delta$ , selection only decides who is in it. The no-selection benchmark is the uniform hazard  $\lambda_u \equiv \delta$  (the  $\xi = 0$  economy); the selective economy has some  $\lambda_s(x)$  with the same flow.

## F.2 Spell values and the stationary density

For a function  $h$  of the gap,

$$\mathbb{E}_f[h] := \int_{-a}^a h(x) f(x) dx$$

is an average over the stationary cross-section of firms,  $f$  being the stationary density of gaps and  $\bar{x} := \mathbb{E}_f[x]$  its mean. For a starting gap  $x$ ,  $\mathbb{E}_x[\cdot]$  is an average over the future sample paths of one firm that starts at  $x$  and follows the process of Section F.1; the first kind integrates over firms at a moment, the second over one firm's future.

Two functions of the initial gap contain the necessary firm-level information. Let  $\tau$  be the firm's spell length, the first time it resets by adjusting or exiting, and define

$$m(x) = \mathbb{E}_x[\tau], \quad \bar{M}(x) = \mathbb{E}_x\left[\int_0^\tau (x_t - \bar{x}) dt\right] :$$

the expected spell length, and the expected cumulated de-meaned gap over the spell. Both are cases of one construction: for a forcing function  $g$ , let  $w(x) = \mathbb{E}_x[\int_0^\tau g(x_t) dt]$ , so that  $m$  is the case  $g = 1$  and  $\bar{M}$  the case  $g = x - \bar{x}$ . The equation for  $w$  comes from conditioning on the first short interval  $[0, dt]$ . Over it the firm accumulates  $g(x) dt$ ; with probability  $\lambda(x) dt$  the exit hazard then fires and the spell is over, so nothing more accumulates; otherwise the firm continues from the displaced gap  $x + dx$ , with continuation value  $w(x + dx)$ :

$$w(x) = g(x) dt + (1 - \lambda(x) dt) \mathbb{E}[w(x + dx)] + O(dt^2).$$

Taylor-expand the continuation to second order and take expectations with the moment facts of Section F.1:

$$\begin{aligned} \mathbb{E}[w(x + dx)] &= w(x) + w'(x) \mathbb{E}[dx] + \frac{1}{2} w''(x) \mathbb{E}[(dx)^2] + O(dt^2) \\ &= w(x) - \bar{\pi} w'(x) dt + \frac{1}{2} \sigma^2 w''(x) dt + O(dt^2). \end{aligned}$$

Substitute, cancel  $w(x)$  from both sides, drop the  $O(dt^2)$  terms (the cross product  $\lambda(x) dt \cdot \mathbb{E}[w(x + dx) - w(x)]$  is among them), and divide by  $dt$ :

$$\begin{aligned} w(x) &= g(x) dt + (1 - \lambda(x) dt) [w(x) - \bar{\pi} w'(x) dt + \frac{1}{2} \sigma^2 w''(x) dt + O(dt^2)] + O(dt^2) \\ &= g dt + (w - \bar{\pi} w' dt + \frac{1}{2} \sigma^2 w'' dt) + (-\lambda w dt + \lambda \bar{\pi} w' dt^2 - \lambda \frac{1}{2} \sigma^2 w'' dt) + O(dt^2) \\ 0 &= g dt - \bar{\pi} w' dt + \frac{1}{2} \sigma^2 w'' dt - \lambda w dt + O(dt^2) \end{aligned}$$

which gives

$$\frac{1}{2} \sigma^2 w'' - \bar{\pi} w' - \lambda(x) w = -g, \quad w(\pm a) = 0, \quad (22)$$

the boundary condition because a firm starting at an edge resets immediately and accumulates nothing. Write  $w = G[g]$  for the solution operator. The case  $g = x - \bar{x}$  is the equation for  $\bar{M}$ ,

$$\frac{1}{2} \sigma^2 \bar{M}'' - \bar{\pi} \bar{M}' - \lambda(x) \bar{M} = -(x - \bar{x}), \quad \bar{M}(\pm a) = 0, \quad (23)$$

and the case  $g = 1$  the equation for  $m$ . A similar equation holds for the stationary density  $f$ , and it follows from stationarity: in a stationary economy the flow of firms across any gap level must be consistent with what happens to firms on either side of it, i.e. the flows must be balanced. First I compute the flow, then impose the balance. Take as given, for now, that  $f$  is continuous on  $[-a, a]$  and differentiable except possibly at the reset point; indeed the reset point is where every adjuster and every entrant re-enters the cross-section, and the injection of mass there kinks the density. The derivation below this shape and determines the kink.

Step one, the flow. Fix an interior gap  $c \neq 0$  and a short interval of length  $h$ , and count the net mass that continuous motion carries across  $c$ , rightward minus leftward. A firm at  $y < c$  ends the interval past  $c$  if its increment  $\delta \sim N(-\bar{\pi}h, \sigma^2 h)$  exceeds the distance  $c - y$ ; a firm at  $y > c$  ends below  $c$  if  $\delta < -(y - c)$ . Writing  $v$  for the distance to  $c$ , the net rightward transport

over the interval is

$$\int_0^\infty f(c-v) \mathbb{P}(\delta > v) dv - \int_0^\infty f(c+v) \mathbb{P}(-\delta > v) dv$$

Expand  $f(c \mp v) = f(c) \mp f'(c)v + O(v^2)$  and use two tail identities, valid for any random variable  $X$  with  $X^+ := \max(X, 0)$ ,

$$\int_0^\infty \mathbb{P}(X > v) dv = \mathbb{E}[X^+], \quad \int_0^\infty v \mathbb{P}(X > v) dv = \frac{1}{2} \mathbb{E}[(X^+)^2]$$

(integrate  $\mathbf{1}\{X > v\}$  and  $v \mathbf{1}\{X > v\}$  over  $v$  and take the expectation outside the integral). The net transport becomes

$$\begin{aligned} & \int f(c) \mathbb{P}(\delta > v) dv - \int f'(c)v \mathbb{P}(\delta > v) dv - \int f(c) \mathbb{P}(-\delta > v) dv - \int f'(c)v \mathbb{P}(-\delta > v) dv = \\ & f(c) (\mathbb{E}[\delta^+] - \mathbb{E}[-\delta^+]) - \frac{1}{2} f'(c) (\mathbb{E}[(\delta^+)^2] + \mathbb{E}[(-\delta^+)^2]) + \\ & + \underbrace{\int_0^\infty O(v^2) \mathbb{P}(\delta > v) dv + \int_0^\infty O(v^2) \mathbb{P}(-\delta > v) dv}_{\text{Remainder}} \end{aligned}$$

The remainder is  $o(h)$ . It is bounded by:

$$\int_0^\infty O(v^2) \mathbb{P}(\delta > v) dv + \int_0^\infty O(v^2) \mathbb{P}(-\delta > v) dv \leq \int_0^\infty O(v^2) \mathbb{P}(|\delta| > v) dv$$

and use the tail identity  $\int v^2 \mathbb{P}(X > v) dv = \frac{1}{3} \mathbb{E}[(X^+)^3]$  to bound it even further by  $\frac{1}{3} \mathbb{E}|\delta|^3$ . Use  $\delta \sim N(-\pi h, \sigma^2 h)$  to write  $\delta = -\pi h + \sigma \sqrt{h} Z$ . Bound the absolute value:

$$\begin{aligned} |\delta| & \leq \pi h + \sigma \sqrt{h} Z \\ |\delta|^3 & \leq (\pi h + \sigma \sqrt{h} Z)^3 = \pi^3 h^3 + 3\pi^2 \sigma h^{5/2} |Z| + 3\pi \sigma^2 h^2 |Z|^2 + \sigma^3 h^{3/2} |Z|^3 \end{aligned}$$

Take expectations to remove the Gaussian variable  $Z$  and, with some abuse of notation, write the leading terms as

$$\mathbb{E}|\delta|^3 = O(h^3) + O(h^{5/2}) + O(h^2) + O(h^{3/2}) = O(h^{3/2}) = o(h)$$

The net transport then becomes

$$\begin{aligned} & f(c) (\mathbb{E}[\delta^+] - \mathbb{E}[(-\delta)^+]) - \frac{1}{2} f'(c) (\mathbb{E}[(\delta^+)^2] + \mathbb{E}[(-\delta)^+)^2]) + o(h) \\ & = f(c) \mathbb{E}[\delta] - \frac{1}{2} f'(c) \mathbb{E}[\delta^2] + o(h), \end{aligned}$$

using  $\delta^+ - (-\delta)^+ = \delta$  and  $(\delta^+)^2 + ((-\delta)^+)^2 = \delta^2$ . With  $\mathbb{E}[\delta] = -\pi h$  and  $\mathbb{E}[\delta^2] = \sigma^2 h + O(h^2)$ , the net rightward flow per unit of time is

$$J(c) := -\pi f(c) - \frac{1}{2} \sigma^2 f'(c) :$$

drift pushes mass downward at speed  $\bar{\pi}$ , and diffusion moves it down the density's slope.

Step two, the balance. Consider the firms with gaps in  $(c, a)$ , first for  $c \in (0, a)$ . Their number changes through exactly three channels: continuous motion across  $c$ , a net inflow of  $J(c)$  per unit of time; hazard firings inside  $(c, a)$ , at rate  $\int_c^a \lambda(x)f(x) dx$ , each relocating one firm to 0, out of the region; and touches of the edge  $a$ , at some rate  $N^+$  per unit of time (this rate is the definition of  $N^+$  but is currently unknown; I solve for it below), each also relocating one firm to 0. Nothing enters through the edge itself. Stationarity means the number of firms in the region is constant, so inflow equals outflow:

$$J(c) = \int_c^a \lambda(x)f(x) dx + N^+, \quad c \in (0, a).$$

For  $c \in (-a, 0)$  the region  $(c, a)$  contains the reset point, so relocations that originate inside it stay inside; instead the region receives every relocation originating to its left, the firings below  $c$  and the touches of  $-a$  (at rate  $N^-$ , the definition of  $N^-$ ):

$$J(c) = - \int_{-a}^c \lambda(x)f(x) dx - N^-, \quad c \in (-a, 0).$$

The right sides of both balances are continuously differentiable in  $c$ , so  $J$  is too, and through  $J$  so is  $f'$  away from 0: the density is assumed smooth except at 0. Differentiate either balance in  $c$ :

$$J'(c) = -\lambda(c)f(c) \iff \frac{1}{2}\sigma^2 f'' + \bar{\pi}f' - \lambda f = 0 \quad \text{at every interior } x \neq 0.$$

Next let  $c \rightarrow 0$  from both sides of the two balances and subtract: the two integrals join into  $\mathbb{E}_f[\lambda]$ , and

$$J(0^+) - J(0^-) = \int_0^a \lambda f + N^+ + \int_{-a}^0 \lambda f + N^- = \mathbb{E}_f[\lambda] + N^+ + N^- = N,$$

the total reset flow, i.e. the sum of its three sources. Using the definition of  $J(c)$ :

$$N = J(0^+) - J(0^-) = -\bar{\pi}(f(0^+) - f(0^-)) - \frac{1}{2}\sigma^2 (f'(0^+) - f'(0^-))$$

Because  $f$  is continuous at 0, the first bracket cancels and the jump in  $J$  is carried by the slope alone:

$$\frac{1}{2}\sigma^2 [f'(0^+) - f'(0^-)] = -N :$$

the density is continuous at the reset point but kinks there, and the size of the kink measures the reinjection flow. Finally let  $c \rightarrow a$  in the first balance:

$$J(a) = \underbrace{\int_a^a \lambda f}_{=0} + N^+ \implies J(a) = N^+$$

and

$$J(a) = - \underbrace{\pi f(a)}_{=0} - \frac{1}{2}\sigma^2 f'(a) \implies N^+ = -\frac{1}{2}\sigma^2 f'(a)$$

assuming for now that from the boundaries absorb, i.e.  $f(a) = 0$ . Symmetrically,  $N^- = \frac{1}{2}\sigma^2 f'(-a)$ . Finally, let's prove that  $f(\pm a) = 0$ .

The edges absorb: a firm is relocated the instant it touches one, and first-touch removal cannot coexist with mass standing at the removal point. Formally, the transition density of driftless Brownian motion absorbed at both edges of an interval is the alternating image series of [Karatzas and Shreve \(1991, Proposition 2.8.10\)](#): free Gaussian kernels centered at the successive mirror images generated by the two edges, with alternating signs. At either edge the images cancel in pairs, so the killed density vanishes there for every starting point and every horizon. Drift tilts and the exit hazard thins the path weights by factors bounded away from zero and infinity over any finite window, and neither revives a zero; the stationary density is an average of these killed kernels over the firms' last reset times, so it inherits the zeros.

Collecting the interior equation, the kink, and the boundary values, and writing  $\delta_0$  for a unit point mass at 0 so that the kink can be absorbed into the second derivative, the density solves

$$\frac{1}{2}\sigma^2 f'' + \pi f' - \lambda f + N\delta_0 = 0, \quad f(\pm a) = 0, \quad (24)$$

which is shorthand for exactly the three statements just derived, with the flux formulas  $N^+ = -\frac{1}{2}\sigma^2 f'(a)$  and  $N^- = \frac{1}{2}\sigma^2 f'(-a)$  and the flow accounting  $N = \mathbb{E}_f[\lambda] + N^+ + N^-$  attached. One remark: the drift enters (24) with the opposite sign to (22). Equation (22) follows one firm forward along its own path, whose gap erodes downward; equation (24) counts the population standing at a fixed gap, which erosion replenishes from above.

One consequence of (24) is used repeatedly. Take any twice continuously differentiable auxiliary function  $g(x)$ , multiply the interior equation  $\frac{1}{2}\sigma^2 f'' + \pi f' - \lambda f = 0$  by it, and integrate over  $[-a, 0]$  and  $[0, a]$  separately, because  $f'$  kinks at 0. The diffusion term moves twice. The first move, done on each piece,

$$\int_{-a}^0 g f'' dx + \int_0^a g f'' dx = g(a)f'(a) - g(-a)f'(-a) + g(0)[f'(0^-) - f'(0^+)] - \int_{-a}^a g' f' dx$$

the first two terms are the edges, and the third collects the two inner endpoints at 0, which do not cancel because of the kink. The second move, now legal over the whole band because  $f$  is continuous,

$$- \int_{-a}^a g' f' dx = -[g' f]_{-a}^a + \int_{-a}^a g'' f dx = \int_{-a}^a g'' f dx,$$

the bracket vanishing because  $f(\pm a) = 0$ . Combine the two moves and multiply by  $\frac{1}{2}\sigma^2$ :

$$\frac{1}{2}\sigma^2 \int_{-a}^a g f'' dx = \frac{1}{2}\sigma^2 \int_{-a}^a g'' f dx + \frac{1}{2}\sigma^2 \left( g(a)f'(a) - g(-a)f'(-a) + g(0)[f'(0^-) - f'(0^+)] \right).$$

The slopes of  $f$  at the edges and at the kink are the reset rates derived above,  $f'(a) = -2N^+/\sigma^2$ ,  $f'(-a) = 2N^-/\sigma^2$ , and  $f'(0^-) - f'(0^+) = 2N/\sigma^2$ , so the boundary terms are  $-N^+g(a) -$

$N^-g(-a) + Ng(0)$ . The drift term moves once and flips sign,  $\bar{\pi} \int_{-a}^a g f' dx = -\bar{\pi} \int_{-a}^a g' f dx$ , again by  $f(\pm a) = 0$ ; the hazard term has nothing to move. Collecting all terms gives the stationarity identity

$$\mathbb{E}_f \left[ \frac{1}{2} \sigma^2 g'' - \bar{\pi} g' - \lambda g \right] + Ng(0) - N^+ g(a) - N^- g(-a) = 0. \quad (25)$$

Note which operator now acts on  $g$ : the single-firm operator of (22), not the population operator of (24) the derivation started from; integration by parts is the bridge between the two, which is what makes the identity useful, since the left side can be evaluated on any twice continuously differentiable  $g$  whose single-firm equation is known. Each choice of  $g$  turns the differential equation (24), a statement holding at every point, into one scalar equation, a statement about one moment. In words: along surviving paths the cross-section drifts, diffuses, and dies, and in a stationary economy the resets must exactly undo all of it, with  $g$  selecting the accounting units in which the books are balanced.

### F.3 The CIR and gap dispersion

Two preliminary facts are used repeatedly, so I state them as lemmas that later proofs can point to.

**Lemma A.1** (A fresh spell carries no deviation).  $\bar{M}(0) = 0$ , at any drift, hazard, and band.

*Proof.* Apply the stationarity identity (25) to  $g = \bar{M}$ , admissible because the solution of (23) is twice continuously differentiable:

$$\mathbb{E}_f \left[ \frac{1}{2} \sigma^2 \bar{M}'' - \bar{\pi} \bar{M}' - \lambda \bar{M} \right] + N \bar{M}(0) - N^+ \bar{M}(a) - N^- \bar{M}(-a) = 0.$$

By (23) the bracket equals  $-(x - \bar{x})$  at every  $x$ , so its cross-sectional average is  $-(\mathbb{E}_f[x] - \bar{x}) = 0$  by the definition of  $\bar{x}$ ; and the last two terms vanish because  $\bar{M}(\pm a) = 0$ . What remains is  $N \bar{M}(0) = 0$ . And  $N > 0$ : the three components of  $N = \mathbb{E}_f[\lambda] + N^+ + N^-$  are each nonnegative ( $f \geq 0$  with  $f(\pm a) = 0$  forces  $f'(a) \leq 0 \leq f'(-a)$ ), so  $N = 0$  would force  $N^+ = 0$ , hence  $f(a) = f'(a) = 0$ , and the only solution of the second-order linear equation (24) with zero value and zero slope at  $a$  is  $f \equiv 0$  on  $(0, a]$ , contradicting  $f > 0$  there. Therefore  $\bar{M}(0) = 0$ : a freshly reset firm carries zero expected cumulated de-measured gap over its spell, precisely because  $\bar{x}$  is the mean gap that spells launched from 0 generate. ■

**Lemma A.2** (Spells end geometrically fast, uniformly in the starting gap). *There is  $p_0 > 0$ , free of the starting gap, such that  $\mathbb{P}_{x_0}(\tau > t) \leq (1 - p_0)^{\lfloor t \rfloor}$  for every  $t \geq 0$ , and hence  $m(x_0) \leq 1/p_0 < \infty$ .*

*Proof.* Suppose no reset happens on the time interval  $[0, 1]$  (one month). Then the path stays inside the band throughout, so  $|x_1 - x_0| < 2a$ ; and absent resets the path is  $x_t = x_0 - \bar{\pi} t + \sigma W_t$ , so  $x_1 - x_0 = -\bar{\pi} + \sigma W_1$ . The event  $\{\tau > 1\}$  therefore forces  $|\sigma W_1 - \bar{\pi}| < 2a$ , and

$$\mathbb{P}_{x_0}(\tau > 1) \leq \mathbb{P}(|\sigma W_1 - \bar{\pi}| < 2a) = 1 - p_0, \quad p_0 := \mathbb{P}(|\sigma W_1 - \bar{\pi}| \geq 2a) > 0,$$

positive because  $W_1$  is a standard normal, and free of  $x_0$ . Iterate with the Markov property at integer dates: conditional on surviving to date  $n - 1$  at gap  $x_{n-1}$ , the next month is a fresh month, so the probability of surviving it is  $\mathbb{P}_{x_{n-1}}(\tau > 1) \leq 1 - p_0$ , and

$$\begin{aligned}\mathbb{P}_{x_0}(\tau > n) &= \mathbb{E}_{x_0}[\mathbf{1}\{\tau > n - 1\} \mathbb{P}_{x_{n-1}}(\tau > 1)] \\ &\leq (1 - p_0) \mathbb{P}_{x_0}(\tau > n - 1) \leq \cdots \leq (1 - p_0)^n.\end{aligned}$$

Hence  $\mathbb{P}_{x_0}(\tau > t) \leq (1 - p_0)^{\lfloor t \rfloor}$  for real  $t$ , and

$$m(x_0) = \mathbb{E}_{x_0}[\tau] = \int_0^\infty \mathbb{P}_{x_0}(\tau > t) dt \leq \sum_{n=0}^\infty (1 - p_0)^n = \frac{1}{p_0} < \infty.$$

■

The central representation result expresses the CIR through the slope of the spell value  $\bar{M}$ .

**Proposition A.1** (CIR representation).

$$\text{CIR} = \mathbb{E}_f[\bar{M}'(x)]. \quad (26)$$

*Proof of Proposition A.1.* The proof is in four steps. Throughout,  $x_t$  is the full single-firm process of Section F.1, resets included: the gap diffuses by  $dx = -\bar{\pi} dt + \sigma dW$ , jumps to 0 when the exit hazard fires or when the path touches  $\pm a$ , and continues from there;  $\mathbb{E}_{x_0}[\cdot]$  is the expectation over these paths for a firm starting at  $x_0$ , and  $\tau$  is its first reset time. A firm starting at  $x_0 \leq -a$  (the shock will create some) resets at time zero, so  $\mathbb{E}_{x_0} = \mathbb{E}_0$  for such points; correspondingly extend  $\bar{M}$  by  $\bar{M}(x_0) := 0$  for  $|x_0| \geq a$ , a continuous extension because  $\bar{M}(\pm a) = 0$ . About  $\bar{M}$  the proof uses only the boundary value problem (23) and the smoothness of its solution, through the two constants

$$K := \max_{[-a,a]} |\bar{M}| < \infty, \quad K' := \max_{[-a,a]} |\bar{M}'| < \infty.$$

The shocked economy at  $t = 0$  is the stationary cross-section with every gap lowered by  $\Delta$ : gaps are  $x - \Delta$  with  $x$  drawn from  $f$ . The economy is a continuum of firms hit by independent idiosyncratic shocks, so its mean gap at date  $t$  is the single-firm expectation averaged over the starting draws,

$$\bar{x}_\Delta(t) := \int_{-a}^a \mathbb{E}_{x-\Delta}[x_t] f(x) dx.$$

Output moves with real balances: its log deviation from trend is proportional to the deviation of nominal demand from the price level (Section 3). The nominal shock is permanent, and the price level is the mean of posted prices, each equal to its target plus the firm's gap, so the deviation of the price level from its post-shock target path equals the deviation of the mean gap from its stationary value, and the output deviation at date  $t$  is proportional to  $-(\bar{x}_\Delta(t) - \bar{x})$ . I normalize the proportionality constant to one, which fixes output units. The definition of the CIR therefore is

$$\text{CIR} = -\lim_{\Delta \downarrow 0} \frac{1}{\Delta} \lim_{T \rightarrow \infty} \int_0^T (\bar{x}_\Delta(t) - \bar{x}) dt, \quad (27)$$

and what is to be proved is that both limits exist and that

$$\text{CIR} = \int_{-a}^a \bar{M}'(x) f(x) dx = \mathbb{E}_f[\bar{M}'].$$

Step 1: an exact identity for one firm at a finite horizon. The claim is that for every starting gap  $x_0$  and every horizon  $T$ ,

$$\int_0^T (\mathbb{E}_{x_0}[x_t] - \bar{x}) dt = \bar{M}(x_0) - \mathbb{E}_{x_0}[\bar{M}(x_T)]. \quad (28)$$

I apply Itô's formula between resets; the reason the resets themselves cause no trouble is the pair of equalities  $\bar{M}(\pm a) = 0 = \bar{M}(0)$ , from the boundary conditions of (23) and from Lemma A.1. When the path jumps from an edge to the reset point,  $\bar{M}$  takes the same value, zero, at both ends of the jump, so the sample path  $t \mapsto \bar{M}(x_t)$  passes through a boundary reset continuously even though  $x_t$  itself does not; only the hazard firings, which jump from an interior gap, move  $\bar{M}(x_t)$ , and those jumps are exactly what the  $\lambda \bar{M}$  term in the equation compensates.

List the reset times  $0 < T_1 < T_2 < \dots$  of the path. There are finitely many in  $[0, T]$ : the spells after the first are, by the strong Markov property, independent copies of the spell launched from 0, whose length is strictly positive with probability one (a finite-rate hazard cannot fire instantly, and reaching an edge from 0 takes positive time), so the  $n$ th reset time grows without bound. Between consecutive resets the path is the diffusion  $dx = -\bar{\pi} dt + \sigma dW$  in the closed band, where  $\bar{M}$  is twice continuously differentiable, so Itô's formula (Øksendal, 2003, Theorem 4.1.2) applies spell by spell. Take any function  $g$ :

$$dg(x_t) = g'(x_t)dx_t + \frac{1}{2}g''(x_t)(dx_t)^2 = (-\bar{\pi}g'(x_t) + \frac{1}{2}\sigma^2g''(x_t)) dt + \sigma g'(x_t)dW_t$$

Integrate from  $u$  to  $v$ :

$$g(x_u) - g(x_v) = \int_u^v (-\bar{\pi}g'(x_t) + \frac{1}{2}\sigma^2g''(x_t)) dt + \int_u^v \sigma g'(x_t)dW_t$$

Use  $g = \bar{M}$  for  $u < v$  inside one spell,

$$\bar{M}(x_v) - \bar{M}(x_u) = \int_u^v \left( \frac{1}{2}\sigma^2\bar{M}'' - \bar{\pi}\bar{M}' \right)(x_t) dt + \int_u^v \sigma \bar{M}'(x_t) dW_t.$$

Denote by  $K_T$  the number of resets in  $[0, T]$ , by  $x_{T_k^-}$  the left limit, the value of the gap the instant before the reset at  $T_k$ , and by  $x_{T_k}$  the value immediately after the reset, which equals zero by construction; set  $T_0 = 0$ . The path of  $x_t$  over  $[0, T]$  then starts at  $x_0$ , moves continuously (diffusion) until  $T_1$ , jumps to 0, moves continuously until  $T_2$ , jumps to 0, and so on through  $T_{K_T}$ , after which it moves continuously until the clock runs out at  $T$ , mid-spell, because  $T$  is a

fixed horizon and not a reset time. Use this to build the telescopic sum

$$\bar{M}(x_T) - \bar{M}(x_0) = \underbrace{\sum_{k=1}^{K_T} [\bar{M}(x_{T_k^-}) - \bar{M}(x_{T_{k-1}})]}_{\text{I: completed spells}} + \underbrace{\sum_{k=1}^{K_T} [\bar{M}(x_{T_k}) - \bar{M}(x_{T_k^-})]}_{\text{II: jumps at resets}} + \underbrace{[\bar{M}(x_T) - \bar{M}(x_{T_{K_T}})]}_{\text{III: running spell}}$$

Apply Itô's lemma to each continuous bracket:

$$\bar{M}(x_{T_k^-}) - \bar{M}(x_{T_{k-1}}) = \int_{T_{k-1}}^{T_k} \left( \frac{1}{2} \sigma^2 \bar{M}'' - \bar{\pi} \bar{M}' \right) (x_t) dt + \int_{T_{k-1}}^{T_k} \sigma \bar{M}'(x_t) dW_t$$

for I and III similarly with limits  $T_k$  and  $T$ . Since the reset times partition the interval, the integral ranges over  $[0, T_1) \cup [T_1, T_2) \cup \dots \cup [T_{K_T}, T)$ , i.e. tiling the whole  $[0, T]$ . They concatenate into a single integral: similarly for the first and third brackets

$$\text{I} + \text{III} = \int_0^T \left( \frac{1}{2} \sigma^2 \bar{M}'' - \bar{\pi} \bar{M}' \right) (x_t) dt + \int_0^T \sigma \bar{M}'(x_t) dW_t$$

Substitute back and keep II:

$$\bar{M}(x_T) = \bar{M}(x_0) + \int_0^T \left( \frac{1}{2} \sigma^2 \bar{M}'' - \bar{\pi} \bar{M}' \right) (x_t) dt + \int_0^T \sigma \bar{M}'(x_t) dW_t + \sum_{k=1}^{K_T} [\bar{M}(x_{T_k}) - \bar{M}(x_{T_k^-})]$$

using the fact that every reset lands at 0, so  $\bar{M}(x_{T_k}) = \bar{M}(0) = 0$ :

$$\bar{M}(x_{T_k}) - \bar{M}(x_{T_k^-}) = \begin{cases} \bar{M}(0) - \bar{M}(\pm a) = 0 & \text{if } T_k \text{ is a boundary touch} \\ \bar{M}(0) - \bar{M}(x_{T_k^-}) = -\bar{M}(x_{T_k^-}) & \text{if } T_k \text{ is a hazard firing} \end{cases}$$

Therefore  $\bar{M}(x_T)$  becomes

$$\bar{M}(x_T) = \bar{M}(x_0) + \int_0^T \left( \frac{1}{2} \sigma^2 \bar{M}'' - \bar{\pi} \bar{M}' \right) (x_t) dt + \int_0^T \sigma \bar{M}'(x_t) dW_t + J_T$$

where  $J_T := \sum_{s \leq T} \bar{M}(x_{s^-})$ . The hazard creates a point process with stochastic intensity  $\lambda(x_t)$ . Define

$$A_T := \int_0^T \lambda(x_t) \bar{M}(x_t) dt$$

By **Brémaud (1981, Chapter II, Theorem T8)**, applicable because  $\bar{M}$  is bounded by  $K$  and  $\lambda$  is a continuous function on a compact set, hence bounded,

$$\mathcal{M}_T^{(2)} := -J_T + A_T \text{ is a zero-mean martingale, } \mathbb{E}_{x_0}[\mathcal{M}_T^{(2)}] = 0$$

Insert  $-J_T = -A_T + \mathcal{M}_T^{(2)}$  into  $\bar{M}(x_T)$  and merge  $-A_T$  into the  $dt$ -integral:

$$\begin{aligned}\bar{M}(x_T) &= \bar{M}(x_0) + \int_0^T \left( \frac{1}{2} \sigma^2 \bar{M}'' - \bar{\pi} \bar{M}' \right) (x_t) dt + \mathcal{M}_T, \\ \mathcal{M}_T &= \underbrace{\int_0^T \sigma \bar{M}'(x_t) dW_t}_{:= \mathcal{M}_T^{(1)}}\end{aligned}\tag{29}$$

The path satisfies  $x_t \in [-a, a]$  for all  $t$ , and on the closed band  $\bar{M}$  solves the ODE (23) at every point,

$$\frac{1}{2} \sigma^2 \bar{M}''(x_t) - \bar{\pi} \bar{M}'(x_t) - \lambda(x_t) \bar{M}(x_t) = -(x_t - \bar{x}),$$

hence (29) becomes

$$\bar{M}(x_T) = \bar{M}(x_0) - \int_0^T (x_t - \bar{x}) dt + \mathcal{M}_T$$

Now take expectations under  $\mathbb{E}_{x_0}$ : the integral of  $\mathcal{M}_T^{(1)}$  is bounded by  $|\sigma \bar{M}'| \leq |\sigma K'|$  so  $\mathcal{M}_T^{(1)}$  is a square-integrable martingale and  $\mathbb{E}_{x_0}[\mathcal{M}_T^{(1)}] = 0$ , similar to  $\mathbb{E}_{x_0}[\mathcal{M}_T^{(2)}]$ . Therefore

$$\mathbb{E}_{x_0}[\bar{M}(x_T)] = \bar{M}(x_0) - \mathbb{E}_{x_0} \int_0^T (x_t - \bar{x}) dt$$

Since  $|x_t - \bar{x}| \leq 2a$  on  $[0, T]$ , Fubini's theorem gives

$$\int_0^T (\mathbb{E}_{x_0}[x_t] - \bar{x}) dt = \bar{M}(x_0) - \mathbb{E}_{x_0}[\bar{M}(x_T)]$$

For  $x_0 \leq -a$  the firm resets at once, so every term in (28) equals its value at  $x_0 = 0$  (using  $\bar{M}(x_0) = 0 = \bar{M}(0)$  for the extension), and the identity holds there as well.

Step 2: aggregate at a finite horizon. Fix  $\Delta > 0$  and  $T < \infty$ . The time integral of the mean-gap deviation is a double integral, over dates and over firms; exchange the order:

$$\begin{aligned}\int_0^T (\bar{x}_\Delta(t) - \bar{x}) dt &= \int_0^T \left[ \int_{-a}^a (\mathbb{E}_{x-\Delta}[x_t] - \bar{x}) f(x) dx \right] dt \\ &= \int_{-a}^a \left[ \int_0^T (\mathbb{E}_{x-\Delta}[x_t] - \bar{x}) dt \right] f(x) dx.\end{aligned}$$

valid because the domain  $[0, T] \times [-a, a]$  is bounded and the integrand is bounded: the process lives in the band, so  $|\mathbb{E}_{x-\Delta}[x_t] - \bar{x}| \leq 2a$ . The inner bracket on the right is the left side of (28) at  $x_0 = x - \Delta$ , so

$$\int_0^T (\bar{x}_\Delta(t) - \bar{x}) dt = \int_{-a}^a [\bar{M}(x - \Delta) - \mathbb{E}_{x-\Delta}[\bar{M}(x_T)]] f(x) dx.$$

Next, use the following property of stationary density in the unshocked economy: started from  $f$ , the cross-section at every later date is again  $f$  (that is the definition of the stationary density; equation (24) is its differential form). The gap  $x_T$  of a firm whose start is drawn from  $f$  is then

itself distributed as  $f$ , so

$$\int_{-a}^a \mathbb{E}_x[\bar{M}(x_T)] f(x) dx = \mathbb{E}_f[\bar{M}] \quad \text{for every } T. \quad (30)$$

Add and subtract  $\bar{M}(x)$  inside the previous integral:

$$\begin{aligned} \int_0^T (\bar{x}_\Delta(t) - \bar{x}) dt &= \int_{-a}^a [\bar{M}(x - \Delta) - \bar{M}(x)] f(x) dx \\ &\quad + \int_{-a}^a [\bar{M}(x) - \mathbb{E}_{x-\Delta}[\bar{M}(x_T)]] f(x) dx, \end{aligned}$$

and since  $\int \bar{M}(x) f(x) dx = \mathbb{E}_f[\bar{M}]$ , by (30) the second integral equals

$$\int_{-a}^a (\mathbb{E}_x[\bar{M}(x_T)] - \mathbb{E}_{x-\Delta}[\bar{M}(x_T)]) f(x) dx.$$

Hence

$$\begin{aligned} \int_0^T (\bar{x}_\Delta(t) - \bar{x}) dt &= \int_{-a}^a [\bar{M}(x - \Delta) - \bar{M}(x)] f(x) dx - \varepsilon_\Delta(T), \\ \varepsilon_\Delta(T) &:= \int_{-a}^a (\mathbb{E}_{x-\Delta}[\bar{M}(x_T)] - \mathbb{E}_x[\bar{M}(x_T)]) f(x) dx : \end{aligned} \quad (31)$$

the shocked-minus-unshocked difference in the horizon term.

Step 3: the horizon limit,  $\varepsilon_\Delta(T) \rightarrow 0$  as  $T \rightarrow \infty$  at fixed  $\Delta$ . This is the only step where the economy's loss of memory enters, in four moves.

(i) A reset erases the past. Let  $F_{x_0}(s) := \mathbb{P}_{x_0}(\tau \leq s)$  be the distribution function of the first reset time, and let  $\psi(u) := \mathbb{E}_0[\bar{M}(x_u)]$  be the horizon- $u$  expectation for a firm launched at the reset point. Split  $\mathbb{E}_{x_0}[\bar{M}(x_T)]$  on whether the first reset has happened by  $T$ , and use the strong Markov property: after a reset at time  $s$  the firm is a fresh firm at 0, so its date- $T$  gap has the law of  $x_{T-s}$  under  $\mathbb{E}_0$ . This gives

$$\mathbb{E}_{x_0}[\bar{M}(x_T)] = \mathbb{E}_{x_0}[\bar{M}(x_T) \mathbf{1}\{\tau > T\}] + \int_0^T \psi(T-s) dF_{x_0}(s), \quad (32)$$

with the first term bounded using Lemma A.2:

$$|\mathbb{E}_{x_0}[\bar{M}(x_T) \mathbf{1}\{\tau > T\}]| \leq K \mathbb{P}_{x_0}(\tau > T) \leq K(1 - p_0)^{\lfloor T \rfloor}.$$

(ii)  $\psi$  converges. Apply (32) at  $x_0 = 0$  and abbreviate  $\zeta(u) := \mathbb{E}_0[\bar{M}(x_u) \mathbf{1}\{\tau > u\}]$ :

$$\psi(u) = \zeta(u) + \int_0^u \psi(u-s) dF_0(s),$$

a renewal equation. The key renewal theorem (Feller, 1971, Volume II, Theorem XI.1.2) applies: if  $F_0$  is a non-lattice proper distribution on  $(0, \infty)$  with finite mean  $m(0)$  and  $\zeta$  is directly

Riemann integrable, the bounded solution converges,

$$\psi(u) \xrightarrow{u \rightarrow \infty} \psi_\infty := \frac{1}{m(0)} \int_0^\infty \zeta(s) ds.$$

- $F_0$  is a proper distribution;
- $F_0$  has a density, hence is non-lattice: conditional on the Brownian path, exit arrives at the first event of a clock with intensity  $\lambda(x_t)$ , an absolutely continuous law, and absent exit the spell ends when the path first touches  $\pm a$ , a Brownian hitting time, which also has a density.
- the mean is positive and finite:  $m(0) = \int_0^\infty (1 - F_0(s)) ds \leq \frac{1}{p_0}$  by Lemma A.2, and  $m(0) > 0$  since

$$\mathbb{P}_0(\tau > t) \geq \mathbb{E}[\mathbf{1}\{\text{no boundary hit by } t\}]e^{-\int_0^t \lambda(x_0) ds} \geq e^{-t \max \lambda} \mathbb{P}_0(\text{no hit by } t) > 0$$

- $\zeta$  is directly Riemann integrable because continuous in  $u$  and dominated by the decreasing integrable function  $K(1 - p_0)^{\lfloor u \rfloor}$

Therefore

$$\eta(u) := \lim_{u \rightarrow \infty} \sup_{v \geq u} |\psi(v) - \psi(u)| = 0 \quad (33)$$

which gives pointwise convergence  $\psi(u) \rightarrow \psi_\infty$ .

(iii) Every starting point inherits the limit. Fix  $x_0$  and let  $T \rightarrow \infty$ . Use  $\int_0^\infty \psi_\infty dF_{x_0}(s) = \psi_\infty F_{x_0}(T) = \psi_\infty - \psi_\infty(1 - F_{x_0}(T))$ . Then write

$$\begin{aligned} \int_0^T \psi(T-s) dF_{x_0}(s) - \psi_\infty &= \underbrace{\int_0^{T/2} (\psi(T-s) - \psi_\infty) dF_{x_0}(s)}_{\leq \eta(T/2) = \sup_{u > T/2} |\psi(u) - \psi_\infty| \rightarrow 0 \text{ since } s \leq T/2} \\ &+ \underbrace{\int_{T/2}^T (\psi(T-s) - \psi_\infty) dF_{x_0}(s)}_{\leq 2K \mathbb{P}_{x_0}(\tau > T/2) \leq 2K(1-p_0)^{\lfloor T/2 \rfloor} \rightarrow 0} \\ &- \underbrace{\psi_\infty(1 - F_{x_0}(T))}_{\leq |\psi_\infty|(1-p_0)^{\lfloor T \rfloor} \leq K(1-p_0)^{\lfloor T \rfloor} \rightarrow 0} \end{aligned}$$

Hence (32) gives

$$\sup_{x_0} |\mathbb{E}_{x_0}[\bar{M}(x_T)] - \psi_\infty| \leq \rho(T) := \eta(T/2) + 2K(1-p_0)^{\lfloor T/2 \rfloor} + K(1-p_0)^{\lfloor T \rfloor} \rightarrow 0,$$

where every term is free of  $x_0$ : every starting gap inherits the limit,  $\mathbb{E}_{x_0}[\bar{M}(x_T)] \rightarrow \psi_\infty$ . Averaging the bound over  $f$  and using (30),

$$|\mathbb{E}_f[\bar{M}] - \psi_\infty| = \left| \int_{-a}^a (\mathbb{E}_x[\bar{M}(x_T)] - \psi_\infty) f(x) dx \right| \leq \rho(T) \xrightarrow{T \rightarrow \infty} 0,$$

and the left side is free of  $T$ , so  $\psi_\infty = \mathbb{E}_f[\bar{M}]$ .

(iv) Apply it to the starting point  $x - \Delta$ :

$$|\varepsilon_\Delta(T)| = \left| \int_{-a}^a (\mathbb{E}_{x-\Delta}[\bar{M}(x_T)] - \psi_\infty) f(x) dx \right| \leq \rho(T) \xrightarrow{T \rightarrow \infty} 0$$

for every fixed  $\Delta$ .

Step 4: the shock limit. By (31) and Step 3, the inner limit of (27) exists and equals

$$\lim_{T \rightarrow \infty} \int_0^T (\bar{x}_\Delta(t) - \bar{x}) dt = \int_{-a}^a [\bar{M}(x - \Delta) - \bar{M}(x)] f(x) dx,$$

so that, carrying the minus sign and the division by  $\Delta$  from (27) inside the integral,

$$\text{CIR} = \lim_{\Delta \downarrow 0} \int_{-a}^a \frac{\bar{M}(x) - \bar{M}(x - \Delta)}{\Delta} f(x) dx.$$

The limit passes inside the integral by dominated convergence. Pointwise convergence: fix  $x$  in the open band; for  $\Delta < x + a$  both points lie in the band, and the mean value theorem gives

$$\frac{\bar{M}(x) - \bar{M}(x - \Delta)}{\Delta} = \bar{M}'(\xi_{x,\Delta}) \text{ for some } \xi_{x,\Delta} \in (x - \Delta, x), \quad \bar{M}'(\xi_{x,\Delta}) \xrightarrow{\Delta \downarrow 0} \bar{M}'(x)$$

by the continuity of  $\bar{M}'$ . Domination: the quotient is bounded by  $K'$  for every  $x$  and every  $\Delta$ , inside the band by the same mean value theorem, and across the edge (the case  $x - \Delta \leq -a$ , where the extension sets  $\bar{M}(x - \Delta) = 0$ ) because  $|\bar{M}(x) - 0| = |\bar{M}(x) - \bar{M}(-a)| \leq K'(x + a) \leq K'\Delta$ ; the dominating function  $K'f(x)$  integrates to  $K'$ . Therefore

$$\text{CIR} = \int_{-a}^a \lim_{\Delta \downarrow 0} \frac{\bar{M}(x) - \bar{M}(x - \Delta)}{\Delta} f(x) dx = \int_{-a}^a \bar{M}'(x) f(x) dx = \mathbb{E}_f[\bar{M}'].$$

■

Proposition 1 in the main text is the zero-drift case of an exact statement that holds at any drift: for any hazard and any band,

$$\text{CIR} = \frac{\text{Var}_f(x) + \bar{\pi} \mathbb{E}_f[\bar{M}]}{\sigma^2}, \tag{34}$$

and at  $\bar{\pi} = 0$  the correction term drops, leaving  $\text{CIR} = \text{Var}_f(x)/\sigma^2$ .

*Proof of Proposition 1, general form (34).* Apply the stationarity identity (25) to  $g(x) = x \bar{M}(x)$ . All three reset terms vanish:  $g(0) = 0$  through the factor  $x$ , and  $g(\pm a) = 0$  through  $\bar{M}(\pm a) = 0$ . With  $g' = \bar{M} + x \bar{M}'$  and  $g'' = 2\bar{M}' + x \bar{M}''$ , the identity reads

$$\mathbb{E}_f[\sigma^2 \bar{M}' + x(\frac{1}{2}\sigma^2 \bar{M}'') - \bar{\pi} \bar{M} - \bar{\pi} x \bar{M}' - \lambda x \bar{M}] = 0.$$

Substitute the ODE (23),  $\frac{1}{2}\sigma^2 \bar{M}'' = \bar{\pi} \bar{M}' + \lambda \bar{M} - (x - \bar{x})$ , inside the expectation: the  $\bar{\pi} x \bar{M}'$

terms cancel, the  $\lambda x \bar{M}$  terms cancel, and

$$\sigma^2 \mathbb{E}_f[\bar{M}'] = \mathbb{E}_f[x(x - \bar{x})] + \bar{\pi} \mathbb{E}_f[\bar{M}] = \text{Var}_f(x) + \bar{\pi} \mathbb{E}_f[\bar{M}].$$

Proposition A.1 identifies the left side as  $\sigma^2 \text{CIR}$ . ■

**Proposition A.2** (Uniform exit dampens). *With  $\lambda(x) \equiv \lambda$ ,  $\bar{\pi} = 0$ , band  $\pm a$ : writing  $q = \sqrt{2\lambda}/\sigma$  and  $y = qa$ ,*

$$\text{CIR}(\lambda) = \frac{1}{\lambda} \left( 1 - \frac{y^2}{2(\cosh y - 1)} \right) = \frac{a^2}{6\sigma^2} - \frac{a^4}{60\sigma^4} \lambda + O(\lambda^2),$$

*strictly decreasing in  $\lambda$ , from the no-exit value  $a^2/(6\sigma^2)$  toward 0.*

*Proof of Proposition A.2.* The proof solves  $f$  first (its symmetry pins  $\bar{x} = 0$ , which fixes  $\bar{M}'$ 's forcing), then  $\bar{M}$ , then computes the integral; the expansion and the monotonicity close the argument.

(i) The density. Away from the reset point  $f$  solves the homogeneous equation  $\frac{1}{2}\sigma^2 f'' = \lambda f$ , that is  $f'' = q^2 f$ . On  $(0, a]$  write the solution in the basis centered at the edge,

$$f(x) = A \sinh(q(a - x)) + B \cosh(q(a - x)),$$

so that the boundary condition  $f(a) = 0$  reads  $B = 0$ . The same argument on  $[-a, 0]$  gives  $f = A' \sinh(q(a + x))$ , and continuity at 0 forces  $A' = A$ . Hence

$$f(x) = A \sinh(q(a - |x|)),$$

with  $A$  pinned by the unit mass of  $f$ ,

$$1 = 2A \int_0^a \sinh(q(a - x)) dx = \frac{2A(\cosh y - 1)}{q} \implies A = \frac{q}{2(\cosh y - 1)},$$

the integral by the substitution  $u = a - x$ . The density is symmetric, as a consequence of the edge conditions and continuity rather than by assumption, and symmetry delivers  $\bar{x} = \int x f dx = 0$ .

(ii) The  $\bar{M}$  function. With  $\bar{x} = 0$ , equation (23) reads  $\frac{1}{2}\sigma^2 \bar{M}'' - \lambda \bar{M} = -x$ ,  $\bar{M}(\pm a) = 0$ . A particular solution: try  $cx$ ; substituting gives  $-\lambda cx = -x$ , so  $c = 1/\lambda$ . The general solution is  $x/\lambda + \alpha \cosh(qx) + \beta \sinh(qx)$ , and the two boundary conditions are a  $2 \times 2$  system,

$$\frac{a}{\lambda} + \alpha \cosh y + \beta \sinh y = 0, \quad -\frac{a}{\lambda} + \alpha \cosh y - \beta \sinh y = 0 :$$

their sum gives  $\alpha = 0$  (the boundary data are antisymmetric, so the even homogeneous term is killed), their difference gives  $\beta = -a/(\lambda \sinh y)$ . Hence

$$\bar{M}(x) = \frac{x}{\lambda} - \frac{a \sinh(qx)}{\lambda \sinh y}, \quad \bar{M}'(x) = \frac{1}{\lambda} - \frac{aq \cosh(qx)}{\lambda \sinh y}.$$

(iii) The integral. Averaging  $\bar{M}'$  over  $f$ ,

$$\text{CIR} = \frac{1}{\lambda} \underbrace{\int_{-a}^a f}_{=1} - \frac{aq}{\lambda \sinh y} \int_{-a}^a f \cosh(qx) dx,$$

and the second term calls for the solved density  $f = A \sinh(q(a - |x|))$ . By the evenness of the integrand and the product-to-sum identity  $\sinh U \cosh V = \frac{1}{2}[\sinh(U + V) + \sinh(U - V)]$  with  $U = q(a - x)$ ,  $V = qx$ , chosen because  $U + V = y$  is constant:

$$\begin{aligned} \int_{-a}^a f \cosh(qx) dx &= 2A \int_0^a \sinh(q(a - x)) \cosh(qx) dx \\ &= A \int_0^a [\sinh y + \sinh(q(a - 2x))] dx = A a \sinh y, \end{aligned}$$

the second piece vanishing because its antiderivative  $-\cosh(q(a - 2x))/(2q)$  takes the value  $-\cosh y/(2q)$  at both endpoints. Substituting  $A$  and  $q^2 a^2 = y^2$ ,

$$\text{CIR} = \frac{1}{\lambda} - \frac{aq}{\lambda \sinh y} \cdot \frac{qa \sinh y}{2(\cosh y - 1)} = \frac{1}{\lambda} \left( 1 - \frac{y^2}{2(\cosh y - 1)} \right),$$

(iv) The local expansion. From  $\cosh y - 1 = \frac{y^2}{2} (1 + \frac{y^2}{12} + \frac{y^4}{360} + \dots)$ ,

$$\frac{y^2}{2(\cosh y - 1)} = 1 - \frac{y^2}{12} + \frac{y^4}{240} + O(y^6),$$

so that

$$\text{CIR} = \frac{1}{\lambda} \left( \frac{y^2}{12} - \frac{y^4}{240} + O(y^6) \right) = \frac{a^2}{6\sigma^2} - \frac{a^4}{60\sigma^4} \lambda + O(\lambda^2),$$

substituting  $y^2 = 2\lambda a^2/\sigma^2$ . This proves the  $\lambda \rightarrow 0$  limit and that the CIR is decreasing near  $\lambda = 0$ .

(v) Global monotonicity. Since  $y$  is increasing in  $\lambda$ , substitute  $1/\lambda = 2a^2/(\sigma^2 y^2)$  and write  $\text{CIR} = (2a^2/\sigma^2) g(y)$  with  $g(y) = y^{-2} - \frac{1}{2}(\cosh y - 1)^{-1}$ ; strict decrease in  $\lambda$  for all  $\lambda > 0$  is  $g'(y) < 0$  for all  $y > 0$ :

$$g'(y) = -\frac{2}{y^3} + \frac{\sinh y}{2(\cosh y - 1)^2} < 0 \iff y^3 \sinh y < 4(\cosh y - 1)^2.$$

The half-angle identities  $\cosh y - 1 = 2 \sinh^2(y/2)$  and  $\sinh y = 2 \sinh(y/2) \cosh(y/2)$ , with  $z = y/2$ , reduce the claim to

$$z^3 \cosh z < \sinh^3 z \quad \text{for every } z > 0.$$

Compare the two power series. From  $\sinh 3z = 3 \sinh z + 4 \sinh^3 z$ , the coefficient of  $z^{2k+1}$  in  $\sinh^3 z$  is  $\frac{3^{2k+1}-3}{4(2k+1)!}$ ; in  $z^3 \cosh z$  it is  $\frac{1}{(2k-2)!}$  for  $k \geq 1$  and zero for  $k = 0$ . The coefficient claim is

therefore

$$\frac{3^{2k+1} - 3}{4} \geq (2k+1)(2k)(2k-1) \quad \text{for every } k \geq 1.$$

Equality holds at  $k = 1$  ( $6 = 6$ ) and  $k = 2$  ( $60 = 60$ ), and the inequality is strict at  $k = 3$  ( $546 > 210$ ) and thereafter: each step  $k \rightarrow k+1$  multiplies the left side by more than 9, since  $3^{2k+3} - 3 > 9(3^{2k+1} - 3)$ , and the right side by  $(2k+3)(2k+2)/(2k(2k-1)) \leq 12/5$ . So every coefficient of  $\sinh^3 z - z^3 \cosh z$  is nonnegative and those from  $z^7$  on are strictly positive: the difference is strictly positive for every  $z > 0$ , hence  $g' < 0$ , and the CIR falls strictly from  $a^2/(6\sigma^2)$ , the  $\lambda \rightarrow 0$  limit of (iv), toward 0, since the bracket in the closed form tends to 1 while  $1/\lambda \rightarrow 0$ . The equalities at  $k = 1, 2$  mean the two sides are tangent to sixth order at  $z = 0$ : the monotonicity is invisible to low-order expansions, which is why step (iv) cannot deliver it and the series comparison is needed. ■

**Proposition A.3** (Censoring is exactly neutral). *The response of the economy to any aggregate shock is invariant to the values of the exit hazard on arrival states outside the inaction band. Only the labels change: each unit of hazard placed there converts price adjustments into exits one for one, lowering the measured frequency of price changes and changing nothing real.*

*Proof of Proposition A.3.* At an arrival state  $y$  outside the band, the event probability (21) is  $e(y) = \lambda(y) + (1 - \lambda(y)) = 1$  for every value of  $\lambda(y)$ : the firm's price is corrected this month with certainty, the hazard only decides whether the correction is called an adjustment or an exit. Both events place a gap of 0 at that slot. The law of motion of the cross-section is therefore independent of  $\lambda$  outside the band. ■

Proposition 2 in the main text states the kernel in spell form. The proof derives it in the equivalent operator form  $\omega = (f/\sigma^2)[m \text{Var}_f - G[(x - \bar{x})^2]]$  and converts at the end; along the way it proves the following lemma, which extends the kernel's two zeros beyond the no-exit baseline.

**Lemma A.3** (Two exact zeros, at any baseline hazard and band).  $\omega(0) = 0$  and  $\omega(\pm a) = 0$ .

Throughout this subsection  $\bar{\pi} = 0$ , and  $G$  is the solution operator of (22):  $w = G[g]$  solves  $\frac{1}{2}\sigma^2 w'' - \lambda w = -g$  with  $w(\pm a) = 0$ , the expected cumulated value of  $g$  over a spell, so that  $m = G[1]$  and  $\bar{M} = G[x - \bar{x}]$ . The solution is unique: a difference of two solutions solves the homogeneous problem with zero boundary values, and with  $\lambda \geq 0$  only the zero function does (at  $\lambda \equiv 0$ ,  $w'' = 0$  with zero boundary values forces  $w$  linear, then zero). Two operator facts carry the proof; later arguments point back to them, so I state them as lemmas.

**Lemma A.4** (Swap identity). *For continuous  $u, v$ ,*

$$\int_{-a}^a u G[v] dx = \int_{-a}^a G[u] v dx.$$

*Proof.* With  $w_1 = G[u]$  and  $w_2 = G[v]$ ,  $u = \lambda w_1 - \frac{1}{2}\sigma^2 w_1''$  and  $v = \lambda w_2 - \frac{1}{2}\sigma^2 w_2''$ . Write

$$\int u w_2 = \int (\lambda w_1 - \frac{1}{2}\sigma^2 w_1'') w_2 = \int \lambda w_1 w_2 - \frac{1}{2}\sigma^2 \int w_1'' w_2.$$

Use integration by parts twice on the last term:

$$-\int w_1'' w_2 = \underbrace{[w_1' w_2]_{-a}^a}_{=0, w_2(\pm a)=0} - \int w_1' w_2' = - \int w_1' w_2' = [w_1 w_2']_{-a}^a + \int w_1 w_2'' = \int w_1 w_2''.$$

Reassemble:

$$\int u w_2 = \int \lambda w_1 w_2 - \frac{1}{2} \sigma^2 \int w_1 w_2'' = \int w_1 (\lambda w_2 - \frac{1}{2} \sigma^2 w_2'') = \int w_1 v.$$

■

**Lemma A.5** (Duality). *For continuous  $w$ ,  $\mathbb{E}_f[w] = N G[w](0)$ ; taking  $w \equiv 1$ ,  $N = 1/m(0)$ .*

*Proof.* Recall the stationarity identity (25) at zero drift,

$$\mathbb{E}_f \left[ \frac{1}{2} \sigma^2 g'' - \lambda g \right] + N g(0) - N^+ g(a) - N^- g(-a) = 0.$$

Take  $g = G[w]$ . By the definition of  $G$ ,  $\frac{1}{2} \sigma^2 g'' - \lambda g = -w$  at every point of the band, so the first term equals  $-\mathbb{E}_f[w]$ ; and  $g(\pm a) = 0$  kills the last two terms. What remains is  $\mathbb{E}_f[w] = N G[w](0)$ . With  $w \equiv 1$ ,  $G[1] = m$  and  $\mathbb{E}_f[1] = 1$ , so  $1 = N m(0)$ , that is  $N = 1/m(0)$ . ■

In words, the duality says that a spell launched from the reset point accumulates, per unit of expected spell length, exactly the cross-sectional average of any function. This is the fact that prices exit at the reset point: killing a firm at 0 exchanges objects of equal value, and Step 4 below turns this into the kernel's zero at  $x = 0$ .

*Proof of Proposition 2 and Lemma A.3.* Throughout, a dot denotes a derivative with respect to  $\theta$ , evaluated at  $\theta = 0$ .

Step 1: the derivative of the density. The pair  $(f_\theta, N_\theta)$  solves a linear two-point boundary value problem whose coefficients contain  $\theta$  linearly, and solutions of such problems are differentiable in the parameter: fundamental solutions of a linear ODE are analytic in a parameter entering the coefficients analytically, and the boundary, matching, and normalization conditions form a finite linear system with nonzero determinant at  $\theta = 0$  (Coddington and Levinson, 1955, Chapter 1). Differentiating (24) at  $\theta = 0$ :

$$\frac{1}{2} \sigma^2 \dot{f}'' - \lambda \dot{f} = h f - \dot{N} \delta_0, \quad \dot{f}(\pm a) = 0. \quad (35)$$

Solve by guess-and-verify: guess that

$$\phi: = -G[hf] + \frac{\dot{N}}{N} f$$

solves (35). Apply the linear operation  $\frac{1}{2} \sigma^2 (\cdot)'' - \lambda (\cdot)$  to each piece. By the definition of  $G$ ,  $w = G[hf]$  solves  $\frac{1}{2} \sigma^2 w'' - \lambda w = -hf$ ; multiplying by  $-1$ , the piece  $-w$  satisfies  $\frac{1}{2} \sigma^2 (-w)'' - \lambda (-w) = hf$ . And  $f$  solves the stationary equation (24) at zero drift,  $\frac{1}{2} \sigma^2 f'' - \lambda f = -N \delta_0$ ; multiplying by the constant  $\frac{\dot{N}}{N}$ , the piece  $\frac{\dot{N}}{N} f$  satisfies  $\frac{1}{2} \sigma^2 (\frac{\dot{N}}{N} f)'' - \lambda \frac{\dot{N}}{N} f = -\dot{N} \delta_0$ . Sum the two:

$\frac{1}{2}\sigma^2\phi'' - \lambda\phi = hf - \dot{N}\delta_0$ , the right side of (35). The boundaries follow,  $\phi(\pm a) = -w(\pm a) + \frac{\dot{N}}{N}f(\pm a) = 0 + 0 = 0$ , and the behavior at the kink follows from  $f$ . Economically: the perturbation kills firms at rate  $hf$  wherever they stand,  $-G[hf]$  deletes the density those firms would have contributed over their remaining spells, and the killed firms reappear at the reset point, re-spreading in the shape of the stationary density.

Step 2: mass conservation pins  $\dot{N}$ . From  $\int f_\theta = 1$ ,

$$0 = \int \dot{f} = - \int G[hf] + \frac{\dot{N}}{N} \int f \implies \frac{\dot{N}}{N} = \int G[hf].$$

Apply Lemma A.4 with  $u = hf$  and  $v = 1$ :

$$\frac{\dot{N}}{N} = \int G[hf] \cdot 1 = \int hf \cdot G[1] = \int h(x)f(x)m(x)dx = \mathbb{E}_f[hm] :$$

killing a firm brings its reset forward by its remaining spell.

Step 3: the kernel. The mean's own motion drops out of the variance derivative. Write the variance as a function of  $\theta$  and of the centering point  $c$ ,  $\text{Var}(\theta, c) = \int (x - c)^2 f_\theta dx$ , so that the total derivative at  $c = \bar{x}$  is

$$\left. \frac{d\text{Var}(\theta, c)}{d\theta} \right|_{c=\bar{x}} = \frac{\partial \text{Var}(\theta, \bar{x})}{\partial \theta} + \frac{\partial \text{Var}(\theta, \bar{x})}{\partial c} \cdot \frac{\partial \bar{x}}{\partial \theta}.$$

The second term vanishes because the mean minimizes the mean squared deviation:

$$\frac{\partial \text{Var}(\theta, c)}{\partial c} = \int (-2)(x - c)f(x)dx = -2 \left( \int xf(x)dx - c \int f(x)dx \right) = -2(\bar{x} - c),$$

which is zero at  $c = \bar{x}$ , and thus  $\frac{d\text{Var}_f}{d\theta} = \int (x - \bar{x})^2 \dot{f} dx$ . Substitute  $\dot{f}$  from Step 1 and  $\dot{N}/N$  from Step 2:

$$\frac{d\text{Var}_f}{d\theta} = \int (x - \bar{x})^2 \left[ -G[hf] + \frac{\dot{N}}{N}f \right] dx = - \int (x - \bar{x})^2 G[hf] dx + \mathbb{E}_f[hm] \text{Var}_f.$$

Apply Lemma A.4 with  $u = hf$  and  $v = (x - \bar{x})^2$  to the first term, and write the second as  $\int hf m \text{Var}_f dx$ :

$$\frac{d\text{Var}_f}{d\theta} = \int hf [m \text{Var}_f - G[(x - \bar{x})^2]] dx.$$

Divide by  $\sigma^2$  and apply Proposition 1, which holds at every  $\theta$  and at zero drift reads  $\text{CIR} = \text{Var}_f/\sigma^2$ :

$$\left. \frac{d \text{CIR}}{d\theta} \right|_{\theta=0} = \int_{-a}^a h(x) \omega(x) dx, \quad \omega(x) = \frac{f(x)}{\sigma^2} [m(x) \text{Var}_f - G[(x - \bar{x})^2](x)].$$

Finally, since  $\text{Var}_f$  is a constant,  $m(x) \text{Var}_f = \mathbb{E}_x \int_0^\tau \text{Var}_f dt$  and  $G[(x - \bar{x})^2](x) = \mathbb{E}_x \int_0^\tau (x_t - \bar{x})^2 dt$ , so the bracket is the spell integral of  $\text{Var}_f - (x_t - \bar{x})^2$  and  $\omega$  takes the form displayed in Proposition 2.

Step 4: Lemma A.3. At  $x = 0$  the bracket itself vanishes: by Lemma A.5 applied to  $(x - \bar{x})^2$  and to 1,  $G[(x - \bar{x})^2](0) = \text{Var}_f/N$  and  $m(0)\text{Var}_f = \text{Var}_f/N$ ; so  $\omega(0) = 0$  through the bracket. At  $\pm a$ ,  $f = 0$  while  $m$  and  $G[(x - \bar{x})^2]$  are bounded, so  $\omega(\pm a) = 0$ ; in fact both terms of the bracket obey the zero boundary conditions, so the bracket vanishes there too and  $\omega$  has a double zero at the band edge, which is the  $(a - |x|)^2$  factor of the closed form.

Step 5: closed form at the no-exit baseline. At  $\lambda \equiv 0$ ,  $\bar{x} = 0$  by symmetry; compute each ingredient of the bracket.

- $f_0$ :  $\frac{1}{2}\sigma^2 f'' = 0 \implies f = c(a - |x|) \implies 1 = 2c \int_0^a (a - x) dx = ca^2 \implies c = \frac{1}{a^2}$ , then  $f_0 = \frac{1}{a^2}(a - |x|)$ .
- $m_0$ :  $m = G[1]$  solves  $\frac{1}{2}\sigma^2 m'' = -1 \implies m'' = -\frac{2}{\sigma^2}$ . Integrate twice to get  $m = -\frac{x^2}{\sigma^2} + c_1 x + c_0$ ; symmetry kills  $c_1$ , and  $m(a) = 0$  gives  $c_0 = \frac{a^2}{\sigma^2}$ , hence  $m_0 = \frac{a^2 - x^2}{\sigma^2}$ .
- $G_0[x^2]$  solves  $\frac{1}{2}\sigma^2 w'' = -x^2$ , or  $w'' = -2\frac{x^2}{\sigma^2}$ . Integrate twice to get  $w = -\frac{x^4}{6\sigma^2} + c_1 x + c_0$ , kill  $c_1$  by symmetry and pin  $c_0$  with  $w(a) = 0$ :  $G_0[x^2] = \frac{a^4 - x^4}{6\sigma^2}$ .
- $\text{Var}_0$ :  $\text{Var}_0 = \int x^2 f_0 = \frac{2}{a^2} \int_0^a x^2 (a - x) dx = \frac{2}{a^2} \left( \frac{a^4}{3} - \frac{a^4}{4} \right) = \frac{a^2}{6}$ .

Putting everything together, the bracket of the kernel is

$$m_0 \text{Var}_0 - G_0[x^2] = \frac{a^2 - x^2}{\sigma^2} \cdot \frac{a^2}{6} - \frac{a^4 - x^4}{6\sigma^2} = -\frac{x^2(a^2 - x^2)}{6\sigma^2} \leq 0,$$

with equality iff  $x = 0$  or  $x = \pm a$ , and the kernel equals

$$\omega_0 = \frac{a - |x|}{a^2} \cdot \frac{1}{\sigma^2} \left( -\frac{x^2(a^2 - x^2)}{6\sigma^2} \right) = -\frac{x^2(a - |x|)^2(a + |x|)}{6\sigma^4 a^2} \leq 0,$$

using  $a^2 - x^2 = (a - |x|)(a + |x|)$ . Its integral: the integrand  $x^2(a - x)^2(a + x)$  expands to  $a^3 x^2 - a^2 x^3 - ax^4 + x^5$  with integral  $a^6(\frac{1}{3} - \frac{1}{4} - \frac{1}{5} + \frac{1}{6}) = a^6/20$  over  $[0, a]$ ; doubling and applying the prefactor  $-1/(6\sigma^4 a^2)$  gives  $\int \omega_0 = -a^4/(60\sigma^4)$ , matching the linear term of the expansion in Proposition A.2, as it must, since a flat unit of hazard costs the integral of the kernel. ■

#### F4 The amplification theorem

The theorem collects the local form stated in the main text and a complete form that removes the small-flow limit. Condition M is stated in Section 6.3; part (ii) uses its placement half, the requirement that the selective hazard not exceed the uniform one away from the band edges.

**Theorem A.1** (Selection amplifies: local and complete form). *Let  $\lambda_s \geq 0$  be bounded with the same stationary exit flow as  $\lambda_u \equiv \delta$ , and let  $\omega_\theta$  be the kernel of Proposition 2 at the baseline hazard  $\lambda_\theta = (1 - \theta)\lambda_u + \theta\lambda_s$ .*

- (i) *Local form (Theorem 1). Write  $\lambda_s = \theta h$  and  $\lambda_u = \theta c$  with  $h \geq 0$  bounded,  $c = \mathbb{E}_{f_0}[h]$ , and intensity  $\theta > 0$ . Then*

$$\text{CIR}_s - \text{CIR}_u = \theta \int_{-a}^a \omega_0 (h - c) dx + O(\theta^2),$$

and under Condition M there is  $\bar{\theta}(h) > 0$  with  $\text{CIR}_s > \text{CIR}_u$  for every  $\theta \in (0, \bar{\theta})$ .

(ii) Complete form.

$$\text{CIR}_s - \text{CIR}_u = \int_0^1 \int_{-a}^a \omega_\theta (\lambda_s - \delta) dx d\theta.$$

Moreover, fix  $\varepsilon > 0$ , write  $M_\varepsilon := \{x : a - |x| \leq \varepsilon\}$  for the set within  $\varepsilon$  of the band edges and  $I_\varepsilon$  for its complement in the band, and assume  $\lambda_s \leq \delta$  on  $I_\varepsilon$ , so that the selective economy's excess exit is confined near the margin. Then

$$\text{CIR}_s - \text{CIR}_u \geq G - \Sigma - K\varepsilon^2 X, \quad \text{where}$$

$$G := \int_0^1 \int_{I_\varepsilon} (\omega_\theta)_- (\delta - \lambda_s) dx d\theta, \quad \Sigma := \int_0^1 \int_{I_\varepsilon} (\omega_\theta)_+ (\delta - \lambda_s) dx d\theta, \quad X := \int_{M_\varepsilon} |\lambda_s - \delta| dx,$$

$(\omega)_\pm$  are the negative and positive parts of the kernel (both  $\geq 0$ ,  $\omega = (\omega)_+ - (\omega)_-$ ),  $G, \Sigma \geq 0$ , and  $K < \infty$  is a constant depending only on  $(\lambda_s, \delta, a, \sigma)$ . In particular  $\text{CIR}_s > \text{CIR}_u$  whenever

$$G > \Sigma + K\varepsilon^2 X.$$

*Proof of Theorem A.1 and Theorem 1.* The uniform hazard  $\theta c$  generates a stationary exit flow of exactly  $\theta c$ : the hazard is constant and the density integrates to one. The selective hazard  $\theta h$  generates the flow  $\int \theta h f_{\theta h} dx$ , and since the stationary density converges to the no-exit density  $f_0$  as  $\theta \rightarrow 0$  (by the parameter smoothness established in (S1) below), that flow equals  $\theta \mathbb{E}_{f_0}[h] + O(\theta^2) = \theta c + O(\theta^2)$ . The choice  $c = \mathbb{E}_{f_0}[h]$ , an expectation under the no-exit density, therefore matches the two exit flows to first order in  $\theta$ , i.e. part (i). Two useful results first.

(S1) For bounded hazards  $\lambda', \lambda'' \geq 0$  let  $\lambda_t := (1 - t)\lambda' + t\lambda''$ ,  $t \in [0, 1]$ , and write  $\omega_{\lambda_t}$  for the kernel at baseline  $\lambda_t$ . Then:  $t \mapsto \text{CIR}(\lambda_t)$  is continuously differentiable with derivative

$$\frac{d \text{CIR}(\lambda_t)}{dt} = \int_{-a}^a \omega_{\lambda_t}(x) (\lambda''(x) - \lambda'(x)) dx, \quad (36)$$

and the kernel is Lipschitz in  $t$  uniformly in  $x$ : there is  $L < \infty$  with

$$\sup_{x \in [-a, a]} |\omega_{\lambda_t}(x) - \omega_{\lambda_\tau}(x)| \leq L |t - \tau| \quad \text{for all } t, \tau \in [0, 1]. \quad (37)$$

The pair  $(f_t, N_t)$  and the images  $m_t = G_t[1]$  and  $G_t[(\cdot - \bar{x}_t)^2]$  solve linear two-point boundary value problems whose coefficients are affine in  $t$ , so the parameter-smoothness argument of Step 1 of the proof of Proposition 2 applies at every  $t$ : the boundary and normalization system stays invertible along the segment, since  $\lambda_t \geq 0$ , and its solutions are analytic in  $t$ . Steps 1 through 3 of that proof then run verbatim at any bounded baseline hazard and any bounded signed direction, here  $\lambda'' - \lambda'$ , which gives (36). Steps 1 through 3 run verbatim at any bounded baseline hazard  $\lambda_t \geq 0$  and any bounded signed direction, here  $\lambda'' - \lambda'$ , which gives (36). For (37):  $\omega_{\lambda_t} = f_t B_t / \sigma^2$  with the factors defined in (S2) below, so  $\partial_t \omega_{\lambda_t}(x)$  exists and is continuous in  $(t, x)$  on a compact set, hence bounded, say by  $L$ ; the mean value theorem in  $t$  then gives (37).

In particular the right side of (36) is continuous in  $t$ , so the fundamental theorem of calculus applies to  $t \mapsto \text{CIR}(\lambda_t)$  on  $[0, 1]$ :

$$\text{CIR}(\lambda'') - \text{CIR}(\lambda') = \int_0^1 \int_{-a}^a \omega_{\lambda_t} (\lambda'' - \lambda') dx dt. \quad (38)$$

(S2) There is  $K < \infty$ , uniform over the segment, with

$$|\omega_{\lambda_t}(x)| \leq K(a - |x|)^2 \quad \text{for all } x \in [-a, a], t \in [0, 1]. \quad (39)$$

Write  $\omega_{\lambda_t}(x) = f_t(x) B_t(x) / \sigma^2$  with

$$B_t(x) := m_t(x) \text{Var}_t - G_t[(\cdot - \bar{x}_t)^2](x),$$

Both factors vanish at the band edges:  $f_t$  by its boundary condition, and  $B_t$  because  $m_t$  and  $G_t[(\cdot - \bar{x}_t)^2]$  both do (Lemma A.3). Set

$$C_1 := \sup_{t \in [0, 1]} \sup_{x \in [-a, a]} |f'_t(x)|, \quad C_2 := \sup_{t \in [0, 1]} \sup_{x \in [-a, a]} |B'_t(x)|.$$

Both are finite: by (S1) each derivative is continuous in  $(t, x)$  on a compact set, except for the single bounded jump of  $f'_t$  at the reset point, where the supremum is taken over the one-sided limits, themselves continuous in  $t$ ; a function continuous on a compact set up to finitely many bounded jumps is bounded, so the suprema exist. Now fix  $x \in [0, a]$ . The interval  $[x, a]$  does not contain the reset point in its interior, so the fundamental theorem of calculus applies on it, and using  $f_t(a) = 0$  and  $B_t(a) = 0$ ,

$$f_t(x) = - \int_x^a f'_t(\iota) d\iota \leq C_1 (a - x), \quad |B_t(x)| = \left| \int_x^a B'_t(\iota) d\iota \right| \leq C_2 (a - x),$$

Since  $f_t \geq 0$ , the first bound is also a bound on  $|f_t|$ . Multiply the two bounds and divide by  $\sigma^2$ :

$$|\omega_{\lambda_t}(x)| = \frac{f_t(x) |B_t(x)|}{\sigma^2} \leq \frac{C_1 C_2}{\sigma^2} (a - x)^2, \quad x \in [0, a],$$

and the mirror argument on  $[-a, 0]$ , integrating from  $-a$  to  $x$ , gives the same bound with  $(a + x)^2$ . So (39) holds with  $K = C_1 C_2 / \sigma^2$ , a constant that depends only on the segment's data  $(\lambda', \lambda'', a, \sigma)$ .

Part (i). Fix the direction  $h$  and its flow-matched level  $c$ . Apply (S1) to the segment from the zero hazard to  $h$ , that is  $\lambda_t = th$ , and separately to the segment from the zero hazard to  $c$ ; write  $\omega_{sh}$  and  $\omega_{sc}$  for the kernels at baselines  $sh$  and  $sc$ ,  $s \in [0, 1]$ . Each application of (37) produces a Lipschitz constant; let  $C_0 = C_0(h)$  be the larger of the two. Two features matter. First,  $C_0$  is attached to the two fixed segments with endpoints  $h$  and  $c$ , in which  $\theta$  appears nowhere, so  $C_0$  does not depend on  $\theta$ ; it does depend on  $h$ , which is why  $\bar{\theta}$  below is a function of  $h$ . Second,

both segments start at the zero hazard, where the kernel is  $\omega_0$ , so (37) at  $\tau = 0$  reads

$$\sup_x |\omega_{sh}(x) - \omega_0(x)| \leq C_0 s, \quad \sup_x |\omega_{sc}(x) - \omega_0(x)| \leq C_0 s, \quad s \in [0, 1]. \quad (40)$$

Apply (38) to the segment from the zero hazard to  $\theta h$ : with  $\lambda' = 0$  and  $\lambda'' = \theta h$ , the baseline at parameter  $t$  is  $t\theta h$ , and since the kernel depends only on the baseline hazard, the kernel there is  $\omega_{sh}$  with  $s = t\theta$ . Substituting  $s = t\theta$ , so that  $\theta dt = ds$ , and repeating the computation along the segment from the zero hazard to  $\theta c$ ,

$$\begin{aligned} \text{CIR}(\theta h) - \text{CIR}(0) &= \int_0^1 \int_{-a}^a \omega_{t\theta h} \theta h dx dt = \int_0^\theta \int_{-a}^a \omega_{sh} h dx ds, \\ \text{CIR}(\theta c) - \text{CIR}(0) &= \int_0^\theta \int_{-a}^a \omega_{sc} c dx ds. \end{aligned}$$

The substitution stays inside the domain of (40) because  $\theta \leq 1$ . Subtract the two lines:

$$\text{CIR}_s - \text{CIR}_u = \int_0^\theta \int_{-a}^a (\omega_{sh} h - \omega_{sc} c) dx ds, \quad (41)$$

isolate the leading term, add and subtract  $\omega_0$  in each product

$$\omega_{sh} h - \omega_{sc} c = \omega_0 (h - c) + (\omega_{sh} - \omega_0) h - (\omega_{sc} - \omega_0) c. \quad (42)$$

and bound the correction terms. First,  $c \leq M_h := \sup h$ , because  $f_0$  is a probability density:  $c = \int h f_0 dx \leq M_h \int f_0 dx = M_h$ . Then, for each  $s \in [0, \theta]$ , by the triangle inequality,

$$\left| \int_{-a}^a [(\omega_{sh} - \omega_0) h - (\omega_{sc} - \omega_0) c] dx \right| \leq \int_{-a}^a |\omega_{sh} - \omega_0| h dx + \int_{-a}^a |\omega_{sc} - \omega_0| c dx;$$

pull the suprema (40) out of each integral to get the bound  $C_0 s \int h dx + C_0 s \int c dx$ ; the first remaining integral is at most  $2aM_h$  and the second equals  $2ac$ , a constant over an interval of length  $2a$ ; so the display is bounded by  $C_0 s \cdot 2a (M_h + c) \leq 4aM_h C_0 s$ , using  $c \leq M_h$  once more.

Substitute (42) into (41) and integrate each term over  $s \in [0, \theta]$ . The leading term does not depend on  $s$ , so its integral is  $\theta \int \omega_0 (h - c) dx$ . The two correction terms define the remainder,

$$R(\theta) := \int_0^\theta \int_{-a}^a [(\omega_{sh} - \omega_0) h - (\omega_{sc} - \omega_0) c] dx ds, \quad |R(\theta)| \leq \int_0^\theta 4aM_h C_0 s ds = 2aM_h C_0 \theta^2,$$

by the previous step. Together,

$$\text{CIR}_s - \text{CIR}_u = \theta \int_{-a}^a \omega_0 (h - c) dx + R(\theta), \quad |R(\theta)| \leq 2aM_h C_0 \theta^2,$$

with an explicit constant on the  $O(\theta^2)$  term.

Finally, define  $I := \int_{-a}^a \omega_0 (h - c) dx$ . Condition M, stated for the pair  $(\lambda_s, \lambda_u) = (\theta h, \theta c)$ , gives  $\int \omega_0 (\theta h - \theta c) dx > 0$ , that is  $\theta I > 0$ ; since  $\theta > 0$ , the condition is equivalent to  $I > 0$ .

Then

$$\text{CIR}_s - \text{CIR}_u = \theta I + R(\theta) \geq \theta I - 2aM_h C_0 \theta^2 = \theta (I - 2aM_h C_0 \theta) > 0$$

for every  $\theta < I/(2aM_h C_0)$ . Intersecting with the restriction  $\theta \leq 1$ ,

$$\bar{\theta}(h) := \min \left\{ 1, \frac{I}{2aM_h C_0} \right\},$$

and  $\text{CIR}_s > \text{CIR}_u$  for every  $\theta \in (0, \bar{\theta})$ , proving part (i) and Theorem 1.

Part (ii). Apply (38) to the segment  $\lambda_\theta = (1 - \theta)\lambda_u + \theta\lambda_s$ , that is  $\lambda' = \lambda_u \equiv \delta$  and  $\lambda'' = \lambda_s$ ; the result is the displayed identity of part (ii).

Split the inaction band:  $[-a, a]$  is the disjoint union of  $M_\varepsilon$  and  $I_\varepsilon$ , so

$$\text{CIR}_s - \text{CIR}_u = \int_0^1 \int_{M_\varepsilon} \omega_\theta (\lambda_s - \delta) dx d\theta + \int_0^1 \int_{I_\varepsilon} \omega_\theta (\lambda_s - \delta) dx d\theta.$$

On  $M_\varepsilon$ ,  $a - |x| \leq \varepsilon$ , so (39) gives  $|\omega_\theta| \leq K\varepsilon^2$  there, uniformly in  $\theta$ . Hence

$$\left| \int_0^1 \int_{M_\varepsilon} \omega_\theta (\lambda_s - \delta) dx d\theta \right| \leq K\varepsilon^2 \int_0^1 \int_{M_\varepsilon} |\lambda_s - \delta| dx d\theta = K\varepsilon^2 \int_{M_\varepsilon} |\lambda_s - \delta| dx,$$

the  $\theta$ -integral collapsing because its integrand no longer depends on  $\theta$ . The surviving integral is  $X$  by definition, so the edge piece lies in  $[-K\varepsilon^2 X, K\varepsilon^2 X]$ .

Define the positive and negative parts of the kernel pointwise,

$$(\omega_\theta)_+(x) := \max\{\omega_\theta(x), 0\}, \quad (\omega_\theta)_-(x) := \max\{-\omega_\theta(x), 0\},$$

so that  $\omega_\theta = (\omega_\theta)_+ - (\omega_\theta)_-$ . Both parts are continuous because  $\omega_\theta$  is, and bounded on the band by (S2), so every integral below is finite. On  $I_\varepsilon$  the assumption  $\lambda_s \leq \delta$  gives  $\delta - \lambda_s \geq 0$ . Multiply the identity by  $(\lambda_s - \delta) = -(\delta - \lambda_s)$ :

$$\omega_\theta (\lambda_s - \delta) = (\omega_\theta)_- (\delta - \lambda_s) - (\omega_\theta)_+ (\delta - \lambda_s),$$

a difference of two functions that are pointwise nonnegative on  $I_\varepsilon$ . Integrating over  $I_\varepsilon \times [0, 1]$ , the first term produces the integral  $G$  in the statement and the second the integral  $\Sigma$ , both nonnegative:

$$\int_0^1 \int_{I_\varepsilon} \omega_\theta (\lambda_s - \delta) dx d\theta = G - \Sigma.$$

Substituting,

$$\text{CIR}_s - \text{CIR}_u \geq -K\varepsilon^2 X + G - \Sigma,$$

which is the bound of part (ii), strictly positive whenever  $G > \Sigma + K\varepsilon^2 X$ . The constant is the  $K = C_1 C_2 / \sigma^2$  of (S2) for the segment from  $\delta$  to  $\lambda_s$ , so it depends only on  $(\lambda_s, \delta, a, \sigma)$ . ■

The two loss terms in part (ii) have clear sources. The margin term  $K\varepsilon^2 X$  prices the fact that in the continuous model the neutrality of the band edge is a double zero rather than an

exact invariance; in the discrete monthly model, where Proposition A.3 makes hazard outside the band exactly free, the corresponding term is zero. The term  $\Sigma$  collects the positive part of the kernel on the interior. At symmetric baselines the kernel is nonpositive and  $\Sigma$  vanishes; at asymmetric baselines a thin positive region can appear just past the reset point on the side opposite the hazard, because one-sided exit drags the cross-sectional mean off the reset point, and a firm sitting between the reset point and the mean then carries less dispersion than average over its remaining spell. This region is second order in the hazard's asymmetry; the theorem carries it as a budgeted term so that nothing about its sign is assumed.

## F.5 The sufficient statistic under exit

This appendix locates the model in the sufficient-statistic literature. Once exits are counted as price changes, the reduced model at zero drift is a generalized-hazard economy in the sense of Alvarez et al. (2022): in their economies every interior hazard event is an observed price adjustment, while here the interior hazard events are exits. The proposition shows that the kurtosis-over-frequency statistic of Alvarez et al. (2016) is exact for the pooled changes, with one correction term for asymmetric hazards; at a symmetric hazard it is Theorem 3 of Alvarez et al. (2022), and the second-moment identity is equation (4) of Alvarez et al. (2016). Everything is at zero drift,  $\bar{\pi} = 0$ .

Count as a pooled price change every reset: an adjuster at gap  $x$  changes its price by  $-x$ , and an exit at gap  $x$  replaced by an entrant at 0 changes the price at that slot by  $-x$  as well. Let  $N$  be the pooled frequency and  $\Delta p$  the pooled size distribution, so that  $N\mathbb{E}[\Delta p^k] = \int (-x)^k \lambda f dx + N^+(-a)^k + N^-a^k$ .

**Proposition A.4** (Sufficient statistic with entry and exit). *At  $\bar{\pi} = 0$ , for any hazard and band:  $\mathbb{E}[\Delta p] = 0$ ,  $N\mathbb{E}[\Delta p^2] = \sigma^2$ ,  $N\mathbb{E}[\Delta p^4] = 6\sigma^2\mathbb{E}_f[x^2]$ , and therefore*

$$\text{CIR} = \frac{\text{Kurt}(\Delta p)}{6N} - \left(\frac{\bar{x}}{\sigma}\right)^2.$$

*Proof of Proposition A.4.* Three applications of (25) at  $\bar{\pi} = 0$ . With  $g = x$ :

$$\mathbb{E}_f[\lambda x] + N^+a - N^-a = 0$$

which is  $N\mathbb{E}[\Delta p] = 0$ . With  $g = x^2$ :

$$\sigma^2 = \mathbb{E}_f[\lambda x^2] + (N^+ + N^-)a^2 = N\mathbb{E}[\Delta p^2].$$

With  $g = x^4$ :

$$6\sigma^2\mathbb{E}_f[x^2] = \mathbb{E}_f[\lambda x^4] + (N^+ + N^-)a^4 = N\mathbb{E}[\Delta p^4]$$

Since  $\mathbb{E}[\Delta p] = 0$  by the first identity, raw and central moments coincide, so  $\text{Kurt}(\Delta p) =$

$\mathbb{E}[\Delta p^4]/\mathbb{E}[\Delta p^2]^2$ ; then

$$\frac{\text{Kurt}}{6N} = \frac{N\mathbb{E}[\Delta p^4]}{6(N\mathbb{E}[\Delta p^2])^2} = \frac{\mathbb{E}_f[x^2]}{\sigma^2} = \frac{\text{Var}_f + \bar{x}^2}{\sigma^2} = \text{CIR} + \frac{\bar{x}^2}{\sigma^2}$$

by Proposition 1. ■